

Preventing School Dropout at Scale: Experimental Evidence from Guatemala*

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Abstract

We evaluate a randomized dropout prevention program across 4,000 schools in Guatemala, where 30% of children leave school during the primary-to-secondary transition. Schools were assigned to receive a guidance manual and training; the manual, the training and a list of high-risk students; the manual, the training, the list, and behavioral nudges; or control. All treatments reduced dropout in the transition from primary to secondary by 1.2 percentage points from a 34% base, but effects faded after two years (i.e., there is no difference in the likelihood of being enrolled in secondary school after the first year). Still, the program increased years of schooling by 0.012, and its low cost (USD 2–3 per student) and successful large-scale implementation make it a promising, cost-effective approach to increasing schooling in resource-constrained contexts.

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1 Introduction

School dropout remains a major challenge in low- and middle-income countries. While primary enrollment increased from 70% in 1970 to 90% in 2020 (World Bank, 2019e), dropout remains widespread at the secondary level in such countries, where enrollment rates are only 66% (World Bank, 2019f). Children who fail to progress to higher grades tend to have lower wages, poorer health outcomes, and higher crime propensities in later life (Lochner & Moretti, 2004; Oreopoulos, 2007; Black, Devereux, & Salvanes, 2008; Oreopoulos & Salvanes, 2011). Dropout is often concentrated in the transition between primary and secondary school (Gibbs & Heaton, 2014; Kattan & Székely, 2015; Adelman & Székely, 2017) and can be largely attributed to structural factors such as credit constraints (Stinebrickner & Stinebrickner, 2008; Kilburn, Handa, Angeles, Mvula, & Tsoka, 2017; Evans, Gale, & Kosec, 2023), low demand for skilled employment (Atkin, 2016; Cascio & Narayan, 2022; Ferriere, Navarro, & Reyes-Heroles, 2018) and the shortage of secondary schools (Campos-Vazquez & Santillan, 2018). However, due to public budgetary constraints, there is a demand for low-cost, high-yield interventions that prevent dropout in the short run.

This paper provides experimental evidence from a low-cost intervention to prevent dropout in Guatemala that is implemented at scale using existing resources. Guatemala has one of the highest school dropout rates in Latin America (and the world) (World Bank, 2019c, 2019d). The main goal of the intervention — developed by the Ministry of Education in collaboration with the World Bank — is to prevent students from leaving school during the transition between primary and secondary school — a time when a third of students currently drop out (Adelman, Haimovich, Ham, & Vazquez, 2018). The program addresses three main drivers that are important causes of dropout during this transition in Guatemala *and* addressable through low-cost interventions: 1) a lack of knowledge among key primary school actors about effective measures to help students stay in school, 2) a lack of knowledge among these actors about which students are most at risk of dropping out, and 3) a failure to prioritize retention, partly because most dropouts occur after students have left primary school. The intervention has three components to address each factor. The first component is a half-day training and a guide for school principals and sixth-grade teachers on evidence-based methods to help students transition from primary to lower secondary school. The guide has strategies to address barriers such as poor academic performance, financial constraints, and lack of motivation, using tools such as tutoring and remedial education, scholarship application guidance, and information on economic returns to education. The second component is a list of students at risk of dropping out based on historical data and a statistical model. The model uses age, sex, GPA, grade repetition history, and school-level historical

dropout as the inputs. This list aims to help schools allocate targeted support to the most vulnerable students. The third component is behavioral nudges to maintain the necessary focus on dropout. These nudges were delivered monthly for five months via an online portal regularly used by principals for administrative tasks.

We evaluate this intervention at scale using a randomized control trial. In the 2018 academic year (which follows the calendar year), 1,000 primary schools were randomly assigned to receive the training and guide on dropout prevention strategies; another 1,000 schools were assigned to receive the list of students in addition to the training and guide; and an additional 1,000 schools received the training, guide, list, and behavioral nudges. The remaining 1,000 schools served as the control group. We use administrative data from all schools in the country to track dropout rates between 2013 and 2022. Since the program was designed by the government and implemented using existing resources, the treatment effects from our experiment likely represent relevant estimates for a program that is scaled nationwide.¹

First, we estimate the treatment’s effect on the transition from primary (sixth grade) to secondary (seventh grade) school in 2018 (i.e., whether students remained enrolled in school in 2019). Children in treated schools were 1.1–1.2 percentage points (p-value 0.055) less likely to drop out of school by 2019 — a change of -3.3 to -3.5% from a base of 34%. The effects are similar across treatment arms (and not statistically different from each other). These results are robust to various controls and econometric specifications.

Next, we estimate whether the positive treatment effect on the transition rate in 2018 between primary and secondary school persists over time. Typically, schools only teach primary or secondary grades. Thus, if students successfully transition from sixth to seventh grade, they need to change schools and enroll in a secondary school. While it is possible that once students successfully transition they remain enrolled over time, the effect may dissipate if students need continued support and secondary schools and teachers — who were not part of the program — are unable to provide it. Specifically, we explore whether students enrolled in grade 6 in 2018 are still in school beyond 2019. By 2020, the treatment effect on the likelihood that students are still enrolled in school is close to zero (0.71 percentage points) and statistically insignificant (p-value = 0.13). The same is true in 2021 and 2022.² Still, the program increased the average

¹By contrast, the treatment effects of programs developed by non-governmental organizations (NGOs) have been found to decrease after the government takes them up (Vivalt, 2020; Bold, Kimenyi, Mwabu, Ng’ang’a, & Sandefur, 2018).

²The difference between the effect in 2019 and 2020 is statistically significant, as is the difference between the 2019 and the 2022 effect, while the difference between 2019 and 2021 is not statistically significant. See Section 4.4 for more details.

years of schooling by 0.012 — that is, for approximately every 1,000 students attending schools served by this intervention, an additional 12 students received an additional year of education. Overall, these results suggest the program’s effects are short-lived: it increased the likelihood that students enrolled in secondary school, but they ended up dropping out before graduating from secondary school.

Comparing our estimates with other dropout prevention strategies is difficult. Many programs measure whether students are still enrolled shortly after the end of the intervention. Our findings advance an emerging literature documenting whether the treatment effects of development interventions fade out in the years after an intervention ends (e.g., [Bouguen, Huang, Kremer, and Miguel \(2019\)](#)). Focusing on the effects at the end of the intervention suggests this program is highly cost effective compared to those evaluated in prior work (e.g., tutoring, health intervention, school feeding, scholarships, and conditional cash transfers). The intervention we test is low cost (2–3 USD per student), which suggests 0.64 additional years of schooling per 100 USD invested. However, since there are high returns to obtaining a degree ([Jaeger & Page, 1996](#); [Ferrer & Riddell, 2002](#); [Jepsen, Troske, & Coomes, 2014](#); [Rodríguez & Muro, 2015](#)) beyond those of completed years of schooling (the “sheepskin effect”), it is unclear how much students will benefit in the long run from this treatment. However, a back-of-the-envelope calculation based on the returns from an additional year of schooling suggests an internal rate of return for the program of 27% (see Section 5 for more details on these calculations).

Our results complement the evidence gathered in prior studies on early warning systems, which tend to employ routinely collected administrative data to identify at-risk students and take preventative actions tailored to students’ needs. At least 30 U.S. states and the majority of European countries use such systems ([European Commission, 2013](#); [O’Cummings & Therriault, 2015](#); [Balfanz & Byrnes, 2019](#)), but the causal evidence on them is scant and comes exclusively from developed settings (e.g., Norway ([Sletten, Tøge, & Malmberg-Heimonen, 2022](#)), the U.S. Midwest ([Faria et al., 2017](#)), and The Netherlands ([Plak, Cornelisz, Meeter, & van Klaveren, 2022](#))). However, as more low- and middle-income countries invest in developing reliable education management information systems, the student data — centralized enrollment records that allow tracking children across schools and time — that forms the foundation of early warning systems is becoming increasingly available. Our results suggest these systems can temporarily boost retention in settings with higher dropout rates, fewer resources, and more limited government capacity, but more support is needed to retain children who successfully transition to secondary school until graduation. However, given that providing the list of students does not yield any additional benefits, our results suggest that identifying at-risk students

does not seem to be the driving mechanism behind the success of early warning systems; rather, it is training teachers to take action to prevent dropout.

Our findings also contribute to research on using machine learning and other statistical tools in education (e.g., grading students, educational software, and improving retention). For example, historical data (and machine learning) are increasingly used to predict who is likely to dropout (e.g., in South Korea (Lee & Chung, 2019; Chung & Lee, 2019), Tanzania (Mnyawami, Maziku, & Mushi, 2022), Mexico (Márquez-Vera et al., 2016), Denmark (Sara, Halland, Igel, & Alstrup, 2015), and the US (Knowles, 2015)). We show that these predictions may not provide new information to teachers and principals, as supplying the list of highest-risk students does not change the treatment effects of the training and guide. More broadly, our results also speak to the use of technology to improve state capacity. Various advancements have been used to improve government capacity to, for example, monitor environmental degradation (Assunção, Gandour, & Rocha, 2023; Greenstone, He, Jia, & Liu, 2022; Saavedra, 2023), track front-line workers' attendance (Duflo, Hanna, & Ryan, 2012; Banerjee, Duflo, & Glennerster, 2008), collect taxes (Bellon, Dabla-Norris, Khalid, & Lima, 2022; Okunogbe & Pouliquen, 2022; Carrillo, Donaldson, Pomeranz, & Singhal, 2023), and target social program beneficiaries (Barnwal, 2024; Muralidharan, Niehaus, & Sukhtankar, 2016). We provide evidence of a simple technological change — using existing administrative data to track individual dropouts and existing government interfaces to provide behavioral nudges.

Finally, our results contribute to the evidence on behavioral nudges and professional development programs. Although behavioral nudges are often seen as a cheap and effective way to motivate workers and achieve results (Chetty, 2015), our results suggest they have no effect on teacher and principal behavior. These front-line workers are responsible for many tasks in our setting, which may explain why the nudges are ineffective. Consistent with recent large-scale evidence of limited effects from nudges (Linos, Prohovsky, Ramesh, Rothstein, & Unrath, 2022; Bird et al., 2021; Page, Sacerdote, Goldrick-Rab, & Castleman, 2023), we find that monthly motivational messages (nudges) to school principals had no additional impact. It is also possible (and principal surveys support this) that the training already prioritized dropout prevention, making the nudges superfluous. While professional development programs are commonly used to upgrade the (poor) skills (Bold et al., 2017; Brunetti, Büchel, Jakob, Jann, & Steffen, 2023) of teachers in developing countries (Popova, Evans, Breeding, & Arancibia, 2021), there is little evidence of their effectiveness (Jacob & Lefgren, 2004; Piper & Korda, 2011; Loyalka, Popova, Li, & Shi, 2019). We present evidence from one such program and demonstrate that it improves students' outcomes.

2 Context and intervention

2.1 Context

Over the past two decades, Guatemala has expanded access to primary and secondary education. From 2000 to 2019, primary and lower secondary school completion rates rose by 43% and 112%, respectively (World Bank, 2019d, 2019c).³

Yet, much remains to be done. School dropout remains an important challenge to increasing educational attainment, especially in early adolescence.⁴ While net enrollment was close to 90% in primary school in 2018, it was 60% in lower secondary and 40% in upper secondary (UNESCO, 2023). Dropouts are concentrated in the transitions from primary to lower secondary (sixth to seventh grade) and from lower to upper secondary (ninth to tenth grade). In each of these transitions, roughly a third of students do not progress to the next level (Adelman et al., 2018).⁵ Typically, schools only teach primary or secondary grades. Thus, if students successfully transition from sixth to seventh grade, they need to change schools and enroll in a secondary school.⁶ The fact that children change schools when transitioning highlights that dropout is a collective action problem between teachers and principals in primary schools, teachers and principals in secondary schools, parents, and students.

The high dropout rates highlight the significant challenges facing the country's education system. With over half the population living below the national poverty line and the nation ranking sixth globally in chronic malnutrition (Marini & Gragnolati, 2003; UNICEF, 2020), many students enter the school system facing multiple deprivations, and public education services are inadequate to address these challenges. The government spends less than 3% of GDP on education, well below its regional peers.⁷ As a result, the quantity and quality of education supply are insufficient to meet existing needs. Preschool starts relatively late, typically around age 6, lasts only 1 year, and covers less than half of the population. There is also a shortage of secondary schools, which tend to be located in urban and

³Learning outcomes have also improved. Guatemalan sixth graders' performance in reading and math improved more between 2006 and 2013 than in the rest of Latin America and the Caribbean (Flotts et al., 2015). However, learning outcomes stagnated between 2013 and 2019 (UNESCO, 2021).

⁴Evidence from Latin America indicates that greater shares of youth who are out of school and out of work contribute to a range of negative social and economic outcomes including higher crime rates, long-term reductions in labor productivity and growth, and rising inequality (De Hoyos Navarro, Popova, & Rogers, 2016).

⁵Primary and lower secondary education are compulsory and free. However, informal fees are common (Bentaouet, 2006). Upper secondary is not free.

⁶Even when secondary schools are placed adjacent to primary schools, they have a different principal and a different set of teachers.

⁷Low government revenues drive low spending on education, as the government devotes over 20% of its budget to education.

more prosperous areas. In the poorer areas, there is only one secondary school for every 400–500 secondary school-age youth; in wealthier areas, the ratio is close to one school for every 100 youth.⁸ Dropout results from the confluence of these structural problems, including poverty, a low supply of public secondary schools, and credit constraints. While fixing these structural problems takes time, we study a low-cost, targeted and scalable intervention to alleviate the high dropout rate in the short term.

2.2 The ENTRE program

In 2017, motivated by the challenges and constraints described above, the Government of Guatemala, with support from the World Bank, designed the ENTRE (*Estrategia Nacional para la Transición Exitosa* or National Strategy for Successful Transitions) program as a scalable and low-cost early warning system. The program was designed to reduce dropout rates during the transition from primary to lower secondary school. The intervention targets primary schools, specifically principals and sixth-grade teachers (the last year of primary schooling).

Like most early warning systems, the key goals of ENTRE are to provide targeted support to at-risk students and take timely, preventive actions. At a minimum, this requires that school principals and teachers: 1) have a basic knowledge of effective dropout prevention strategies, 2) can identify at-risk students, and 3) can implement these strategies in a timely manner. However, the available quantitative and qualitative data — generated by focus groups with teachers and school principals, as well as interviews with public officials conducted to inform the design of ENTRE — revealed important gaps in all three dimensions.

In the first dimension, school principals and teachers have limited knowledge of dropout prevention strategies. School principals — who play a leading role in these initiatives — are usually regular teachers without specific leadership or management training (Bloom, Lemos, Sadun, Scur, & Van Reenen, 2014; Bloom, Lemos, Sadun, & Van Reenen, 2015; Romero, Bedoya, Yanez-Pagans, Silveyra, & de Hoyos, 2022). Second, principals and teachers have limited information on who is at risk of dropping out. Data on mandatory “dropout risk flags” reported by principals and teachers indicates that they identify fewer than 5% of the students who drop out in the transition from primary to lower secondary.⁹ Third, several parallel demands and tasks compete for the attention of principals and teachers. Dropout prevention may thus not be their top priority, particularly since the

⁸In contrast, there is a primary school in almost every settlement.

⁹This data is collected by asking principals and teachers to identify at-risk students. When they do, they are required to undertake several administrative tasks for each student, which may cause them to under-report students at risk of dropping out.

period under study occurs after students are no longer their responsibility. According to our surveys of school principals, on average, they rank it third out of five options (below teaching/curriculum and internal administrative tasks).

ENTRE combines three key components to address these gaps: 1) a brief training on actionable, evidence-based strategies to prevent students from dropping out; 2) information on sixth-grade students at high risk of dropping out; and 3) small behavioral nudges to increase the urgency of dropout prevention for principals and teachers. The remainder of the section addresses each component in turn.

2.2.1 Component 1: Short training and guide on actionable strategies to prevent dropout

The first component consisted of a half-day training and a practical guide for principals and sixth-grade teachers, focused on evidence-based methods to improve student retention during the primary-to-secondary transition. Given the limited resources available in public schools, the intervention prioritized low-cost, actionable recommendations that teachers and principals could implement with minimal administrative burden.

Each participant received a short and user-friendly guidance manual at the beginning of the training session (13 pages of text and informative illustrations in the main manual, plus 16 pages of additional resources in an annex) that summarizes the training content (Appendix C reproduces the full guide).¹⁰ Along with the guide, the school principals also received a personalized letter from the Minister of Education that emphasized the importance of every Guatemalan child completing at least 9 years of basic education and urged the principal to lead her or his staff to implement what they learned in the training session.

The training and guide included a section on the importance of focusing efforts on students at risk of dropping out. To reduce the risk of negative labeling, the letter from the Minister of Education and the training content cautioned against stigmatizing at-risk students. Participants were encouraged to use their judgment to identify students who were more likely to leave school. They were also informed that some schools would receive complementary information on at-risk students (component 2, explained below) as part of the ministry's efforts to identify the most effective content for ENTRE.

The half-day sessions culminated in school-specific action plans, with principals and teachers jointly committing to at least one strategy to prevent dropout. Schools were guided to develop a simple, step-by-step plan with key milestones for tracking student progress

¹⁰After the experiment, which took place during the 2018 academic year, the guide was widely distributed to all schools. The second edition of the guide is available online at <https://www.mineduc.gob.gt/entre/>.

toward secondary enrollment. Teachers and principals were encouraged to collaborate, ensuring that dropout prevention was a shared responsibility rather than an isolated effort.

The training took place in two phases. On May 8–9, 2018, approximately 70 pedagogical support staff from the Ministry of Education received a 2-day training session delivered by core members of the ministry’s Technical Working Group on Dropout, with support from World Bank staff. Between May 29 and June 8, these 70 staff members traveled across Guatemala to train sixth-grade teachers and principals from 3,000 schools.¹¹

The training and the guide emphasized five concrete behavioral changes for school staff, supported by structured activities and resources:

1. Proactive student motivation strategies The training and the guide provided tools to address motivational barriers, including self-affirmation exercises and communication of education’s economic returns (à la [Jensen \(2010\)](#); [Nguyen \(2008\)](#)). Teachers practiced role-playing scenarios where they conveyed the long-term benefits of secondary education using real-world examples (e.g., projected lifetime earnings differences between primary and secondary school graduates). Another exercise involved students writing “future self” letters articulating personal goals, which teachers reviewed periodically to reinforce commitment. Finally, the guide also suggested organizing visits from successful alumni to serve as role models.

2. Scholarship support The Ministry of Education offered three types of scholarships at the time. In line with empirical evidence that many eligible low-income families fail to apply for such benefits (e.g., [Bettinger, Long, Oreopoulos, and Sanbonmatsu \(2012\)](#); [Bhargava and Manoli \(2015\)](#)), the guide provided clear instructions on how teachers and principals could help students apply (see Appendix C). The training and the guide also instructed teachers to identify families who might be unaware of these programs and to proactively reach out to offer support.

The three scholarships were:

- *Bolsa de Estudio*: This scholarship targeted economically disadvantaged students in public institutions. Eligibility criteria included a sworn statement from the parents that they are in economic need, not repeating a grade and achieving a minimum academic average of 70/100. The annual benefit was 1,350 quetzals (approximately USD 184 in

¹¹A total of ~ 4,100 principals and teachers were trained. The number is below 6,000 as in many small, rural schools, the principal also teaches sixth grade. Due to a volcanic eruption, training for principals and teachers in the departments of Escuintla and Chimaltenango was postponed until June.

2018 or 4% of Guatemala's per capita GDP (World Bank, 2018)). For comparison, in 2018, the annual government expenditure per student in primary was 11.9% (World Bank, 2019a) and in secondary was 5.4% (World Bank, 2019b).

- *Beca para Estudiantes con Discapacidad*: This scholarship provided financial assistance to low-income students with documented physical, visual, auditory, intellectual, or multiple disabilities. Eligibility criteria included proof of disability, a sworn statement from the parents that they are in economic need and a minimum attendance rate. The annual benefit was 1,000 quetzals (approximately USD 136 in 2018), representing 3% of per capita GDP (World Bank, 2018).
- *Bono de Transporte*: This benefit, limited to students in Guatemala City, was intended to cover public transportation costs. Students had to document the distance between their residence and the school and maintain at least 80% attendance. It provided 2.2 quetzals per day for round-trip fares. However, by 2018, the amount was insufficient to cover both trips due to inflation and rising transportation costs.

Overall, the training and the guide aimed to enhance access to these scholarships by training school staff to assist eligible students with applications, reducing informational and procedural barriers, and encouraging participation among economically disadvantaged families. However, as discussed in Section 4.3, the low supply of scholarships suggests this was not a primary driver of the program's impact.¹²

3. Academic remediation protocols Based on a large literature documenting that struggling students often disengage and drop out, the training introduced the idea of peer mentoring where high-achieving students assist those at risk. The training and the guide also encouraged teachers to use simple formative assessments to identify learning gaps early and offer remedial classes for students needing additional support.

4. Family engagement Many parents, especially those with lower educational attainment, are not engaged in their children's education and this has been associated with higher dropout rates (e.g., Huisman and Smits (2015)). The training recommended: a) conducting

¹²While there are no formal impact evaluations of these scholarship programs, an internal World Bank study of scholarships awarded between 2013 and 2015 with similar eligibility criteria and higher amounts (approximately USD 330 annually) found that dropout rates were 2-3 percentage points lower among recipients compared to applicants with similar characteristics after one year. We do not have data to see if applications to these scholarships were affected by our treatment, but evidence from the qualitative focus groups suggests scholarship applications are not a main driver behind the treatment effects (see Appendix B).

home visits to discuss the importance of secondary education with families; b) framing conversations positively by emphasizing students' potential rather than the risk of dropout; and c) organizing parent meetings where school staff explain secondary school enrollment procedures and financial aid options.

5. Enrollment logistics support The transition process itself can deter students from enrolling in secondary school. To address this, the training suggested organizing school visits so students can familiarize themselves with the secondary school environment before enrollment. It also suggested assigning staff members as transition coordinators to follow up with students and ensure they complete the enrollment process, as well as ensuring students and families are aware of secondary school enrollment deadlines and requirements.

2.2.2 Component 2: Information on students at high risk of dropping out

The second component of ENTRE is a list provided to each principal of sixth-grade students at high risk of dropping out during the transition to lower secondary school. The lists contained each student's name and identification number (see Figure A.1 for an example). The probability of dropout for each student was estimated using a linear probability model with dropout as a binary outcome using data on each student's sex, age, grades, whether they had repeated a grade, and school fixed effects. This simple model can correctly identify 82% of the sixth-grade students who will drop out within the next year. It performs better than other commonly used targeting approaches and is comparable to models used in early warning systems elsewhere (Adelman et al., 2018).¹³ The main objective of this component was to help school actors target their support to the students most in need of it, given limited time and resources.

2.2.3 Component 3: Behavioral nudges

The third component consisted of five monthly reminders sent to school principals to keep dropout at the top of their agendas and motivate them to act. The messages were sent through an online portal (known as SIRE) that school principals regularly use to perform administrative tasks and exchange information with the Ministry of Education.¹⁴ Each month, between June 20 and October 1, these messages appeared on the portal's homepage as soon as the principals logged in. The reminders used various behavioral insights to motivate action, namely:

¹³For more details on the methodology, please see Adelman et al. (2018).

¹⁴About 85% of school principals reported in the baseline questionnaire that they regularly use SIRE; only 15% said they used it rarely or never (14.3% and 0.25%, respectively).

- *Social recognition* (message 1): Social incentives can have a powerful effect on behavior, given the human desire for status and recognition (Frank, 1985; Cassar & Meier, 2018). Social rewards, such as status and recognition, can motivate people to exert effort and even serve as a substitute for monetary rewards in some situations (e.g., Kosfeld and Neckermann (2011); Blanes i Vidal and Nossol (2011); Ashraf, Bandiera, and Lee (2014); Bradler, Dur, Neckermann, and Non (2016); Ager, Bursztyn, Leucht, and Voth (2022)). The first message was inspired by this theory, informing school principals about an official certificate to recognize the schools that have made the most progress in reducing the dropout rate and the publication of the list of recognized schools in a popular national newspaper (see Figure A.2).
- *Salience* (messages 2–4): Individuals are more likely to respond to stimuli that are timely and accessible, and are more likely to do something that their attention is drawn toward (e.g., Castleman and Page (2015, 2016); Escueta, Nickow, Oreopoulos, and Quan (2020); Fishbane, Ouss, and Shah (2020); Dai et al. (2021); Zhang, Hemmeter, Kessler, Metcalfe, and Weathers (2023)). Even if school principals have concrete ideas about how to address the problem, dropout prevention may not be their top priority, especially given the number of other tasks they are responsible for (e.g., leading the day-to-day operation of the school, managing staff, and engaging with the community). Salient reminders about the need to help at-risk students and the tools and resources they have available can draw their attention to the task and inspire them to act. Messages 2–4 were therefore structured to increase the salience of dropout prevention; they included motivational phrases, reminders of key actions, and references to the guidance manual (see Figure A.3).
- *Loss frame* (message 5): People sometimes place more weight on potential losses than on potential gains. This tendency can also affect people’s level of effort in response to various incentives. Message framing is based on prospect theory (Tversky & Kahneman, 1981), which posits that describing a behavior in terms of its prospective costs or benefits can significantly affect individuals’ decision-making. The final reminder utilized a “loss frame” to communicate the urgency of acting now since failing to do so could result in losing the chance for social recognition (see Figure A.4).

The first message announcing social recognition appeared about 6 weeks after the training, on June 20. Reminders were then updated in SIRE monthly (July 16, August 6, September 3, and October 1). School principals were informed by their supervisors about the presence of a new reminder in SIRE and were invited to enter the online portal each time.

2.3 COVID-19 pandemic

While our experiment took place in 2018 and our main outcome is whether students are still enrolled in April of 2019, well before the beginning of the COVID-19 pandemic, we also study the effects of the treatment on the likelihood children are enrolled in 2020–2022. However, these estimates may be affected by the COVID-19 pandemic. Thus, in this section we provide a brief overview of the effect of the pandemic on the educational system.

In response to the pandemic, schools closed for in-person classes in March 2020, shifting entirely to virtual instruction. Schools partially reopened in 2021 and 2022, depending on the epidemiological risk in each municipality. The government adopted a municipal-level warning index to monitor infections, implementing a ‘stop-light’ system to determine school operations. Districts in the green category could resume in-person instruction, while yellow and orange districts operated in hybrid formats with varying social distancing requirements, and red districts were limited to virtual schooling. By the start of 2023, schools fully reopened for in-person classes.

Andres Ham and Yanez-Pagans (2024) find that dropout rates increased by 8.1% relative to pre-pandemic levels, with younger children and those in rural areas disproportionately affected. There was also a 10% increase in students switching from private to public schools, driven by economic pressures and the adaptability challenges private institutions face.

3 Research design and data

3.1 Sampling and randomization

Since ENTRE was not designed to address the supply-side constraint of access to secondary schools, the universe of eligible schools was restricted to public schools in the top 30% of municipalities in terms of supply of secondary schools per student. Within those municipalities, primary schools in the bottom 15% of school size were excluded from the sample, as they are likely located in rural areas and have a limited supply of secondary schools nearby. Finally, to target schools that could benefit more from the type of information provided by the program, we excluded schools that — according to our predictive model — had no students at risk of dropping out. A total of 6,080 public primary schools met all of these criteria, representing 43.7% of sixth-grade enrollment in the country at the time.¹⁵ The sample schools are, on average, larger, have slightly older students, are more likely to be bilingual (Spanish and an indigenous

¹⁵Guatemala has 15,930 public primary schools. We excluded all 2,699 private primary schools in the country from the sample.

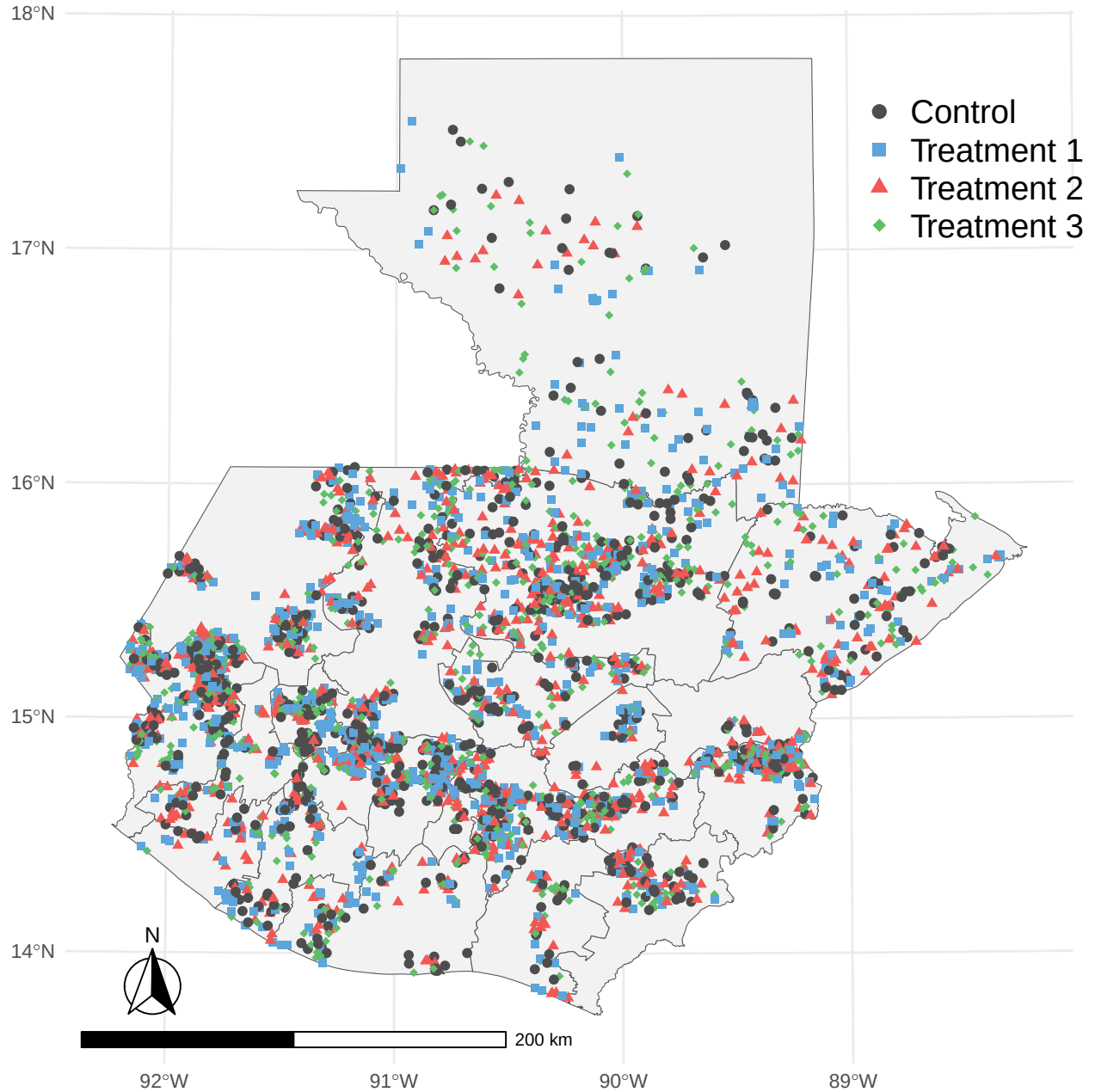
language), and have a higher dropout rate (historically) than the rest of the public schools in the country (see Tables A.1–A.2 and Figure A.5).¹⁶

Of these 6,080 schools, 4,000 were randomly assigned to four groups (1,000 each) via a simple random draw, without stratification. First, 1,000 were randomly allocated to receive component 1 (the half-day training, user-friendly guidance manual, and a letter from the Minister of Education). Another 1,000 schools were randomly selected to receive components 1 and 2 (providing the list of students at risk of dropping out). An additional 1,000 were randomly selected to receive all three components (i.e., the first two components, plus the behavioral nudges). Finally, 1,000 schools were kept as controls, and the Ministry of Education was asked not to implement any of the components in these schools, or any other dropout programs they would not implement in the 3,000 treatment schools.¹⁷ Schools in the four groups are distributed throughout the country (see Figure 1).

¹⁶In Guatemala, ~90% of public primary schools are rural, as urban enrollment is concentrated in a small number of large schools. Thus, even after excluding the smallest schools, most sample schools are still rural.

¹⁷Since the ministry implemented other programs to prevent dropout in some of the other 2,080 schools, we do not use them as controls. Indeed, while the 2,080 schools are similar to the 1,000 we used as controls in 2017 (see Table A.3), in 2018, they have lower dropout rates (see Table A.4), suggesting they are no longer valid controls likely due to other programs implemented by the government in those schools.

Figure 1: Distribution of treatment schools across the country



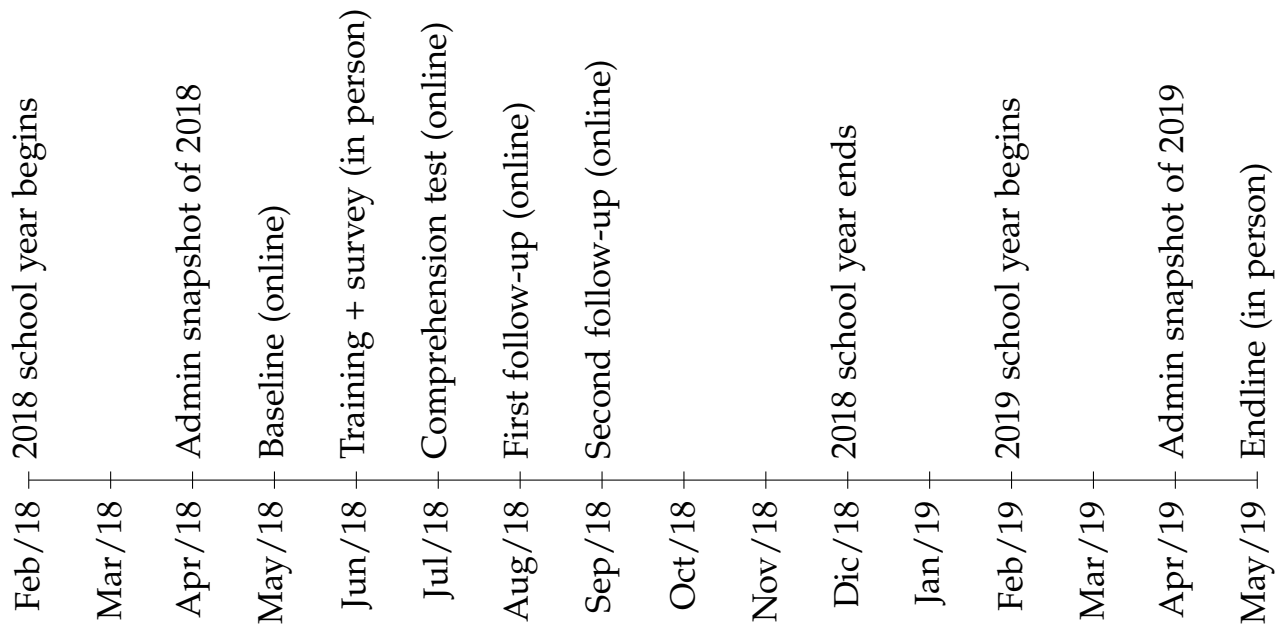
3.2 Data

Our main data source is individual-level student academic records. The Ministry of Education gathers annual enrollment data from every school which includes unique student identifiers that enable longitudinal tracking of students. The records include basic demographic information and enrollment status (from which we can infer dropout). Thus, we can track student-level dropout and estimate school-level dropout rates since 2013.

For 2017 and 2018 we have detailed grades in each subject. This data was also the main input for estimating the dropout prediction model (Adelman et al., 2018).

We also collected data on the quality of the program’s implementation and several intermediate (self-reported) outcomes through paper-based and online questionnaires integrated into the SIRE online platform. All 4,000 schools involved in the experiment were requested to complete a baseline questionnaire online, two follow-up online surveys, and a final, paper-based follow-up survey. At the end of the half-day training (related to component 1), the 3,000 treatment schools were also requested to complete an online comprehension test related to the contents of the manual (Figure 2 presents the full timeline).

Figure 2: Timeline



We focus on the administrative records, baseline online survey, and the paper-based final follow-up since the online follow-ups had high (over 60%) and differential attrition rates across the treatments (see Table A.5). Attrition is around 30% for the baseline survey and attrition does not vary across treatment arms. Attrition was much lower (below 20%) during the final follow-up, but there is some evidence of differential attrition: control schools were 2–4 percentage points more likely to complete the survey.

The baseline survey asked principals about which tasks they spend time on (and how they prioritize them) and their perceptions of school dropout, returns to education, and influences on students’ decisions to drop out. The endline survey asked similar questions

as well as those about the actual take-up of the treatment (i.e., whether they received the guide, training, and list of students at risk).

The main outcome indicator is a dummy variable capturing whether individual students who were enrolled in school in April of a given school year (the school year follows the calendar year) had dropped out by the following April. To construct this variable, we use administrative enrollment data to track students’ academic trajectories over time using their unique national identification numbers. If a student is not found in the enrollment records for the next year, we assume they dropped out. While potential errors in linking students’ administrative records across years or students migrating to another country would also be interpreted as dropouts in our dataset, we do not expect these to affect our results for two reasons. First, these “linkage errors” are likely second order as the individual-level enrollment records replicate the dropout patterns observed in Guatemalan household surveys (Adelman et al., 2018). Second, there is no reason to believe this potential measurement error would differ across randomly selected control and treatment schools. In short, we expect relatively small and random measurement error in dropout, leading (if anything) to slightly larger standard errors.

3.2.1 Qualitative analysis

We also held 12 focus groups to gather qualitative information on the challenges of program implementation. We interviewed almost 100 people involved in the program, including 63 teachers and principals from treatment schools and 34 trainers.¹⁸ Participants also completed short paper-based questionnaires during these sessions. Appendix B provides more details on how these focus groups were conducted and the main results from the analysis.

3.3 Balance

The observable characteristics of students and schools are broadly the same across the four experimental groups (Table 1). The average primary school in our sample had ~150 students enrolled, including 19 in grade 6. The average sixth-grade student in the sample was almost 13 years old (by the beginning of the school year) and about 50% of them were male. The year before our intervention, the schools in our sample had a dropout rate of 34% in the transition between primary (sixth grade) and secondary school (seventh grade). The statistical model predicted that about half of these students were at risk of dropping out.¹⁹

¹⁸The interviewed teachers and principals come from several departments, including Quetzaltenango, Quiché, Totonicapán, Chimaltenango, Sololá, Sacatepéquez, Jalapa, and Jutiapa.

¹⁹The number of students is also balanced in the years after the treatment (Table A.7). Further, the share of male students — the one observable characteristic we have in the data across years — remains constant after

the treatment began (Table [A.8](#)). Thus, students do not seem to have sorted across schools in response to the treatment.

Table 1: Balance across experimental groups

	Mean/SD for each group				p-value (equality)
	Control	T1	T2	T3	
	(1)	(2)	(3)	(4)	
Panel A: School level (2017)					
% rural	91.16 (28.41) [995]	93.03 (25.48) [990]	93.88 (23.99) [996]	92.54 (26.29) [992]	0.138
% morning shift	92.46 (26.41) [995]	92.42 (26.47) [990]	94.18 (23.43) [996]	94.05 (23.66) [992]	0.218
Total enrollment	155.86 (142.17) [995]	153.14 (137.90) [987]	155.50 (132.89) [992]	152.65 (135.21) [992]	0.936
Grd 6 enrollment	19.14 (20.73) [1,000]	19.01 (20.02) [1,000]	18.84 (18.29) [1,000]	18.58 (20.22) [1,000]	0.937
Number of teachers	6.21 (5.14) [995]	6.11 (4.79) [987]	6.06 (4.76) [992]	6.04 (4.73) [992]	0.870
Panel B: Student level (2017)					
% male	51.29 (49.98) [19,138]	51.89 (49.97) [19,006]	52.34 (49.95) [18,838]	51.77 (49.97) [18,583]	0.751
Age	12.87 (1.26) [18,905]	12.87 (1.24) [18,662]	12.90 (1.24) [18,630]	12.89 (1.27) [18,346]	0.714
GPA	7.59 (0.85) [18,905]	7.60 (0.85) [18,662]	7.57 (0.84) [18,630]	7.61 (0.84) [18,346]	0.515
% at-risk (statistical model)	48.96 (49.99) [18,904]	50.99 (49.99) [18,662]	51.65 (49.97) [18,630]	49.97 (50.00) [18,346]	0.781
Dropout	33.58 (47.23) [19,138]	35.48 (47.85) [19,006]	35.57 (47.87) [18,838]	35.10 (47.73) [18,583]	0.557

Notes: Columns 1–4 show the mean, standard deviation (in parentheses), and number of observations (in square brackets) for each of the following groups: control, students in schools that received only the first component (T1), students in schools with components 1 and 2 (T2), and those in schools with all three components (T3). Column 5 displays the p-value of tests of whether the mean is the same across all groups. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. N varies slightly across observations as the information on some variables is missing for some schools/students. Table A.6 presents more detailed data on the differences across groups.

3.4 Empirical strategy

We estimate the intent-to-treat (ITT) effect of the three treatments vis-à-vis the control group by ordinary least squares. First, we focus on the cross-section of students who were enrolled in the schools in our experiment in 2018:

$$Y_{ist} = \alpha_1 T1_s + \alpha_2 T2_s + \alpha_3 T3_s + \lambda X_i + \phi X_s + \varepsilon_{is} \quad (1)$$

where Y_{ist} denotes the outcome of student i in school s at time t (i.e., whether the student dropped out between t and $t + 1$), and X_i and X_s are individual- (e.g., sex, age, predicted probability of dropping out) and school-level time-invariant characteristics (e.g., province dummies, baseline enrollment, and historical dropout rate). We cluster standard errors at the school level (the level at which the treatment was assigned). Since the random assignment was not stratified, we do not include any strata fixed effects. We vary the outcome to measure dropout at the end of the experiment (in 2019) and 1, 2, and 3 years later. This allows us to assess whether any treatment effects persist at the student level by studying enrollment rates over time for the cohort that was treated during the original experiment.

While equation 1 allows us to study whether the treatment effects persist at the *student* level by following the same cohort over time, we can also determine whether the treatment effects persist at the *school* level by studying the dropout rates between grades 6 and 7 for different cohorts. We construct a school-level panel (using the student-level data) that tracks transition dropout rates between grades 6 and 7 for the cohorts between 2013 and 2022. As a robustness check, we use this panel to estimate the treatment effect on the cohort that was treated during the original experiment.²⁰ We restrict the study period to 2013–2018 and estimate a two-way fixed-effects model with school and year fixed effects as follows:

$$Y_{ist} = \alpha_1 T1_{st} \times \mathbb{1}_{t=2018} + \alpha_2 T2_{st} \times \mathbb{1}_{t=2018} + \alpha_3 T3_{st} \times \mathbb{1}_{t=2018} + \lambda X_i + \gamma_t + \gamma_s + \varepsilon_{ist} \quad (2)$$

where Y_{ist} is the outcome of student i in school s at time t (whether a student who was enrolled in grade 6 in year t is enrolled in grade 7 in $t + 1$), X_i are individual (e.g., sex and age) characteristics, γ_t are year fixed effects, and γ_s are school fixed effects. Finally,

²⁰Our main estimates are based on an ANCOVA-style specification (as opposed to a difference-in-differences design) as it has been shown to have more power in certain settings (McKenzie, 2012).

$T1_s$ indicates whether the school is in treatment 1 (i.e., training), $T2_s$ whether it is in treatment 2 (i.e., training and list), and $T3_s$ whether it is in treatment 3 (i.e., training, list, and nudges). $\mathbb{1}_{t=2018}$ equals 1 if the year is 2018.

Our coefficients of interest are α_1 – α_3 , which measure the effect of the different treatments vis-à-vis the control group in 2018 (when the treatment took place). We can also estimate the additional effect (above the training) of providing schools with the list (by estimating the difference $\alpha_2 - -\alpha_1$), the additional effect of providing the nudges and the list over the training (by estimating the difference $\alpha_3 - -\alpha_1$), and the effect of providing the nudges over the training and the list (by estimating the difference $\alpha_3 - -\alpha_2$). These estimates should be similar to those from equation 1 when the outcome is dropping out by the end of the experiment.

However, we can also use the full panel (2013–2022) to study the evolution of differences between the treatment and control schools using an event-study specification:

$$Y_{ist} = \sum_{\tau \neq 2017} (\alpha_{1\tau} T1_s \times \mathbb{1}_{t=\tau} + \alpha_{2\tau} T2_s \times \mathbb{1}_{t=\tau} + \alpha_{3\tau} T3_s \times \mathbb{1}_{t=\tau}) + \lambda X_i + \gamma_t + \gamma_s + \varepsilon_{ist} \quad (3)$$

where $\alpha_{j\tau}$ measures the difference between schools in treatment j and the control group at time τ , relative to the difference in 2017 (the year before the treatment). The $\alpha_{j\tau}$ coefficients for the years prior to the treatment (i.e., $\tau \leq 2016$) should be close to zero given the random assignment of the treatments. The coefficients α_{j2018} measure the treatment effect on the transition rate for the cohort that was treated in 2018 when the experiment took place. The coefficients $\alpha_{j\tau}$ for $\tau \geq 2019$ estimate whether the cohorts in treated schools experienced any treatment effects after the experiment ended.

3.5 Imperfect compliance

Our estimates represent intent-to-treat effects as the take-up of the treatment was imperfect (i.e., not all treated schools were treated as intended, and some control schools received some treatment). To assess the take-up of the program, we use multiple data sources. For components 1 and 2, we use school principals' reports on whether they received the dropout prevention guide (which is also a proxy for participating in the training) and the list of at-risk students. This information was collected during the online and in-person follow-up (there is no differential attrition in this measurement across the groups, see Column 1, Table 2). We measure the take-up of component 3 directly from administrative records, which capture whether the school principals logged into the online system at least once when the reminders appeared on the platform.

A third of the control schools (34%) reported receiving the guide (Table 2). While it is possible that the guide was copied and passed around among principals, control schools are unlikely to have received the guide (or a copy) from nearby treated schools, as evidenced by a stable proportion of control schools reporting receipt of the guide regardless of their distance from treatment schools. Specifically, among control schools located more than 1 km from the nearest treated school, 33.5% report receiving the guide, and this figure is nearly identical (34.5%) for those more than 2 km away. These shares align closely with the overall proportion (34%) of control schools reporting guide receipt, suggesting minimal diffusion of the guide between treatment and control schools. Instead, some control schools are likely misreporting having received the guide which is further supported by the fact that 13% of principals from control schools reported receiving the list of at-risk students, which is impossible since we did not create such lists for these schools (Column 2, Table 2).²¹ Still, the likelihood that principals in any treatment group reported receiving the guide increased by ~50 percentage points (to roughly 84%).

Principals in the treatment groups that were meant to receive the list of students at risk were also more likely to report receiving it, although some principals in the treatment group that only received the guide and training also reported receiving a list — which, again, is impossible since we never generated it. Finally, about 64% of principals in the treatment group with the nudges logged in while at least one of the nudges was displayed (Column 4, Table 2).

²¹Since the lists were tailored to each school, a list from a different school would be irrelevant.

Table 2: Imperfect take-up

	(1)	(2)	(3)	(4)
	No attrition	Guide	List	Message #1
Training (α_1)	.001 (.015)	.45*** (.021)	.16*** (.019)	
Training+List (α_2)	-.02 (.015)	.46*** (.021)	.53*** (.02)	
Training+List+Nudge (α_3)	-.003 (.015)	.5*** (.02)	.59*** (.019)	.64*** (.015)
Control mean	.88	.34	.13	0
R^2	.00066	.2	.25	.57
N. of obs.	4,000	3,498	3,498	4,000

Notes: This table reports the difference between the treatment groups and the control, with standard errors in parentheses. The specification is similar to that of equation 1, but using school-level data. The outcome for Column 1 is an indicator for whether we have some measure of attrition from the survey data collected from principals. For Column 2, the outcome is whether the principal reported receiving a copy of the guide (i.e., component 1). For Column 3, the outcome is whether the principal reported receiving a list (i.e., component 2). For Column 4, the outcome is whether the principal logged into the system when the messages from component 3 were displayed. Since the messages were not displayed for anyone in the first two treatment groups, the coefficients are zero by construction and omitted from the table. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Non-classical measurement error in compliance leads to a situation in which the first-stage coefficient is biased, although it remains indicative of treatment compliance. For this reason, we focus on the reduced-form effects of the treatment.

4 Results

4.1 Effect on transition rates

First, we estimate the treatment's effect on the transition from primary (sixth grade) to secondary (seventh grade) schools using data from the cohort of students enrolled in sixth grade in 2018 (see Table 3). Students in schools that received the training and guide were 1.1 percentage points (p-value 0.055) less likely to drop out of school by 2019 — a change of -3.3% from a base of 34%. Students in schools that also received a list experienced a similar decrease in the likelihood of dropping out of 1.2 percentage points (p-value 0.030). The same is true for students in schools that received the training, the list and the nudge, who experienced a decrease of 1.2 percentage points (p-value 0.035).

These results are robust to different controls and to estimating the treatment effects using the two-way fixed-effects model; the coefficients are almost identical (see Table A.9).²² For example, while the estimated treatment effect on dropout of providing any of the treatments using the 2018 cross-section is 1.2 percentage points (p-value 0.012), the estimate using the two-way fixed-effects models is 1.2 percentage points (p-value 0.011).²³

These findings suggest that the training and guide effectively increased the transition rate, and that there are no additional benefits of providing the list or from the nudges.

²²We also investigate the possibility of spillovers between schools by leveraging natural variation in the proportion of nearby schools that were treated, following the approach of Miguel and Kremer (2004). Specifically, we regress the outcome variable not only on the treatment status of the school but also on the number of nearby schools that were treated, along with the total number of nearby public schools. For robustness, we include an alternative specification that controls for the fraction of nearby public schools that were treated. To define 'nearby,' we consider distances of 1, 2, 3, 4, 5, and 10 kilometers. Across all specifications, we find no evidence of spillover effects (see Tables A.11 and A.12). These analyses are conducted on a subset of the experimental sample due to missing GPS coordinates for some schools; however, the main results remain robust in this restricted sample (see Table A.10).

²³Since treated principals and teachers work at primary schools, and students transfer (if they do not drop out) to secondary schools, it is unlikely that any treatment effect on dropout, which comes from administrative enrollment records, is affected by misreporting due to the treatment.

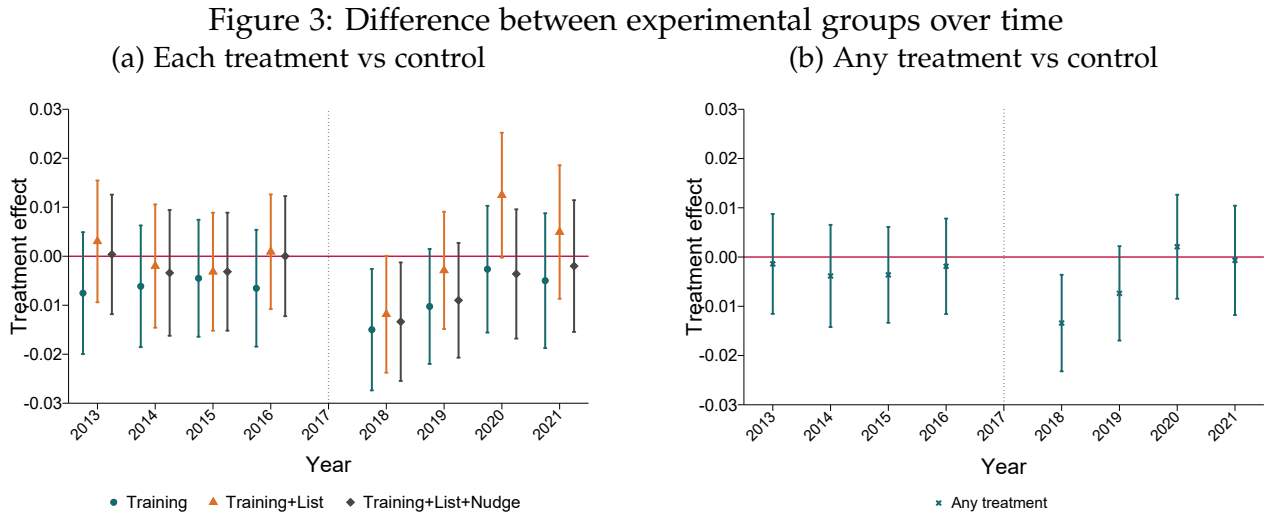
Table 3: Effect on dropout using the 2018 cross-section

	Not enrolled in 2019		
	(1)	(2)	(3)
Panel A: Effects for each treatment			
Training (α_1)	-.011* (.006)	-.01* (.0061)	-.011* (.0059)
Training+List (α_2)	-.0092 (.0056)	-.01* (.0057)	-.012** (.0056)
Training+List+Nudge (α_3)	-.012** (.0058)	-.012** (.0058)	-.012** (.0057)
Control mean	.34	.34	.34
$\alpha_2 - \alpha_1$	.0013	-1.9e-07	-.00068
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.83	1	.91
$\alpha_3 - \alpha_2$	-.0025	-.0021	.0002
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.66	.71	.97
$\alpha_3 - \alpha_1$	-.0012	-.0021	-.00049
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.84	.73	.93
R^2	.24	.3	.3
N. of obs.	77,094	77,094	77,094
Panel B: Effects for any treatment			
Any treatment	-.01** (.0047)	-.011** (.0048)	-.012** (.0047)
Control mean	.34	.34	.34
R^2	.24	.3	.3
N. of obs.	77,094	77,094	77,094
Student controls	No	Yes	Yes
School dropout	No	No	Yes

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). The specification follows Equation 1. Panel A presents the effects for the three treatment groups separately, while Panel B presents the effects of receiving any treatment. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

We can further leverage the panel nature of the data to study how the difference between experimental groups evolves over time (see Figure 3). As expected, the four experimental groups had a similar dropout rate in the year prior to the experiment. In line with the results presented in Table 3, in 2018, when the experiment took place, students in the three treatment groups were 1–1.2 percentage points less likely to drop out in 2019.

As mentioned above, starting in 2019, the government distributed the training guide to every school in the country. If the guide and training have the most impact, the differences across the treatment groups should disappear after this point. This is indeed what happened. For students enrolled in grade 6 in 2019, the difference in dropout (in 2020) is no longer statistically significant. The same applies to students enrolled in sixth grade in 2020 or 2021.²⁴ However, in 2020 the emergency measures associated with the COVID-19 pandemic — including prolonged school closures — negatively impacted the economy (see Section 2.3). Thus, these results should be interpreted with caution as the pandemic may have led to changes in the conditions of schools and families that muted the treatment effects.²⁵



Note: This figure displays the estimates from equation 3, as well as 90% confidence intervals. Each estimate is the difference between treatment and control schools in the transition between sixth and seventh grade (relative to the difference in 2017). Thus, each estimate reflects the treatment effect for a different cohort. The estimates for 2013–2016 show pre-treatment differences. The 2018 estimate is for the cohort that was treated during the experiment. The 2019–2021 estimates reflect the persistence of treatment effects at the school level after the experiment ended. Figure 3a presents the effects for the three treatment groups separately, while Figure 3b presents the effects of receiving any treatment.

²⁴The p-value for the difference between the 2018 and 2019 estimates is 0.0790. Comparisons of 2018 with 2020 and 2021 yield p-values of 0.0182 and 0.0443, respectively.

²⁵Even the estimates for the 2019 cohort may be affected by the pandemic since we use data from April every year. Across both treatment and control schools, the dropout rate is lower for the 2019 cohort than for other cohorts (see Figure A.6), while dropout rates are significantly higher for the 2020 and 2021 cohorts. In 2020, at the onset of the pandemic, with schools closed and classes conducted virtually, it may have been challenging to accurately verify enrollment, potentially resulting in lower recorded dropout rates for the 2019 cohort. In contrast, the higher dropout rates for the 2020 and 2021 cohorts are likely attributable to the prolonged school closures, associated learning losses, and the economic effects of the pandemic (Andres Ham & Yanez-Pagans, 2024).

4.2 Heterogeneity

Next, we explore whether the treatment effects differed by student or school characteristics. We find little evidence of heterogeneity by school and student characteristics (e.g., school size, age, gender, and GPA). This is true even if we pool all the treatments together, which suggests the effects are indeed relatively homogeneous, as opposed to being underpowered to detect differences across groups (see Tables A.15-A.18).

Further, there is no heterogeneity by the ex-ante risk (estimated by the model) that students will drop out (see Tables A.19), even though one of the program’s cornerstones was targeting support to at-risk students. This is also true if we pool all the treatments (Table A.20) and for the treatment groups that received the list of students at risk of dropping out (based on the model).²⁶ There are at least three possible explanations for this. First, teachers may not have used the list to target their efforts to prevent dropouts. Second, it is possible the list does not provide any additional information to improve targeting. Finally, additional interventions may be needed to prevent high-risk students from dropping out. Survey data from principals and teachers suggest that teachers found the lists useful and accurate. For example, 68% of those who reported having received a list rated it over 4 (out of 5) in how well it identified students at risk, and 66% rated it above 4 on how useful it is for preventing dropouts. Thus, it is unlikely the lists were not used. However, these responses are consistent with the possibility that principals were already able to identify the at-risk students; the lists may, therefore, not have provided any additional information. The results in Section 4.4, which show that the treatment effects (on the cohort that transitioned to secondary school in 2018) fade out quickly, also suggest that addressing other barriers is key to permanently preventing dropout.

4.3 Qualitative insights and mechanisms

There are three main qualitative insights from the focus groups (for a more detailed analysis, see Appendix B). First, the program signaled that dropout prevention was a top priority for the government, motivating educators to focus more on at-risk students. Most teachers did not view the program as an extra burden — rather, it reinforced practices they believed in, boosting their confidence to intervene.

Second, the guide provided a practical framework that empowered teachers — by giving them concrete strategies and an ‘official’ mandate — to intervene with struggling students. Teachers found the guide’s tips and activities useful and easy to integrate into their classes, and they appreciated having an official “tool” to back up their interventions with

²⁶This is not because the model lacks predictive power: the estimated probability of dropout is highly correlated with the likelihood of dropout.

students and parents. Following the guide's suggestions, teachers reported implementing several motivational strategies for students, from creative classroom activities (like goal-setting exercises and reward charts) to sharing personal success stories. Teachers reported that, as a result, previously unmotivated sixth-graders became excited about finishing the year and enrolling in secondary school.

Third, persistent structural challenges (poverty, cultural norms, limited scholarships) remained, which the ENTRE intervention could not fully overcome. Teachers noted that long-standing barriers at home persisted: economic hardships still pressured families to pull children out for work, and cultural norms (especially gender biases against girls' schooling) often limited how far parents were willing to go in their children's education. A common perception among teachers was that the supply of scholarships was insufficient relative to the demand. While the guide provided detailed information on scholarships and emphasized their importance, many teachers found this section frustrating: They reported investing considerable effort in helping families apply, only to face frequent disappointment due to the limited availability of scholarships. Some teachers pointed out that a single year of encouragement could only go so far if broader structural issues remained unaddressed.

Further, the results from the focus groups align with the main quantitative outcomes described above. The dropout prevention guide was the most frequently praised element of the intervention, with an excellent reception from teachers and principals, highlighting the importance of the core intervention in preventing dropouts. Teachers and principals from all treatment arms reported an increase in their targeted efforts.²⁷ Principals in the third treatment arm reported that the automated reminders were too frequent and confusing, which may explain why this treatment was indistinguishable in its impact.

In addition to the above-mentioned general claim of a greater emphasis on students who needed more support, principals and teachers in multi-grade schools reported having no time for ENTRE activities, which is consistent with the program's effect being slightly smaller in small schools (see Table A.15).²⁸ When asked about proposals to improve the program, many school actors suggested that the guide should include differentiated strategies for boys and girls — although we do not find statistically significant differences in treatment effects by gender (see Table A.18), both the baseline dropout rate and the magnitude of the estimated treatment effect are higher for female students.

²⁷The principal surveys also demonstrated a systematic increase in the likelihood that treated principals make dropout their top priority (see Table A.21).

²⁸In multi-grade schools, students from two or more grades are put together in a classroom. They are typically prevalent among semi-urban and rural areas, and they are significantly smaller than average.

Self-reported data from open questions in the focus groups also provide information on mechanisms and compliance. Teachers and principals claimed to have focused their strategies on motivating students to continue to secondary school. When the interviews were conducted (about 4 months after the training sessions were held), school actors showed strong confidence in the program's positive results, which is consistent with the short-term impact on dropout (see Section 4.1). Finally, trainers reported some logistical problems distributing the guide and lists, which confirms that compliance was imperfect.

4.4 Fading effects

Finally, we explore whether the positive treatment effect on the transition rate between primary and secondary school in 2018 persisted over time or faded out by determining whether students who were enrolled in grade 6 in 2018 are still enrolled in school in the subsequent 4 years (see Table 4). As shown in Section 4.1, by 2019 — when students should have transitioned to lower secondary school — the likelihood that students were still enrolled in school increased by 1.2 percentage points (p-value 0.011). However, by 2020 the treatment effect was closer to zero (0.71 percentage points) and statistically insignificant (p-value 0.13). The same was true in 2021 — the last year of lower secondary school — when the treatment effect was 0.82 percentage points (p-value 0.1). By 2022 — when students should have transitioned to upper secondary school — the treatment effect remained close to zero (-0.42 percentage points) and statistically insignificant (p-value 0.43).²⁹

Overall, these results suggest that the treatment effects fade over time. The most likely explanation is that the students who were induced by the treatment to transition to secondary school ended up dropping out in the following years. However, we do not know whether this is the case. Since there are high returns to obtaining a degree (i.e., the “sheepskin effect”), and not just from completed years of schooling (Jaeger & Page, 1996; Ferrer & Riddell, 2002; Jepsen et al., 2014; Rodríguez & Muro, 2015), it is unclear how much students will benefit in the long run from this treatment.

A possible explanation for the fade-out is that the treatment involved primary school principals and teachers. Once these at-risk students reached secondary school, where teachers were not trained on dropout prevention strategies, they were left without the necessary support to stay in school. These results also align with the qualitative evidence suggesting the program was unable to overcome structural issues like poverty and limited secondary school supply (see Section 4.3 and Appendix B). However, the results from 2021 and 2022 were likely heavily influenced by the COVID-19

²⁹The p-value for the difference between the 2019 and 2020 estimates is 0.0523. Comparisons of 2019 with 2021 and 2022 yield p-values of 0.3344 and 0.0065, respectively. The differences between the 2020 and 2021 estimates and the 2022 estimate are statistically significant, with p-values of 0.0399 and 0.0134, respectively.

pandemic. By March 2020, schools had closed for in-person classes, which affected enrollment and dropout rates (see Section 2.3).

Table 4: Fade-out of treatment effects

	Enrolled in school			
	2019 (1)	2020 (2)	2021 (3)	2022 (4)
Panel A: Effects for each treatment				
Training (α_1)	.011* (.0059)	.0074 (.0059)	.0059 (.0063)	-.0089 (.0064)
Training+List (α_2)	.012** (.0056)	.0068 (.0056)	.0071 (.0059)	-.0056 (.0065)
Training+List+Nudge (α_3)	.012** (.0057)	.0072 (.0058)	.012* (.0062)	.0021 (.0066)
Control mean	.66	.6	.53	.32
$\alpha_2 - \alpha_1$	.00079	-.0006	.0012	.0033
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.89	.92	.84	.6
$\alpha_3 - \alpha_2$	-.00035	.00044	.0046	.0077
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.95	.94	.44	.24
$\alpha_3 - \alpha_1$	.00043	-.00016	.0058	.011
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.94	.98	.36	.089
R^2	.3	.3	.28	.24
N. of obs.	77,083	77,083	77,083	77,083
Panel B: Effects for any treatment				
Any treatment	.012** (.0047)	.0071 (.0047)	.0082 (.005)	-.0042 (.0053)
Control mean				
R^2	.3	.3	.28	.24
N. of obs.	77,083	77,083	77,083	77,083

Notes: This table presents the effects of the different treatments on the likelihood that students enrolled in 2018 are still enrolled in school in the next four academic years. The specification follows Equation 1. Panel A presents the effects for the three treatment groups separately, while Panel B presents the effects of receiving any treatment. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Scalability and cost-benefit analysis

As the size of the experiment suggests, the program was designed to be implemented at scale by relying on existing staff from the Ministry of Education, including trainers, teachers, and school principals, and at a very low cost per student (USD 2–3). The training and overall implementation of the program were chiefly in the hands of the Ministry of

Education, with limited support from the World Bank. This minimizes the risk that the program's impact will diminish as it is scaled up, unlike programs tested by NGOs in small pilots (Vivalt, 2020; Bold et al., 2018).

The cost-benefit analysis was conducted following Dhaliwal, Duflo, Glennerster, and Tulloch (2013). Roughly a third (36%) of the program's per-student cost of USD 2.91 corresponds to the design of the guide and training and 45% to the opportunity cost of principals and teachers involved in it. Setting aside the one-off expenditures in the design of the intervention, the cost of implementing the program amounts to USD 1.85 per student.³⁰ Total per-student spending in Guatemala was ~530 USD in 2018, suggesting that the costs of a program like ENTRE could be absorbed into the national education budget.

Our estimated treatment effect suggests an increase of 0.64 years of schooling per 100 USD spent ($0.012 \times 100 / 1.85$).³¹ As a comparison, in a recent review from J-PAL, the impact of all the scholarships, conditional cash transfers, and subsidies analyzed range between 0.01 and 0.17 additional years of education per USD 100 spent (including the well-known PROGRESA in Mexico, which has an impact of 0.01 additional years per USD 100 spent). However, comparing cost-effective estimates across studies is always difficult (e.g., because of differences in institutional settings or the time horizon used to estimate outcomes). Further, our estimates have several limitations. For example, if the government spends more money expanding the program, it would increase the likelihood that students transition, but at some point, once all schools are reached, it would be unable to increase the number of years of schooling using this program. In contrast, other programs may be able to continue increasing average years of schooling as their budget increases, especially if they can increase not only the likelihood that students transition but also that they remain in school.

An alternative measure of the program's success is its internal rate based on the labor market returns to an additional year of schooling. We calculate the return of each additional year of schooling to be 8.8% using a Mincer regression and the *Encuesta Nacional de Condiciones de Vida* (ENCOVI) 2014 dataset.³² The expected labor income without the

³⁰Expanding ENTRE to additional grades would reduce the costs per student by approximately 76% since the training and materials for principals (which represent roughly half of the budget for an average school) are a fixed cost.

³¹Students in treated groups have, on average, 0.012 additional years of schooling by 2022 (see Tables A.13-A.14), which is consistent with finding a significant increase in the probability of still being enrolled in school in 2019 of 1.2 percentage points, and close to zero in 2020–2022 (Table 4).

³²The dependent variable in the Mincer equations is the logarithm of the hourly wage of individuals aged 25–55. The explanatory variables include regional dummies and six sectoral binary variables. The regression uses a Heckman correction, including the same variables in the selection equation plus the number of children, the number of children interacted with a gender dummy, a marriage indicator, and a school attendance binary variable.

project is computed as the average monthly labor income in nominal Quetzales (computed from ENCOVI - second semester 2014). We assume, conservatively, that students exposed to the program first benefit from the increased formal education at age 21 (after finishing, potentially, tertiary education) and retire at the age of 60.³³ Under these assumptions and a treatment effect of additional years of schooling of 0.012, the internal rate of return of the program is 27%. Further, using a discount rate of 10% yields a cost-benefit ratio over 7, while a 5% discount rate yields a cost-benefit ratio over 17.

6 Discussion and conclusions

This paper presents experimental evidence that a low-cost intervention that provides a half-day training to teachers and principals on how to prevent dropout and an accompanying guide with strategies to reduce dropout can significantly reduce school dropout rates during the transition from primary to lower secondary school. The program reduced the probability of dropout in treated schools by 1.2 percentage points (the effect is about twice as large when we focus on compliers), roughly 3% of the baseline dropout rate of 34%. Additional components, such as a list of students at risk of dropping out and behavioral nudges, provided no additional benefits.

The program's low implementation cost — 2–3 USD per student — makes it highly cost effective. The intervention's (conservative) rate of return is estimated to be 27%, with a cost-benefit ratio of approximately USD 19 for each 1 USD invested. The evaluation was embedded in a large-scale implementation that included almost 17% of all primary schools in the country, which makes the results more informative for understanding the impacts of the program at scale.

However, this success is qualified by the observation that, 2 years after implementation, treated students are no longer more likely to be enrolled than their peers in the control group. The short-lived nature of the program's impacts are perhaps not surprising given the multiple structural challenges students continue to face even after successfully transitioning to secondary school, and suggests that complementary interventions are needed to help them persist.

7 Data Availability

Code replicating the tables and figures in this article can be found in [Romero \(2025\)](#) in the Harvard Dataverse <https://doi.org/10.7910/DVN/WHPHER>.

³³Students exposed to the program at older ages and those who do not complete tertiary education would probably enter the labor market earlier and, therefore, receive a greater discounted benefit.

References

- Adelman, M., Haimovich, F., Ham, A., & Vazquez, E. (2018). Predicting school dropout with administrative data: new evidence from Guatemala and Honduras. *Education Economics*, 26(4), 356-372. Retrieved from <https://doi.org/10.1080/09645292.2018.1433127> doi: 10.1080/09645292.2018.1433127
- Adelman, M., & Székely, M. (2017). An overview of school dropout in Central America: Unresolved issues and new challenges for education progress. *European Journal of Educational Research*, 6(3), 235-259.
- Ager, P., Bursztyn, L., Leucht, L., & Voth, H.-J. (2022, 10). Killer Incentives: Rivalry, Performance and Risk-Taking among German Fighter Pilots, 1939-45. *The Review of Economic Studies*, 89(5), 2257-2292. Retrieved from <https://doi.org/10.1093/restud/rdab085> doi: 10.1093/restud/rdab085
- Andres Ham, E. V., & Yanez-Pagans, M. (2024). The short-term effects of differential exposure to COVID-19 on educational outcomes in Guatemala. *Education Economics*, 0(0), 1-22. Retrieved from <https://doi.org/10.1080/09645292.2024.2320786> doi: 10.1080/09645292.2024.2320786
- Ashraf, N., Bandiera, O., & Lee, S. S. (2014). Awards unbundled: Evidence from a natural field experiment. *Journal of Economic Behavior & Organization*, 100, 44-63. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0167268114000079> doi: <https://doi.org/10.1016/j.jebo.2014.01.001>
- Assunção, J., Gandour, C., & Rocha, R. (2023). DETER-ing deforestation in the Amazon: Environmental monitoring and law enforcement. *American Economic Journal: Applied Economics*, 15(2), 125-156.
- Atkin, D. (2016, August). Endogenous skill acquisition and export manufacturing in Mexico. *American Economic Review*, 106(8), 2046-85. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aer.20120901> doi: 10.1257/aer.20120901
- Balfanz, R., & Byrnes, V. (2019). Chapter 4 - early warning indicators and intervention systems: State of the field. In J. A. Fredricks, A. L. Reschly, & S. L. Christenson (Eds.), *Handbook of student engagement interventions* (p. 45-55). Academic Press. Retrieved from <https://www.sciencedirect.com/science/article/pii/B9780128134139000048> doi: <https://doi.org/10.1016/B978-0-12-813413-9.00004-8>
- Banerjee, A. V., Duflo, E., & Glennerster, R. (2008, 05). Putting a Band-Aid on a Corpse: Incentives for Nurses in the Indian Public Health Care System. *Journal of the European Economic Association*, 6(2-3), 487-500. Retrieved from <https://doi.org/10.1162/JEEA.2008.6.2-3.487> doi: 10.1162/JEEA.2008.6.2-3.487
- Barnwal, P. (2024, December). Curbing leakage in public programs: Evidence from India's direct benefit transfer policy. *American Economic Review*, 114(12), 3812-46. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aer.20161864> doi: 10.1257/aer.20161864
- Bellon, M., Dabla-Norris, E., Khalid, S., & Lima, F. (2022). Digitalization to improve tax compliance: Evidence from VAT e-invoicing in Peru. *Journal of Public Economics*, 210, 104661. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0047272722000639> doi: <https://doi.org/10.1016/j.jpubeco.2022.104661>
- Bentaouet, R. K. (2006). *Implementation of Free Basic Education Policy* (Tech. Rep. No. 7). Washington, D.C.: World Bank. Retrieved from <https://documents.worldbank.org/>

- en/publication/documents-reports/documentdetail/368451468340870142/implementation-of-free-basic-education-policy
- Bettinger, E. P., Long, B. T., Oreopoulos, P., & Sanbonmatsu, L. (2012, 07). The Role of Application Assistance and Information in College Decisions: Results from the H&R Block FAFSA Experiment. *The Quarterly Journal of Economics*, 127(3), 1205-1242. Retrieved from <https://doi.org/10.1093/qje/qjs017> doi: 10.1093/qje/qjs017
- Bhargava, S., & Manoli, D. (2015, November). Psychological frictions and the incomplete take-up of social benefits: Evidence from an IRS field experiment. *American Economic Review*, 105(11), 3489-3529. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aer.20121493> doi: 10.1257/aer.20121493
- Bird, K. A., Castleman, B. L., Denning, J. T., Goodman, J., Lamberton, C., & Rosinger, K. O. (2021). Nudging at scale: Experimental evidence from FAFSA completion campaigns. *Journal of Economic Behavior & Organization*, 183, 105-128. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0167268120304819> doi: <https://doi.org/10.1016/j.jebo.2020.12.022>
- Black, S. E., Devereux, P. J., & Salvanes, K. G. (2008). Staying in the classroom and out of the maternity ward? the effect of compulsory schooling laws on teenage births. *The Economic Journal*, 118(530), 1025-1054.
- Blanes i Vidal, J., & Nossol, M. (2011). Tournaments without prizes: Evidence from personnel records. *Management science*, 57(10), 1721-1736.
- Bloom, N., Lemos, R., Sadun, R., Scur, D., & Van Reenen, J. (2014). The new empirical economics of management. *Journal of the European Economic Association*, 12(4), 835-876.
- Bloom, N., Lemos, R., Sadun, R., & Van Reenen, J. (2015). Does management matter in schools? *The Economic Journal*, 125(584), 647-674.
- Bold, T., Filmer, D., Martin, G., Molina, E., Stacy, B., Rockmore, C., ... Wane, W. (2017, November). Enrollment without learning: Teacher effort, knowledge, and skill in primary schools in Africa. *Journal of Economic Perspectives*, 31(4), 185-204. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/jep.31.4.185> doi: 10.1257/jep.31.4.185
- Bold, T., Kimenyi, M., Mwabu, G., Ng'ang'a, A., & Sandefur, J. (2018). Experimental evidence on scaling up education reforms in Kenya. *Journal of Public Economics*, 168, 1-20. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0047272718301518> doi: <https://doi.org/10.1016/j.jpubeco.2018.08.007>
- Bouguen, A., Huang, Y., Kremer, M., & Miguel, E. (2019). Using randomized controlled trials to estimate long-run impacts in development economics. *Annual Review of Economics*, 11, 523-561.
- Bradler, C., Dur, R., Neckermann, S., & Non, A. (2016). Employee recognition and performance: A field experiment. *Management Science*, 62(11), 3085-3099.
- Brunetti, A., Büchel, K., Jakob, M., Jann, B., & Steffen, D. (2023). Inadequate teacher content knowledge and what could be done about it: evidence from El Salvador. *Journal of Development Effectiveness*, 0(0), 1-24. Retrieved from <https://doi.org/10.1080/19439342.2023.2213681> doi: 10.1080/19439342.2023.2213681
- Campos-Vazquez, R. M., & Santillan, A. S. (2018). Supply of schooling and dropout rates: Evidence from the oportunidades programme in Mexico. *Development Policy Review*, 36(4), 445-464. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/dpr.12238> doi: <https://doi.org/10.1111/dpr.12238>

- Carrillo, P., Donaldson, D., Pomeranz, D., & Singhal, M. (2023, December). Ghosting the tax authority: Fake firms and tax fraud in ecuador. *American Economic Review: Insights*, 5(4), 427-44. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aeri.20220321> doi: 10.1257/aeri.20220321
- Cascio, E. U., & Narayan, A. (2022). Who needs a fracking education? the educational response to low-skill-biased technological change. *ILR Review*, 75(1), 56-89. Retrieved from <https://doi.org/10.1177/0019793920947422> doi: 10.1177/0019793920947422
- Cassar, L., & Meier, S. (2018, August). Nonmonetary incentives and the implications of work as a source of meaning. *Journal of Economic Perspectives*, 32(3), 215-38. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/jep.32.3.215> doi: 10.1257/jep.32.3.215
- Castleman, B. L., & Page, L. C. (2015). Summer nudging: Can personalized text messages and peer mentor outreach increase college going among low-income high school graduates? *Journal of Economic Behavior & Organization*, 115, 144-160. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0167268114003217> (Behavioral Economics of Education) doi: <https://doi.org/10.1016/j.jebo.2014.12.008>
- Castleman, B. L., & Page, L. C. (2016). Freshman year financial aid nudges: An experiment to increase FAFSA renewal and college persistence. *Journal of Human Resources*, 51(2), 389-415. Retrieved from <https://jhr.uwpress.org/content/51/2/389> doi: 10.3368/jhr.51.2.0614-6458R
- Chetty, R. (2015, May). Behavioral economics and public policy: A pragmatic perspective. *American Economic Review*, 105(5), 1-33. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aer.p20151108> doi: 10.1257/aer.p20151108
- Chung, J. Y., & Lee, S. (2019). Dropout early warning systems for high school students using machine learning. *Children and Youth Services Review*, 96, 346-353. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0190740918309721> doi: <https://doi.org/10.1016/j.childyouth.2018.11.030>
- Dai, H., Saccardo, S., Han, M. A., Roh, L., Raja, N., Vangala, S., ... Croymans, D. M. (2021). Behavioural nudges increase COVID-19 vaccinations. *Nature*, 597(7876), 404-409.
- De Hoyos Navarro, R., Popova, A., & Rogers, H. (2016). Out of school and out of work: a diagnostic of ninis in latin america. *World Bank Policy Research Working Paper*(7548).
- Dhaliwal, I., Duflo, E., Glennerster, R., & Tulloch, C. (2013). Comparative cost-effectiveness analysis to inform policy in developing countries: a general framework with applications for education. *Education policy in developing countries*, 17, 285-338.
- Duflo, E., Hanna, R., & Ryan, S. P. (2012). Incentives work: Getting teachers to come to school. *American economic review*, 102(4), 1241-1278.
- Escueta, M., Nickow, A. J., Oreopoulos, P., & Quan, V. (2020, December). Upgrading education with technology: Insights from experimental research. *Journal of Economic Literature*, 58(4), 897-996. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/jel.20191507> doi: 10.1257/jel.20191507
- European Commission. (2013). Reducing early school leaving: key messages and policy support: final report of the thematic group on early school leaving.
- Evans, D. K., Gale, C., & Kosec, K. (2023). The educational impacts of cash transfers in Tanzania. *Economics of Education Review*, 92, 102332. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0272775722001054> doi: <https://doi.org/10.1016/j.econedurev.2022.102332>

doi.org/10.1016/j.econedurev.2022.102332

- Faria, A.-M., Sorensen, N., Heppen, J., Bowdon, J., Taylor, S., Eisner, R., & Foster, S. (2017). Getting students on track for graduation: Impacts of the early warning intervention and monitoring system after one year. rel 2017-272. *Regional Educational Laboratory Midwest*.
- Ferrer, A. M., & Riddell, W. C. (2002). The role of credentials in the canadian labour market. *Canadian Journal of Economics/Revue canadienne d'économie*, 35(4), 879-905. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/0008-4085.00158> doi: <https://doi.org/10.1111/0008-4085.00158>
- Ferriere, A., Navarro, G., & Reyes-Heroles, R. (2018). *Escaping the Losses from Trade: The Impact of Heterogeneity on Skill Acquisition* (2018 Meeting Papers No. 1248). Society for Economic Dynamics. Retrieved from <https://ideas.repec.org/p/red/sed018/1248.html>
- Fishbane, A., Ouss, A., & Shah, A. K. (2020). Behavioral nudges reduce failure to appear for court. *Science*, 370(6517), eabb6591. Retrieved from <https://www.science.org/doi/abs/10.1126/science.abb6591> doi: 10.1126/science.abb6591
- Flotts, M. P., Manzi, J., Jiménez, D., Abarzúa, A., Cayuman, C., & García, M. J. (2015). *Informe de resultados TERCE: logros de aprendizaje*. UNESCO Publishing.
- Frank, R. (1985). *Choosing the right pond: Human behavior and the quest for status*. Oxford University Press. Retrieved from <https://books.google.com.mx/books?id=us3rAAAAAAAJ>
- Gibbs, B. G., & Heaton, T. B. (2014). Drop out from primary to secondary school in Mexico: A life course perspective. *International Journal of Educational Development*, 36, 63–71.
- Greenstone, M., He, G., Jia, R., & Liu, T. (2022). Can technology solve the principal-agent problem? evidence from China's war on air pollution. *American Economic Review: Insights*, 4(1), 54–70.
- Huisman, J., & Smits, J. (2015). Keeping children in school: Effects of household and context characteristics on school dropout in 363 districts of 30 developing countries. *SAGE Open*, 5(4), 2158244015609666. Retrieved from <https://doi.org/10.1177/2158244015609666> doi: 10.1177/2158244015609666
- Jacob, B. A., & Lefgren, L. (2004). The impact of teacher training on student achievement quasi-experimental evidence from school reform efforts in Chicago. *Journal of Human Resources*, 39(1), 50–79.
- Jaeger, D. A., & Page, M. E. (1996). Degrees matter: New evidence on sheepskin effects in the returns to education. *The Review of Economics and Statistics*, 78(4), 733–740. Retrieved 2024-01-19, from <http://www.jstor.org/stable/2109960>
- Jensen, R. (2010, 05). The (Perceived) Returns to Education and the Demand for Schooling. *The Quarterly Journal of Economics*, 125(2), 515-548. Retrieved from <https://doi.org/10.1162/qjec.2010.125.2.515> doi: 10.1162/qjec.2010.125.2.515
- Jepsen, C., Troske, K., & Coomes, P. (2014). The labor-market returns to community college degrees, diplomas, and certificates. *Journal of Labor Economics*, 32(1), 95-121. Retrieved from <https://doi.org/10.1086/671809> doi: 10.1086/671809
- Kattan, R. B., & Székely, M. (2015). Analyzing the dynamics of school dropout in upper secondary education in Latin America: a cohort approach. *World Bank Policy Research Working Paper*(7223).
- Kilburn, K., Handa, S., Angeles, G., Mvula, P., & Tsoka, M. (2017). Short-term impacts of

- an unconditional cash transfer program on child schooling: Experimental evidence from Malawi. *Economics of Education Review*, 59, 63-80. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0272775716305416> doi: <https://doi.org/10.1016/j.econedurev.2017.06.002>
- Knowles, J. E. (2015). Of needles and haystacks: Building an accurate statewide dropout early warning system in Wisconsin. *Journal of Educational Data Mining*, 7(3), 18–67.
- Kosfeld, M., & Neckermann, S. (2011, August). Getting more work for nothing? symbolic awards and worker performance. *American Economic Journal: Microeconomics*, 3(3), 86-99. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/mic.3.3.86> doi: 10.1257/mic.3.3.86
- Lee, S., & Chung, J. Y. (2019). The machine learning-based dropout early warning system for improving the performance of dropout prediction. *Applied Sciences*, 9(15). Retrieved from <https://www.mdpi.com/2076-3417/9/15/3093> doi: 10.3390/app9153093
- Linos, E., Prohofskey, A., Ramesh, A., Rothstein, J., & Unrath, M. (2022, November). Can nudges increase take-up of the eite? evidence from multiple field experiments. *American Economic Journal: Economic Policy*, 14(4), 432-52. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/pol.20200603> doi: 10.1257/pol.20200603
- Lochner, L., & Moretti, E. (2004). The effect of education on crime: Evidence from prison inmates, arrests, and self-reports. *American Economic Review*, 94(1), 155–189.
- Loyalka, P., Popova, A., Li, G., & Shi, Z. (2019, July). Does teacher training actually work? evidence from a large-scale randomized evaluation of a national teacher training program. *American Economic Journal: Applied Economics*, 11(3), 128-54. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/app.20170226> doi: 10.1257/app.20170226
- Marini, A., & Gragnolati, M. (2003). Malnutrition and poverty in Guatemala. Available at SSRN 636329.
- Márquez-Vera, C., Cano, A., Romero, C., Noaman, A. Y. M., Mousa Fardoun, H., & Ventura, S. (2016). Early dropout prediction using data mining: a case study with high school students. *Expert Systems*, 33(1), 107-124. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/exsy.12135> doi: <https://doi.org/10.1111/exsy.12135>
- McKenzie, D. (2012). Beyond baseline and follow-up: The case for more T in experiments. *Journal of Development Economics*, 99(2), 210-221. Retrieved from <https://www.sciencedirect.com/science/article/pii/S030438781200003X> doi: <https://doi.org/10.1016/j.jdeveco.2012.01.002>
- Miguel, E., & Kremer, M. (2004). Worms: Identifying impacts on education and health in the presence of treatment externalities. *Econometrica*, 72(1), 159-217. Retrieved from <https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1468-0262.2004.00481.x> doi: <https://doi.org/10.1111/j.1468-0262.2004.00481.x>
- Mnyawami, Y. N., Maziku, H. H., & Mushi, J. C. (2022). Enhanced model for predicting student dropouts in developing countries using automated machine learning approach: A case of Tanzanian's secondary schools. *Applied Artificial Intelligence*, 36(1), 2071406. Retrieved from <https://doi.org/10.1080/08839514.2022.2071406> doi: 10.1080/08839514.2022.2071406
- Muralidharan, K., Niehaus, P., & Sukhtankar, S. (2016, October). Building state

- capacity: Evidence from biometric smartcards in India. *American Economic Review*, 106(10), 2895-2929. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/aer.20141346> doi: 10.1257/aer.20141346
- Nguyen, T. (2008). Information, role models and perceived returns to education: Experimental evidence from Madagascar. *Unpublished manuscript*.
- O'Cummings, M., & Therriault, S. B. (2015). From accountability to prevention: Early warning systems put data to work for struggling students. *American Institutes for Research*.
- Okunogbe, O., & Pouliquen, V. (2022, February). Technology, taxation, and corruption: Evidence from the introduction of electronic tax filing. *American Economic Journal: Economic Policy*, 14(1), 341-72. Retrieved from <https://www.aeaweb.org/articles?id=10.1257/pol.20200123> doi: 10.1257/pol.20200123
- Oreopoulos, P. (2007). Do dropouts drop out too soon? wealth, health and happiness from compulsory schooling. *Journal of Public Economics*, 91(11), 2213-2229. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0047272707000138> doi: <https://doi.org/10.1016/j.jpubeco.2007.02.002>
- Oreopoulos, P., & Salvanes, K. G. (2011). Priceless: The nonpecuniary benefits of schooling. *Journal of Economic perspectives*, 25(1), 159-184.
- Page, L. C., Sacerdote, B. I., Goldrick-Rab, S., & Castleman, B. L. (2023). Financial aid nudges: A national experiment with informational interventions. *Educational Evaluation and Policy Analysis*, 45(2), 195-219. Retrieved from <https://doi.org/10.3102/01623737221111403> doi: 10.3102/01623737221111403
- Piper, B., & Korda, M. (2011). EGRA Plus: Liberia. program evaluation report. *RTI International*.
- Plak, S., Cornelisz, I., Meeter, M., & van Klaveren, C. (2022). Early warning systems for more effective student counselling in higher education: Evidence from a dutch field experiment. *Higher Education Quarterly*, 76(1), 131-152.
- Popova, A., Evans, D. K., Breeding, M. E., & Arancibia, V. (2021, 06). Teacher Professional Development around the World: The Gap between Evidence and Practice. *The World Bank Research Observer*, 37(1), 107-136. Retrieved from <https://doi.org/10.1093/wbro/lkab006> doi: 10.1093/wbro/lkab006
- Rodríguez, J. J. M., & Muro, J. (2015). On the size of sheepskin effects: A meta-analysis. *Economics*, 9(1), 20150037.
- Romero, M. (2025). *Replication Data for: "Preventing School Dropout at Scale: Experimental Evidence from Guatemala"*. Harvard Dataverse. Retrieved from <https://doi.org/10.7910/DVN/WHPHER> doi: 10.7910/DVN/WHPHER
- Romero, M., Bedoya, J., Yanez-Pagans, M., Silveyra, M., & de Hoyos, R. (2022). Direct vs indirect management training: Experimental evidence from schools in Mexico. *Journal of Development Economics*, 154, 102779. Retrieved from <https://www.sciencedirect.com/science/article/pii/S0304387821001371> doi: <https://doi.org/10.1016/j.jdeveco.2021.102779>
- Saavedra, S. (2023). *Technology and state capacity: Experimental evidence from illegal mining in colombia*. Retrieved from https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3933128
- Sara, N.-B., Halland, R., Igel, C., & Alstrup, S. (2015). High-school dropout prediction using machine learning: A Danish large-scale study. In *Esann* (Vol. 2015, p. 23rd).

- Sletten, M. A., Tøge, A. G., & Malmberg-Heimonen, I. (2022). Effects of an early warning system on student absence and completion in Norwegian upper secondary schools: a cluster-randomised study. *Scandinavian Journal of Educational Research*, 0(0), 1-15. Retrieved from <https://doi.org/10.1080/00313831.2022.2116481> doi: 10.1080/00313831.2022.2116481
- Stinebrickner, R., & Stinebrickner, T. (2008). The effect of credit constraints on the college drop-out decision: A direct approach using a new panel study. *American Economic Review*, 98(5), 2163–2184.
- Tversky, A., & Kahneman, D. (1981). The framing of decisions and the psychology of choice. *Science*, 211(4481), 453-458. Retrieved from <https://www.science.org/doi/abs/10.1126/science.7455683> doi: 10.1126/science.7455683
- UNESCO. (2021). *Resultados de logros de aprendizaje y factores asociados del estudio regional comparativo y explicativo (ERCE 2019)* (Tech. Rep.). Author.
- UNESCO. (2023). *Sdg 4 - country profiles: Guatemala*. (data retrieved from <http://sdg4-data.uis.unesco.org/> on April 13, 2023)
- UNICEF. (2020). *The state of the world's children 2019: Children, food and nutrition-growing well in a changing world*. United Nations.
- Vivalt, E. (2020). How much can we generalize from impact evaluations? *Journal of the European Economic Association*, 18(6), 3045–3089.
- World Bank. (2018). *Gdp per capita (current us\$) - Guatemala*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/NY.GDP.PCAP.CD?locations=GT>)
- World Bank. (2019a). *Government expenditure per student, primary (gdp per capita) - Guatemala*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/SE.XPD.PRIM.PC.ZS?locations=GT>)
- World Bank. (2019b). *Government expenditure per student, secondary (gdp per capita) - Guatemala*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/SE.XPD.SECO.ZS?locations=GT>)
- World Bank. (2019c). *Lower secondary completion rate, total (% of relevant age group) - Guatemala*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/SE.SEC.CMPT.L0.ZS?locations=GT>)
- World Bank. (2019d). *Primary completion rate, total (% of relevant age group) - Guatemala*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/SE.PRM.CMPT.ZS?locations=GT>)
- World Bank. (2019e). *School enrollment, primary (% net)*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/SE.PRM.NENR>)
- World Bank. (2019f). *School enrollment, secondary (% net)*. (data retrieved from World Development Indicators on April 23, 2023, <https://data.worldbank.org/indicator/SE.SEC.NENR>)
- Zhang, C. Y., Hemmeter, J., Kessler, J. B., Metcalfe, R. D., & Weathers, R. (2023). Nudging timely wage reporting: Field experimental evidence from the us supplemental security income program. *Management Science*, 69(3), 1341–1353.

A Additional tables and figures

Figure A.1: List of students at risk of dropping out

Estrategia Nacional para la Transición Exitosa

I. DATOS DEL ESTABLECIMIENTO

Nombre del establecimiento: [REDACTED] L'

Código: [REDACTED]

Departamento: GUATEMALA Municipio: GUATEMALA

Cantidad de estudiantes que requieren apoyo: 5

II. LISTA DE ESTUDIANTES QUE REQUIEREN APOYO PARA SU TRANSICIÓN AL CICLO BÁSICO

No.	NOMBRE DEL ESTUDIANTE	CÓDIGO PERSONAL	FECHA DE NACIMIENTO	SEXO
1	[REDACTED]	[REDACTED]	24/09/2004	MUJER
2	[REDACTED]	[REDACTED]	12/01/2004	HOMBRE
3	[REDACTED]	[REDACTED]	2/11/2000	HOMBRE
4	[REDACTED]	[REDACTED]	27/12/2001	HOMBRE
5	[REDACTED]	[REDACTED]	11/07/2005	MUJER

Note: This figure shows a typical list given to principals of students at risk of dropping out according to the statistical model outlined in [Adelman et al. \(2018\)](#).

Figure A.2: First message (social recognition)

(a) Main screen with short message

Sistema de Registros Educativos

Inicio de Sesión: 07:20
Tiempo de Sesión: 0 : 2 : 33

INICIO CICLO LECTIVO DOCENTES ESTUDIANTES REGISTROS REPORTES INFRAESTRUCTURA Estrategia ENTRE CERRAR SESIÓN ?

BIENVENIDO AL SISTEMA NACIONAL DE REGISTRO EDUCATIVO

CODIGO		NIVEL	43-PRIMARIA
DEPARTAMENTO	PETEN	SECTOR	OFICIAL
MUNICIPIO	LA LIBERTAD	JORNADA	MATUTINA
NOMBRE		PLAN	DIARIO(REGULAR)
DIRECCION		AREA	RURAL
TELEFONO		CICLO	1-ANUAL
CORREO	NO HA REGISTRADO CORREO	ESTADO	ABIERTA
NOMBRE DIRECTOR		MODALIDAD	BILINGUE

ALERTAS
El Centro Educativo está pendiente de registrar
- Mobiliario y Equipo
- Ambientes/Áreas

¡Gane un Reconocimiento! ¡Aumente la inscripción en Primero Básico!
Para mayor información haga click en el link [ENTRE](#)

(b) Long message

Sistema de Registros Educativos

17-05-0083-43 — PARCELAMIENTO LA LLORONA

Inicio de Sesión: 07:20
Tiempo de Sesión: 0 : 4 : 31

INICIO CICLO LECTIVO DOCENTES ESTUDIANTES REGISTROS REPORTES INFRAESTRUCTURA Estrategia ENTRE

Su establecimiento puede ser distinguido con el "Reconocimiento Honorífico por su Acción Efectiva para prevenir el abandono escolar". Esta distinción se otorgará a quienes logren reducir más el abandono escolar de sexto grado e incrementar la inscripción en primero básico. El reconocimiento consiste en un diploma para su escuela extendido por el Ministro de Educación, el cual podrá ser visible a la comunidad educativa y el nombre de los establecimientos que obtengan mejores resultados serán publicados en uno de los principales diarios de Guatemala.

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Note: Figure A.2a shows the homepage of the system principals used when the first message of component 3 was active. A short message about the possibility of getting an award appeared, and if principals clicked on the short message (which had a link embedded), they were shown a longer message with more information (Figure A.2b).

Figure A.3: Second message (salience)

(a) Main screen with short message

Sistema de Registros Educativos

Inicio de Sesión: 17:49
Tiempo de Sesión: 0 : 1 : 53



INICIO CICLO LECTIVO DOCENTES ESTUDIANTES REGISTROS REPORTES INFRAESTRUCTURA Estrategia ENTRE CERRAR SESIÓN

BIENVENIDO AL SISTEMA NACIONAL DE REGISTRO EDUCATIVO

CODIGO		NIVEL	43-PRIMARIA
DEPARTAMENTO	PETEN	SECTOR	OFICIAL
MUNICIPIO	LA LIBERTAD	JORNADA	MATUTINA
NOMBRE		PLAN	DIARIO(REGULAR)
DIRECCION		AREA	RURAL
TELEFONO		CICLO	1-ANUAL
CORREO	NO HA REGISTRADO CORREO	ESTADO	ABIERTA
NOMBRE DIRECTOR		MODALIDAD	BILINGUE

ALERTAS

El Centro Educativo está pendiente de registrar
- Mobiliario y Equipo
- Ambientes/Áreas

¡Ayude a sus estudiantes que necesitan su apoyo para continuar en Primero Básico!, para ver la lista completa de recursos disponibles ingrese al link [ENTRE](#)

(b) Long message

Sistema de Registros Educativos

Inicio de Sesión: 17:49
Tiempo de Sesión: 0 : 3 : 34



INICIO CICLO LECTIVO DOCENTES ESTUDIANTES REGISTROS REPORTES INFRAESTRUCTURA Estrategia ENTRE

Estrategia ENTRE->Alertas

¡Ayude a sus estudiantes que necesitan su apoyo para continuar en Primero Básico!

La "Guía para promover la transición exitosa" cuenta con acciones de apoyo que contribuyen a realizar la transición a Primero Básico. Realice un diagnóstico de las principales barreras que impiden que sus estudiantes continúen con sus estudios. Luego, implemente las acciones de apoyo que mejor se adapten a sus estudiantes:

1. Ejercicios cortos y prácticos de autoafirmación (pág. 9 de la Guía)
2. Proveer información sobre los beneficios económicos de la educación media (pág. 11)
3. Apoyar a las familias en el proceso de postulación para ayuda financiera (pág. 13)
4. Motivar a las familias (pág. 14)
5. Implementar adecuadamente el proceso de mejoramiento de los aprendizajes (pág. 15)
6. Implementar tutorías (pág. 16)
7. Apoyo en el proceso de inscripción (pág. 17)

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Note: Figure A.2a shows the homepage of the system principals used when the second message of component 3 was active. A short message about the possibility of getting an award appeared. If principals clicked on the embedded link, they were shown a longer message with more information (Figure A.3b). Messages 3 and 4 are similar.

Figure A.4: Fifth message (loss frame)

(a) Main screen with short message

Sistema de Registros Educativos

Inicio de Sesión: 06:29
Tiempo de Sesión: 0 : 1 : 20



INICIO CICLO LECTIVO DOCENTES ESTUDIANTES REGISTROS REPORTES INFRAESTRUCTURA Estrategia ENTRE CERRAR SESIÓN

BIENVENIDO AL SISTEMA NACIONAL DE REGISTRO EDUCATIVO

CODIGO		NIVEL	43- PRIMARIA
DEPARTAMENTO	PETEN	SECTOR	OFICIAL
MUNICIPIO	LA LIBERTAD	JORNADA	MATUTINA
NOMBRE		PLAN	DIARIO(REGULAR)
DIRECCION		AREA	RURAL
TELEFONO		CICLO	1-ANUAL
CORREO		ESTADO	ABIERTA
NOMBRE DIRECTOR		MODALIDAD	BILINGUE

ALERTAS

El Centro Educativo está pendiente de confirmar información.

El establecimiento esta pendiente de registrar información sobre inasistencia de: **ENERO, FEBRERO, MARZO, ABRIL, MAYO, JUNIO, JULIO, AGOSTO, SEPTIEMBRE**

¡Esta es su última oportunidad para ganar una distinción!
 Actúe ahora para aumentar la inscripción en primero básico.
 Para obtener información sobre acciones de último momento para aumentar la inscripción en primero básico Vaya al link [ENTRE](#)

(b) Long message

Sistema de Registros Educativos

Inicio de Sesión: 06:29
Tiempo de Sesión: 0 : 4 : 55



INICIO CICLO LECTIVO DOCENTES ESTUDIANTES REGISTROS REPORTES INFRAESTRUCTURA Estrategia ENTRE

¡Urgente! Se aproxima el fin del ciclo escolar y es la última oportunidad de actuar. Aún tiene tiempo para apoyar a sus estudiantes de sexto grado a inscribirse en primero básico y no abandonar la escuela. ¿Cómo ayudará a que sus estudiantes de sexto primaria se inscriban? ¿Les ha dado a conocer los programas de ayudas económicas del Ministerio de Educación? En la "Guía para promover la transición exitosa" puede encontrar información para aquellos estudiantes con falta de recursos económicos, por ejemplo las bolsas de estudio, el bono de transporte (ciudad de Guatemala) y las becas para estudiantes con discapacidad. Recuerde orientar a los padres de familia en el proceso de inscripción y contribuir con el llenado de formularios, esto podría cambiar la vida de sus estudiantes. El "Reconocimiento Honorífico por su Acción Efectiva para prevenir el abandono escolar" lo está esperando. ¡Aún hay tiempo! ¡Actúe ahora!

Ministerio de Educación, Copyright © 2,015

Note: Figure A.2a shows the homepage of the system principals used when the second message of component 3 was active. A short message about the possibility of getting an award appeared. If principals clicked on the embedded link, they were shown a longer message with more information (Figure A.3b).

Figure A.5: Distribution of eligible schools across the country

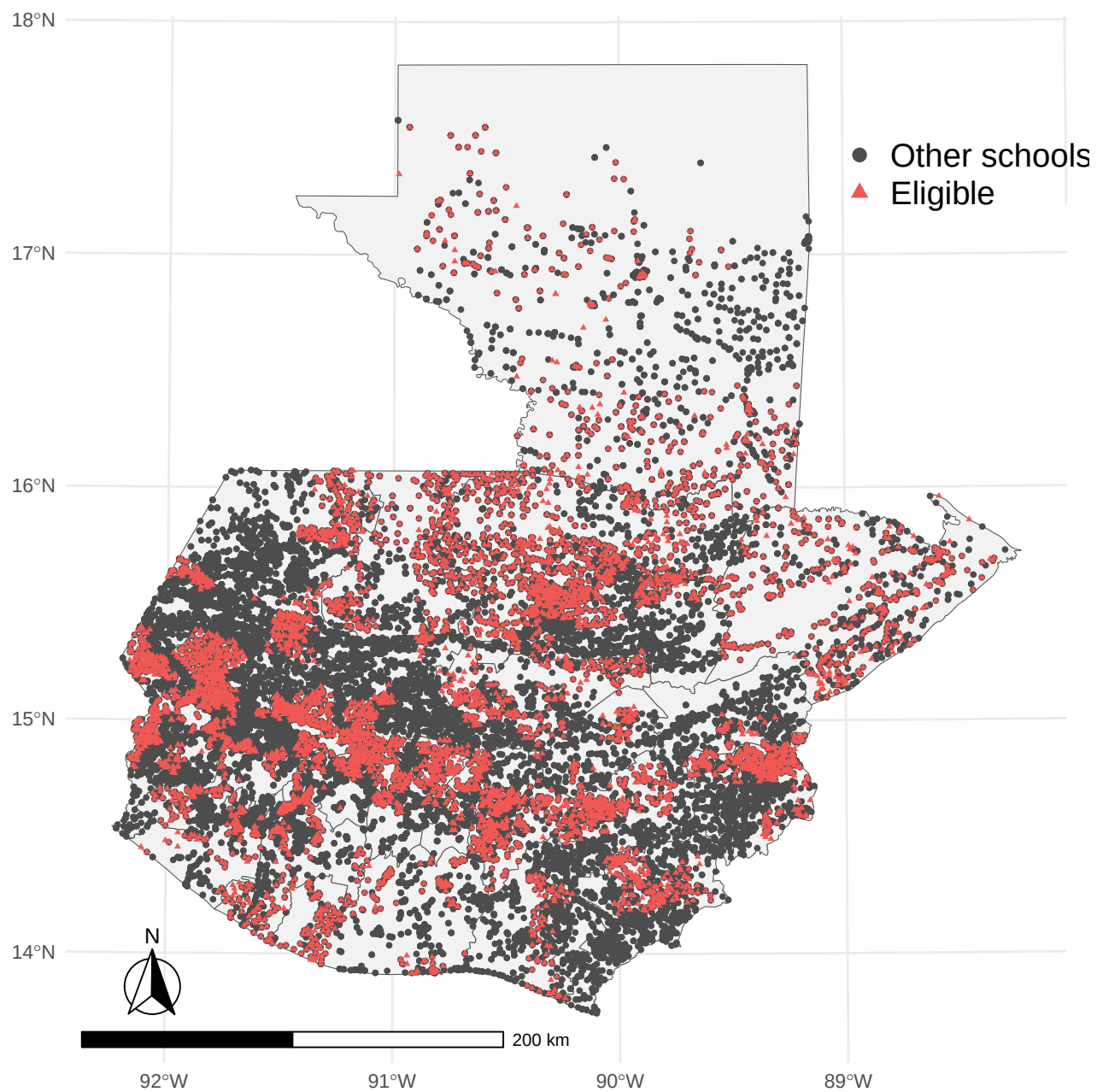
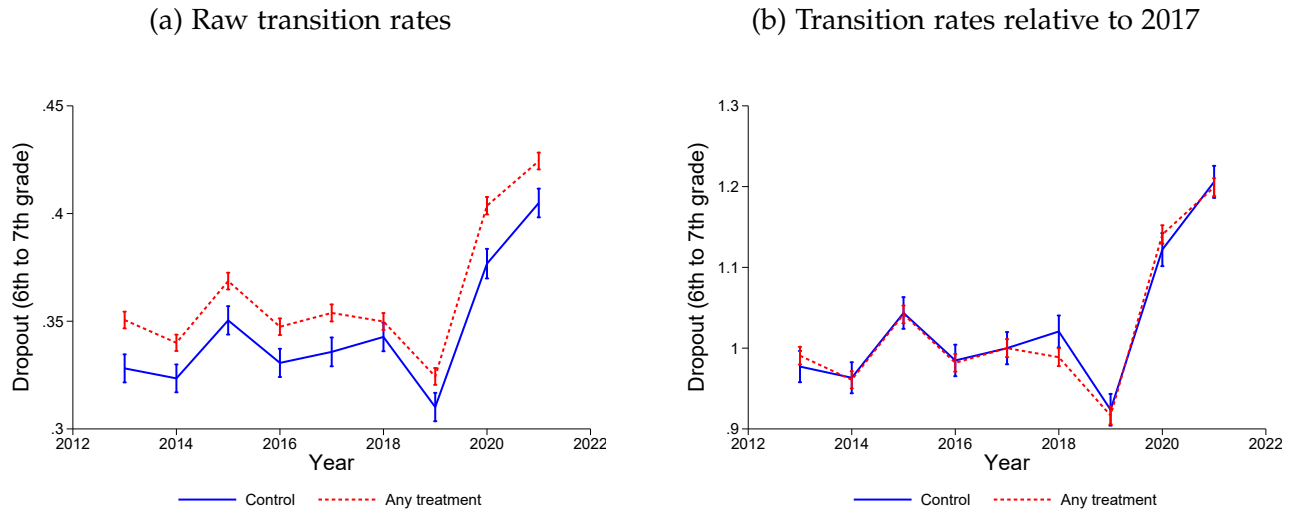


Figure A.6: Raw dropout rate between 6th and 7th grade across treatment and control schools



Note: This figure shows the share of students that drop out between 6th and 7th grade each year for treatment and control schools (as well as 95% confidence intervals). For example, the share of students that drops out in 2018 reflects the percentage of students that do not enroll in 7th grade in 2019. Figure A.6a presents the raw dropout rates, while Figure A.6b presents the rates relative to the 2017 share (e.g, a value of 1.2 represents a 20% higher dropout rate).

Table A.1: Differences between schools in the experiment and those not participating

	Other schools (1)	In the experiment (2)	Difference (2)-(1) (3)
% rural	90.09 (29.88) [9,312]	92.36 (26.57) [6,046]	2.27*** (0.46)
% morning shift	95.36 (21.03) [9,312]	93.35 (24.92) [6,046]	-2.01*** (0.39)
% bilingual	45.33 (49.78) [9,506]	51.22 (49.99) [6,080]	5.89*** (0.82)
Total enrollment	127.07 (133.08) [9,276]	153.96 (136.03) [6,037]	26.88*** (2.23)
Grd 6 enrollment	15.71 (18.79) [9,506]	18.89 (19.71) [6,080]	3.18*** (0.32)
Number of teachers	5.50 (4.98) [9,276]	6.10 (4.84) [6,037]	0.59*** (0.08)
% with a school board	85.40 (35.31) [9,283]	86.63 (34.03) [6,038]	1.23** (0.57)

Notes: Column 1 displays the mean, standard deviation (in parentheses), and number of observations (in square brackets) for schools not in the experimental sample. Column 2 shows the mean, standard deviation (in parentheses), and number of observations (in square brackets) for schools in the sample. Column 3 reports the differences between the two, as well as the standard error of the difference (in parentheses). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: Students enrolled in grade 6 in 2017 in the experiment vs. non-experiment schools

	Other schools (1)	In the experiment (2)	Difference (2)-(1) (3)
% male	51.34 (49.98) [149,307]	51.68 (49.97) [114,860]	0.35 (0.39)
Age (Jan 1st, 2017)	12.72 (1.20) [146,608]	12.88 (1.25) [113,299]	0.15*** (0.01)
GPA	7.62 (0.85) [146,608]	7.59 (0.84) [113,299]	-0.03*** (0.01)
% repeat grade	1.53 (12.27) [149,307]	1.30 (11.32) [114,860]	-0.23*** (0.08)
% at-risk (statistical model)	34.01 (47.37) [146,603]	50.66 (50.00) [113,298]	16.65*** (1.00)
% dropout	26.32 (44.04) [149,307]	35.13 (47.74) [114,860]	8.81*** (0.58)

Notes: Column 1 displays the mean, standard deviation (in parentheses), and number of observations (in square brackets) for schools not in the experimental sample. Column 2 shows the mean, standard deviation (in parentheses), and number of observations (in square brackets) for schools in the sample. Column 3 reports the differences between the two, as well as the standard error of the difference (in parentheses). Standard errors, clustered at the school level, are in parentheses. The dropout rate corresponds to students enrolled in sixth grade in 2017 and measures whether they are enrolled in school in 2018. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.3: Balance between control schools in the experimental sample and the rest of eligible schools not selected to be in any treatment group or the control

	Other eligible schools (1)	Control schools in the experiment (2)	Difference (2)-(1) (3)
Panel A: School level (2017)			
% rural	91.80 (27.44) [2,073]	91.16 (28.41) [995]	-0.64 (1.08)
% morning shift	93.49 (24.68) [2,073]	92.46 (26.41) [995]	-1.03 (1.00)
Total enrollment	153.32 (134.11) [2,071]	155.86 (142.17) [995]	2.54 (5.38)
Grd 6 enrollment	18.89 (19.48) [2,080]	19.14 (20.73) [1,000]	0.25 (0.78)
Number of teachers	6.08 (4.81) [2,071]	6.21 (5.14) [995]	0.13 (0.19)
Panel B: Student level (2017)			
% male	51.42 (49.98) [39,295]	51.29 (49.98) [19,138]	-0.13 (0.89)
Age	12.87 (1.23) [38,756]	12.87 (1.26) [18,905]	0.00 (0.02)
GPA	7.59 (0.83) [38,756]	7.59 (0.85) [18,905]	0.01 (0.02)
% at-risk (statistical model)	51.18 (49.99) [38,756]	48.96 (49.99) [18,904]	-2.21 (2.43)
Dropout	35.51 (47.85) [39,295]	33.58 (47.23) [19,138]	-1.93 (1.38)

Notes: Column 1 displays the mean, standard deviation (in parentheses), and number of observations (in square brackets) for schools not in the experimental sample but that are eligible. Column 2 shows the mean, standard deviation (in parentheses), and number of observations (in square brackets) for control schools in the experimental sample. Column 3 reports the differences between the two, as well as the standard error of the difference (in parentheses). Standard errors, clustered at the school level, are in parentheses. The dropout rate corresponds to students enrolled in sixth grade in 2017 and measures whether they are enrolled in school in 2018. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.4: Effect of being selected to be a control school in the experiment on the dropout using the 2018 cross-section

	Not enrolled in 2019		
	(1)	(2)	(3)
Pure controls	.011** (.005)	.0098* (.005)	.011** (.005)
Control mean	.35	.35	.35
R^2	.24	.3	.3
N. of obs.	59,954	59,954	59,954
Student controls	No	Yes	Yes
School dropout	No	No	Yes

Notes: This table presents the effects of being selected to be a control in the experiment (compared to the other 2,080 schools that were eligible and were not selected to be in any treatment or the control) on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). Pure control is equal to one if the school is selected to be a control school in the experiment (and is zero for the eligible schools not selected to be in the experimental sample). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.5: Proportion of schools surveyed in each round

	Mean/SD for each group				p-value (equality)	Treatment effect (vis-à-vis control)				Comparing treatments		
	Control	T1	T2	T3		Any	T1	T2	T3	T2 vs T1	T3 vs T1	T2-T3
	(1)	(2)	(3)	(4)		(2-4)vs(1)	(2)vs(1)	(3)vs(1)	(4)vs(1)	(3)vs(2)	(4)vs(2)	(4)vs(3)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Baseline (%)	71.90 (44.97) [1,000]	67.40 (46.90) [1,000]	69.40 (46.11) [1,000]	67.70 (46.79) [1,000]	0.106	-3.73** (1.66) [4,000]	-4.50** (2.05) [2,000]	-2.50 (2.04) [2,000]	-4.20** (2.05) [2,000]	2.00 (2.08) [2,000]	0.30 (2.09) [2,000]	-1.70 (2.08) [2,000]
Guide comprehension test (%)	8.40 (27.75) [1,000]	34.90 (47.69) [1,000]	35.30 (47.81) [1,000]	53.30 (49.92) [1,000]	0.000***	32.77*** (1.26) [4,000]	26.50*** (1.74) [2,000]	26.90*** (1.75) [2,000]	44.90*** (1.81) [2,000]	0.40 (2.14) [2,000]	18.40*** (2.18) [2,000]	18.00*** (2.19) [2,000]
First online survey (%)	30.60 (46.11) [1,000]	28.90 (45.35) [1,000]	29.00 (45.40) [1,000]	45.80 (49.85) [1,000]	0.000***	3.97** (1.70) [4,000]	-1.70 (2.05) [2,000]	-1.60 (2.05) [2,000]	15.20*** (2.15) [2,000]	0.10 (2.03) [2,000]	16.90*** (2.13) [2,000]	16.80*** (2.13) [2,000]
Second online survey (%)	15.70 (36.40) [1,000]	21.70 (41.24) [1,000]	21.10 (40.82) [1,000]	37.30 (48.38) [1,000]	0.000***	11.00*** (1.41) [4,000]	6.00*** (1.74) [2,000]	5.40*** (1.73) [2,000]	21.60*** (1.91) [2,000]	-0.60 (1.84) [2,000]	15.60*** (2.01) [2,000]	16.20*** (2.00) [2,000]
In-person endline (%)	85.90 (34.82) [1,000]	83.20 (37.41) [1,000]	81.70 (38.69) [1,000]	82.10 (38.35) [1,000]	0.041**	-3.57*** (1.30) [4,000]	-2.70* (1.62) [2,000]	-4.20** (1.65) [2,000]	-3.80** (1.64) [2,000]	-1.50 (1.70) [2,000]	-1.10 (1.69) [2,000]	0.40 (1.72) [2,000]

Notes: Columns 1–4 show the mean, standard deviation (in parentheses), and number of observations (in square brackets) for each of the experimental groups: Control, Treatment 1 (only component 1), Treatment 2 (components 1 and 2), and Treatment 3 (all three components). Column 5 displays the p-value of testing whether the mean is the same across all groups. Columns 6–9 report the difference between the three treatment groups and the control, the standard error of the difference (in parentheses), and the number of observations used for estimation (in square brackets). Column 6 shows the difference between all the treatment groups pooled together and the control, Column 7 the difference between the group with only the first component (T1) and the control, Column 8 the difference between the group with the first two components (T2) and the control, and Column 9 the difference between the group with all three components (T3) and the control. Columns 10–12 present the difference between different treatment groups, the standard error of the difference (in parentheses), and the number of observations used for estimation (in square brackets). Column 10 reports the difference between T2 and T1, Column 11 the difference between T3 and T1, and Column 12 the difference between T2 and T3. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.6: Balance across experimental groups with multiple comparisons

	Mean/SD for each group				Treatment effect (vis-à-vis control)				Comparing treatments		
	Control	T1	T2	T3	Any	T1	T2	T3	T2 vs T1	T3 vs T1	T2-T3
	(1)	(2)	(3)	(4)	(2-4)vs(1)	(2)vs(1)	(3)vs(1)	(4)vs(1)	(3)vs(2)	(4)vs(2)	(4)vs(3)
Panel A: School level											
% rural	91.16 (28.41) [995]	93.03 (25.48) [990]	93.88 (23.99) [996]	92.54 (26.29) [992]	1.99** (1.01) [3,973]	1.87 (1.21) [1,985]	2.72** (1.18) [1,991]	1.38 (1.23) [1,987]	0.85 (1.11) [1,986]	-0.49 (1.16) [1,982]	-1.34 (1.13) [1,988]
% morning shift	92.46 (26.41) [995]	92.42 (26.47) [990]	94.18 (23.43) [996]	94.05 (23.66) [992]	1.09 (0.95) [3,973]	-0.04 (1.19) [1,985]	1.71 (1.12) [1,991]	1.59 (1.12) [1,987]	1.75 (1.12) [1,986]	1.63 (1.13) [1,982]	-0.12 (1.06) [1,988]
Total enrollment	155.86 (142.17) [995]	153.14 (137.90) [987]	155.50 (132.89) [992]	152.65 (135.21) [992]	-2.10 (5.14) [3,966]	-2.73 (6.29) [1,982]	-0.36 (6.17) [1,987]	-3.22 (6.22) [1,987]	2.36 (6.09) [1,979]	-0.49 (6.14) [1,979]	-2.85 (6.02) [1,984]
Grd 6 enrollment	19.14 (20.73) [1,000]	19.01 (20.02) [1,000]	18.84 (18.29) [1,000]	18.58 (20.22) [1,000]	-0.33 (0.75) [4,000]	-0.13 (0.91) [2,000]	-0.30 (0.87) [2,000]	-0.55 (0.92) [2,000]	-0.17 (0.86) [2,000]	-0.42 (0.90) [2,000]	-0.25 (0.86) [2,000]
Number of teachers	6.21 (5.14) [995]	6.11 (4.79) [987]	6.06 (4.76) [992]	6.04 (4.73) [992]	-0.14 (0.18) [3,966]	-0.10 (0.22) [1,982]	-0.15 (0.22) [1,987]	-0.17 (0.22) [1,987]	-0.05 (0.21) [1,979]	-0.07 (0.21) [1,979]	-0.02 (0.21) [1,984]
Panel B: Student level											
% male	51.29 (49.98) [19,138]	51.89 (49.97) [19,006]	52.34 (49.95) [18,838]	51.77 (49.97) [18,583]	0.71 (0.84) [75,565]	0.60 (0.98) [38,144]	1.05 (0.98) [37,976]	0.48 (0.96) [37,721]	0.45 (0.86) [37,844]	-0.13 (0.85) [37,589]	-0.57 (0.85) [37,421]
Age	12.87 (1.26) [18,905]	12.87 (1.24) [18,662]	12.90 (1.24) [18,630]	12.89 (1.27) [18,346]	0.01 (0.02) [74,543]	-0.00 (0.03) [37,567]	0.03 (0.03) [37,535]	0.02 (0.03) [37,251]	0.03 (0.03) [37,292]	0.02 (0.03) [37,008]	-0.01 (0.03) [36,976]
GPA	7.59 (0.85) [18,905]	7.60 (0.85) [18,662]	7.57 (0.84) [18,630]	7.61 (0.84) [18,346]	-0.00 (0.02) [74,543]	0.00 (0.03) [37,567]	-0.02 (0.03) [37,535]	0.02 (0.03) [37,251]	-0.03 (0.03) [37,292]	0.01 (0.03) [37,008]	0.04 (0.03) [36,976]
% at-risk (statistical model)	48.96 (49.99) [18,904]	50.99 (49.99) [18,662]	51.65 (49.97) [18,630]	49.97 (50.00) [18,346]	1.91 (2.30) [74,542]	2.03 (2.82) [37,566]	2.69 (2.76) [37,534]	1.01 (2.82) [37,250]	0.66 (2.76) [37,292]	-1.02 (2.81) [37,008]	-1.68 (2.75) [36,976]
Dropout	33.58 (47.23) [19,138]	35.48 (47.85) [19,006]	35.57 (47.87) [18,838]	35.10 (47.73) [18,583]	1.81 (1.29) [75,565]	1.91 (1.57) [38,144]	1.99 (1.54) [37,976]	1.52 (1.58) [37,721]	0.08 (1.49) [37,844]	-0.38 (1.53) [37,589]	-0.46 (1.50) [37,421]

Notes: Columns 1–4 show the mean, standard deviation (in parentheses), and number of observations (in square brackets) for each of the experimental groups: Control, Treatment 1 (only component 1), Treatment 2 (components 1 and 2), and Treatment 3 (all three components). Columns 5–8 present the difference between the three treatment groups and the control, the standard error of the difference (in parentheses), and the number of observations used for estimation (in square brackets). Column 5 reports the difference between all the treatment groups pooled together and the control, Column 6 the difference between the group with only the first component (T1) and the control, Column 7 the difference between the group with the first two components (T2) and the control, and Column 8 the difference between the group with all three components (T3) and the control. Columns 9–11 show the difference between different treatment groups, the standard error of the difference (in parentheses), and the number of observations used for estimation (in square brackets). Column 9 shows the difference between T2 and T1, Column 10 the difference between T3 and T1, and Column 11 the difference between T2 and T3. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.7: Students enrolled in grade 6 in each year across treatment groups

	Mean/SD for each group				p-value (equality)
	Control	T1	T2	T3	
	(1)	(2)	(3)	(4)	(5)
2013	20.57 (22.86) [974]	20.29 (21.46) [968]	20.32 (20.01) [967]	19.78 (21.66) [965]	0.882
2014	20.47 (21.99) [985]	20.52 (22.01) [978]	20.39 (20.33) [979]	20.48 (22.03) [980]	0.999
2015	20.38 (21.61) [989]	20.12 (20.59) [991]	20.26 (19.47) [987]	19.63 (19.96) [985]	0.853
2016	19.94 (20.54) [994]	19.60 (20.48) [994]	19.63 (19.29) [994]	19.57 (20.64) [990]	0.975
2017	19.14 (20.73) [1,000]	19.01 (20.02) [1,000]	18.84 (18.29) [1,000]	18.58 (20.22) [1,000]	0.937
2018	19.67 (20.89) [1,000]	19.21 (20.09) [1,000]	19.29 (18.67) [1,000]	18.92 (19.77) [1,000]	0.876
2019	19.05 (20.34) [995]	18.73 (19.25) [994]	18.59 (17.68) [998]	18.32 (19.00) [996]	0.870
2020	19.19 (21.04) [998]	19.12 (19.71) [988]	19.02 (18.21) [997]	18.91 (19.73) [991]	0.990
2021	20.94 (22.73) [997]	20.60 (20.93) [991]	20.84 (19.88) [992]	20.53 (20.75) [997]	0.969

Notes: Columns 1–4 show the mean, standard deviation (in parentheses), and number of observations (in square brackets) for each of the following groups: Control, students in schools with only the first component (T1), students in schools with components 1 and 2 (T2), and students in schools with all three components (T3). Column 5 displays the p-value of testing whether the mean is the same across all groups. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Share of male students enrolled in grade 6 in each year across treatment groups

	Mean/SD for each group				p-value (equality)
	Control	T1	T2	T3	
	(1)	(2)	(3)	(4)	
2013	0.51 (0.50) [20,039]	0.51 (0.50) [19,643]	0.52 (0.50) [19,646]	0.51 (0.50) [19,088]	0.223
2014	0.52 (0.50) [20,166]	0.52 (0.50) [20,066]	0.54 (0.50) [19,959]	0.51 (0.50) [20,072]	0.010***
2015	0.50 (0.50) [20,154]	0.52 (0.50) [19,935]	0.52 (0.50) [20,000]	0.52 (0.50) [19,332]	0.221
2016	0.51 (0.50) [19,824]	0.52 (0.50) [19,478]	0.53 (0.50) [19,515]	0.52 (0.50) [19,377]	0.238
2017	0.51 (0.50) [19,138]	0.52 (0.50) [19,006]	0.52 (0.50) [18,838]	0.52 (0.50) [18,583]	0.751
2018	0.51 (0.50) [19,673]	0.51 (0.50) [19,212]	0.52 (0.50) [19,286]	0.51 (0.50) [18,923]	0.445
2019	0.51 (0.50) [18,955]	0.51 (0.50) [18,622]	0.52 (0.50) [18,551]	0.51 (0.50) [18,248]	0.618
2020	0.50 (0.50) [19,149]	0.51 (0.50) [18,894]	0.51 (0.50) [18,958]	0.50 (0.50) [18,735]	0.637
2021	0.51 (0.50) [20,882]	0.52 (0.50) [20,414]	0.52 (0.50) [20,672]	0.51 (0.50) [20,464]	0.454

Notes: Columns 1–4 report the mean, standard deviation (in parentheses), and number of observations (in square brackets) for each of the following groups: Control, students in schools with only the first component (T1), students in schools with components 1 and 2 (T2), and students in schools with all three components (T3). Column 5 shows the p-value of testing whether the mean is the same across all groups. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.9: ITT effect on dropout using the 2013–2018 panel

	Dropout		
	(1)	(2)	(3)
Panel A: Effects for each treatment			
Training $\times$ 2018 (α_1)	-.0081 (.006)	-.0087 (.006)	-.0097* (.0059)
Training+List $\times$ 2018 (α_2)	-.011* (.0056)	-.011** (.0056)	-.013** (.0055)
Training+List+Nudge $\times$ 2018 (α_3)	-.012** (.0058)	-.012** (.0058)	-.013** (.0057)
Control mean	.34	.34	.34
$\alpha_2 - \alpha_1$	-.0026	-.0024	-.0034
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.66	.68	.55
$\alpha_3 - \alpha_2$	-.0011	-.00089	.00047
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.85	.88	.93
$\alpha_3 - \alpha_1$	-.0037	-.0033	-.0029
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.54	.59	.62
R^2	.25	.25	.28
N. of obs.	468,953	468,953	468,953
Panel B: Effects for any treatment			
Any treatment $\times$ 2018	-.01** (.0047)	-.011** (.0047)	-.012** (.0046)
Control mean	.34	.34	.34
R^2	.25	.25	.28
N. of obs.	468,953	468,953	468,953
Sex/school size controls	No	Yes	Yes
Additional controls	No	No	Yes

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). The specification follows Equation 2. Panel A presents the effects for the three treatment groups separately, while Panel B presents the effects of receiving any treatment. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.10: Effect on dropout using the 2018 cross-section — restricted to schools with location data

	Not enrolled in 2019		
	(1)	(2)	(3)
Panel A: Effects for each treatment			
Training (α_1)	-.0092 (.0063)	-.0091 (.0064)	-.011* (.0062)
Training+List (α_2)	-.0093 (.0058)	-.0098* (.0059)	-.012** (.0058)
Training+List+Nudge (α_3)	-.011* (.0061)	-.011* (.0061)	-.012* (.006)
Control mean	.34	.34	.34
$\alpha_2 - \alpha_1$	-.00014	-.00065	-.00086
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.98	.92	.89
$\alpha_3 - \alpha_2$	-.0012	-.0016	.00038
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.83	.78	.95
$\alpha_3 - \alpha_1$	-.0014	-.0023	-.00047
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.83	.72	.94
R^2	.24	.3	.3
N. of obs.	70,336	70,336	70,336
Panel B: Effects for any treatment			
Any treatment	-.0097* (.005)	-.01** (.005)	-.011** (.0049)
Control mean	.34	.34	.34
R^2	.24	.3	.3
N. of obs.	70,336	70,336	70,336
Student controls	No	Yes	Yes
School dropout	No	No	Yes

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). The specification follows Equation 1. Panel A presents the effects for the three treatment groups separately, while Panel B presents the effects of receiving any treatment. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.11: Assessing possible spillovers using the number of nearby schools that are treated

	(1)	Dropout (2)	(3)	(4)	(5)	(6)	(7)
Any treatment	-.011** (.0049)	-.012** (.0049)	-.012** (.0049)	-.012** (.0049)	-.012** (.0049)	-.012** (.0049)	-.012** (.0049)
Treated schools within 1 km		.00019 (.00024)					
Experimental schools within 1 km		-.000047 (.000035)					
Public schools within 1 km		2.6e-06 (1.8e-06)					
Treated schools within 2 km			-.0023 (.0051)				
Experimental schools within 2 km			.0045 (.0049)				
Public schools within 2 km			-.00032 (.00047)				
Treated schools within 3 km				-.0033 (.0028)			
Experimental schools within 3 km				.0049* (.0027)			
Public schools within 3 km				-.00038* (.00022)			
Treated schools within 4 km					-.00077 (.0022)		
Experimental schools within 4 km					.0023 (.002)		
Public schools within 4 km					-.00021 (.00015)		
Treated schools within 5 km						.00063 (.0017)	
Experimental schools within 5 km						.00073 (.0016)	
Public schools within 5 km						-.00016 (.00011)	
Treated schools within 10 km							.000088 (.0015)
Experimental schools within 10 km							.001 (.0013)
Public schools within 10 km							-.00016* (.000085)
Control mean							
R ²	.3	.3	.3	.3	.3	.3	.3
N. of obs.	70,336	70,336	70,336	70,336	70,336	70,336	70,336

Notes: This table presents the effects of receiving any of the treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., not enrolled in 2019). The specification follows Equation 1. Column 1 replicates the results from Column 3 in Table 3. The other columns control for the number of treated schools, experimental schools, and public schools within a certain distance. All regressions control for student controls (sex, age, GPA, and relative ranking) and for historical dropout rates at the school level. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.12: Assessing possible spillovers using the fraction of nearby public schools that are treated

	(1)	Dropout (2)	(3)	(4)	(5)	(6)	(7)
Any treatment	-.011** (.0049)	-.01* (.0053)	-.012** (.0051)	-.012** (.005)	-.012** (.0049)	-.012** (.0049)	-.012** (.0049)
% of public schools treated within 1 km		.000045 (.00014)					
% of public schools treated within 2 km			-.000014 (.00016)				
% of public schools treated within 3 km				-.000047 (.00019)			
% of public schools treated within 4 km					3.9e-06 (.00025)		
% of public schools treated within 5 km						-.00011 (.00029)	
% of public schools treated within 10 km							.000051 (.00033)
Control mean							
R^2	.3	.29	.3	.3	.3	.3	.3
N. of obs.	70,336	63,661	66,987	69,521	70,121	70,287	70,319

Notes: This table presents the effects of receiving any of the treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., not enrolled in 2019). The specification follows Equation 1. Column 1 replicates the results from Column 3 in Table 3. The other columns control for the fraction of public schools within a certain distance that are treated. All regressions control for student controls (sex, age, GPA, and relative ranking) and for historical dropout rates at the school level. Standard errors, clustered at the school level, are in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.13: Effect on the number of years of schooling by 2022

	Years of schooling in 2022		
	(1)	(2)	(3)
Panel A: Effects for each treatment			
Training (α_1)	-.00019 (.014)	-.0026 (.013)	.0022 (.013)
Training+List (α_2)	-.0082 (.014)	-.0025 (.013)	.0053 (.013)
Training+List+Nudge (α_3)	.031** (.014)	.032** (.013)	.027** (.013)
Control mean	8.2	8.2	8.2
$\alpha_2 - \alpha_1$	-.008	.000065	.0031
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.58	1	.82
$\alpha_3 - \alpha_2$	.04	.035	.022
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.0063	.0097	.1
$\alpha_3 - \alpha_1$	.031	.035	.025
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.029	.0091	.061
R^2	.26	.36	.37
N. of obs.	77,094	77,094	77,094
Panel B: Effects for any treatment			
Any treatment	.0075 (.012)	.0089 (.011)	.012 (.011)
Control mean	8.2	8.2	8.2
R^2	.26	.36	.37
N. of obs.	77,094	77,094	77,094
Student controls	No	Yes	Yes
School dropout	No	No	Yes

Notes: This table presents the effects of the different treatments on the number of years of schooling students have by 2022. We calculate the years of schooling as the number of years they were enrolled in school from 2019 to 2022, plus 6 (since they were enrolled in Grade 6 in 2018). The specification follows Equation 1. Panel A presents the effects for the three treatment groups separately, while Panel B presents the effects of receiving any treatment. Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.14: Effect on the highest grade attained by 2022

	Highest grade attained by 2022		
	(1)	(2)	(3)
Panel A: Effects for each treatment			
Training (α_1)	.0021 (.014)	.0003 (.013)	.0048 (.013)
Training+List (α_2)	-.014 (.014)	-.0087 (.013)	-.0034 (.013)
Training+List+Nudge (α_3)	.037*** (.014)	.038*** (.013)	.034** (.013)
Control mean	8.1	8.1	8.1
$\alpha_2 - \alpha_1$	-.016	-.009	-.0082
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.27	.49	.53
$\alpha_3 - \alpha_2$	.051	.047	.037
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.00038	.00041	.0052
$\alpha_3 - \alpha_1$	.035	.038	.029
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.015	.0043	.029
R^2	.23	.35	.36
N. of obs.	77,094	77,094	77,094
Panel B: Effects for any treatment			
Any treatment	.0082 (.011)	.0097 (.01)	.011 (.011)
Control mean	8.1	8.1	8.1
R^2	.23	.35	.36
N. of obs.	77,094	77,094	77,094
Student controls	No	Yes	Yes
School dropout	No	No	Yes

Notes: This table presents the effects of the different treatments on the highest grade attained by 2022. The specification follows Equation 1. Standard errors, clustered at the school level, are in parentheses. Panel A presents the effects for the three treatment groups separately, while Panel B presents the effects of receiving any treatment. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.15: Heterogeneity by school characteristics

	Dropout in 2019			
	(1)	(2)	(3)	(4)
Training (α_1)	-.0094 (.0074)	-.018** (.007)	-.023 (.014)	-.032* (.016)
Training+List (α_2)	-.013* (.0077)	-.013* (.007)	-.014 (.013)	-.0067 (.015)
Training+List+Nudge (α_3)	-.0095 (.0077)	-.014** (.0068)	-.023* (.014)	-.02 (.014)
Training (α_1) $\times$ Covariate	-.0042 (.0094)	.016 (.012)	.013 (.015)	.023 (.017)
Training+List (α_2) $\times$ Covariate	.0015 (.0086)	.0035 (.012)	.0021 (.015)	-.0057 (.016)
Training+List+Nudge (α_3) $\times$ Covariate	-.0049 (.0089)	.0051 (.012)	.014 (.015)	.0094 (.016)
Control mean	.34	.34	.34	.34
$\alpha_1 + \text{Covariate}$	-.014	-.0022	-.0092	-.0087
p-value ($H_0 : \alpha_1 + \text{Covariate} = 0$)	.081	.84	.16	.17
$\alpha_2 + \text{Covariate}$	-.011	-.01	-.012	-.012
p-value ($H_0 : \alpha_2 + \text{Covariate} = 0$)	.076	.28	.057	.037
$\alpha_3 + \text{Covariate}$	-.014	-.0088	-.0096	-.011
p-value ($H_0 : \alpha_3 + \text{Covariate} = 0$)	.033	.37	.12	.077
R^2	.3	.3	.3	.3
N. of obs.	77,094	77,094	77,094	77,094
Covariate	Large school	Bilingual	School board	Morning shift

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). Each column interacts the treatment dummies with different covariates (presented at the bottom of the table). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.16: Heterogeneity by school characteristics — pooling treatments

	Dropout in 2019			
	(1)	(2)	(3)	(4)
Any treatment	-.011* (.0058)	-.015*** (.0057)	-.02* (.011)	-.021* (.012)
Any treatment $\times$ Covariate	-.0025 (.006)	.0081 (.0098)	.0094 (.012)	.01 (.013)
Control mean	.34	.34	.34	.34
<i>Treatment + Covariate</i>	-.013	-.007	-.01	-.011
p-value ($H_0 : \textit{Treatment} + \textit{Covariate} = 0$)	.014	.38	.048	.035
R^2	.3	.3	.3	.3
N. of obs.	77,094	77,094	77,094	77,094
Covariate	Large school	Bilingual	School board	Morning shift

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). Each column interacts the treatment dummy with different covariates (presented at the bottom of the table). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.17: Heterogeneity by student characteristics

	Dropout in 2019			
	(1)	(2)	(3)	(4)
Training (α_1)	-.013** (.0065)	-.0032 (.0078)	-.014** (.0074)	-.012** (.006)
Training+List (α_2)	-.015** (.0062)	-.0064 (.0075)	-.016** (.0071)	-.012** (.0056)
Training+List+Nudge (α_3)	-.0083 (.0064)	-.014* (.0074)	-.0096 (.0071)	-.012** (.0057)
Training (α_1) $\times$ Covariate	.0049 (.0099)	-.016* (.0097)	.0059 (.0074)	.03 (.046)
Training+List (α_2) $\times$ Covariate	.0072 (.0097)	-.011 (.0096)	.0085 (.0076)	-.036 (.042)
Training+List+Nudge (α_3) $\times$ Covariate	-.011 (.0098)	.0043 (.0097)	-.0053 (.0072)	.0099 (.045)
Control mean	.34	.34	.34	.34
$\alpha_1 + \text{Covariate}$	-.0081	-.019	-.0085	.018
p-value ($H_0 : \alpha_1 + \text{Covariate} = 0$)	.38	.011	.2	.7
$\alpha_2 + \text{Covariate}$	-.0078	-.018	-.0078	-.048
p-value ($H_0 : \alpha_2 + \text{Covariate} = 0$)	.38	.016	.22	.25
$\alpha_3 + \text{Covariate}$	-.019	-.0098	-.015	-.0021
p-value ($H_0 : \alpha_3 + \text{Covariate} = 0$)	.033	.19	.021	.96
R^2	.3	.3	.3	.3
N. of obs.	76,445	77,094	75,580	77,094
Covariate	Overage	Male	High GPA	Repeated grade

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., not enrolled in 2019). Each column interacts the treatment dummies with different covariates (presented at the bottom of the table). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.18: Heterogeneity by student characteristics — pooling treatments

	Dropout in 2019			
	(1)	(2)	(3)	(4)
Any treatment	-.012** (.0052)	-.0079 (.006)	-.013** (.0055)	-.012** (.0047)
Any treatment $\times$ Covariate	.0006 (.008)	-.0077 (.0078)	.0031 (.005)	.00034 (.036)
Control mean	.34	.34	.34	.34
<i>Treatment + Covariate</i>	-.012	-.016	-.01	-.011
p-value ($H_0 : \textit{Treatment} + \textit{Covariate} = 0$)	.11	.011	.046	.75
R^2	.3	.3	.3	.3
N. of obs.	76,445	77,094	75,580	77,094
Covariate	Overage	Male	High GPA	Repeated grade

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). Each column interacts the treatment dummy with different covariates (presented at the bottom of the table). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.19: Heterogeneity by risk of dropping out

Covariate	Dropout in 2019				
	(1)	(2)	(3)	(4)	(5)
	At-risk (model)	Probability (model)	Top tercile probability	Top quintile probability	Top decile probability
Training (α_1)	-.014** (.0068)	-.0081 (.0081)	-.0083 (.0066)	-.0066 (.0064)	-.0093 (.0062)
Training+List (α_2)	-.011* (.0066)	-.01 (.0077)	-.01 (.0063)	-.0098 (.006)	-.011* (.0058)
Training+List+Nudge (α_3)	-.018*** (.0067)	-.016** (.0078)	-.015** (.0063)	-.011* (.0061)	-.0098* (.0059)
Training (α_1) $\times$ Covariate	.004 (.011)	-.0093 (.02)	-.0071 (.013)	-.022 (.014)	-.019 (.018)
Training+List (α_2) $\times$ Covariate	-.0019 (.01)	-.0052 (.019)	-.0054 (.012)	-.01 (.014)	-.011 (.019)
Training+List+Nudge (α_3) $\times$ Covariate	.011 (.01)	.012 (.02)	.013 (.012)	.00011 (.015)	-.02 (.019)
Covariate	.032*** (.0088)	.15*** (.026)	.068*** (.011)	.067*** (.012)	.039*** (.014)
Control mean	.34	.34	.34	.34	.34
$\alpha_1 + \text{Covariate}$	-.0096	-.017	-.015	-.028	-.028
p-value ($H_0 : \alpha_1 + \text{Covariate} = 0$)	.28	.26	.18	.033	.11
$\alpha_2 + \text{Covariate}$	-.013	-.015	-.016	-.02	-.022
p-value ($H_0 : \alpha_2 + \text{Covariate} = 0$)	.12	.3	.13	.12	.22
$\alpha_3 + \text{Covariate}$	-.0067	-.0037	-.0016	-.011	-.03
p-value ($H_0 : \alpha_3 + \text{Covariate} = 0$)	.43	.81	.88	.42	.096
R^2	.3	.3	.3	.3	.3
N. of obs.	77,094	77,094	77,094	77,094	77,094

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). Each column interacts the treatment dummies with different covariates (presented at the top of the table). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.20: Heterogeneity by risk of dropping out — pooling treatments

Covariate	Dropout in 2019				
	(1)	(2)	(3)	(4)	(5)
	At-risk (model)	Probability (model)	Top tercile probability	Top quintile probability	Top decile probability
Any treatment	-.014** (.0056)	-.011* (.0065)	-.011** (.0053)	-.0092* (.005)	-.01** (.0049)
Any treatment $\times$ Covariate	.0044 (.0084)	-.0007 (.016)	.00014 (.01)	-.011 (.012)	-.017 (.015)
Covariate	.032*** (.0088)	.15*** (.026)	.068*** (.011)	.066*** (.012)	.039*** (.014)
Control mean	.34	.34	.34	.34	.34
<i>Treatment + Covariate</i>	-.0097	-.012	-.011	-.02	-.027
p-value ($H_0 : \text{Treatment} + \text{Covariate} = 0$)	.16	.33	.21	.067	.067
R^2	.3	.3	.3	.3	.3
N. of obs.	77,094	77,094	77,094	77,094	77,094

Notes: This table presents the effects of the different treatments on the likelihood of students enrolled in 2018 dropping out in the next academic year (i.e., they are not enrolled in 2019). Each column interacts the treatment dummy with different covariates (presented at the bottom of the table). Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.21: Reports from principals surveys

	% students dropout (1)	Returns to secondary (USD) (2)	Returns to secondary (%) (3)	Dropout top priority (4)	Knows most families (5)
Training (α_1)	.35 (1.7)	-197 (185)	.56 (2.2)	.064*** (.022)	-.02 (.017)
Training+List (α_2)	1.9 (1.6)	121 (362)	.49 (2.1)	.073*** (.022)	-.023 (.017)
Training+List+Nudge (α_3)	-.21 (1.7)	82 (263)	1.2 (2.3)	.046** (.022)	-.00031 (.017)
Control mean	.28	2,050	.52	.25	.86
$\alpha_2 - \alpha_1$	1.6	318	-.074	.0091	-.0025
p-value ($H_0 : \alpha_2 - \alpha_1 = 0$)	.34	.31	.97	.69	.89
$\alpha_3 - \alpha_2$	-2.1	-39	.7	-.027	.022
p-value ($H_0 : \alpha_3 - \alpha_2 = 0$)	.2	.92	.73	.23	.2
$\alpha_3 - \alpha_1$	-2.1	-39	.7	-.027	.022
p-value ($H_0 : \alpha_3 - \alpha_1 = 0$)	.2	.92	.73	.23	.2
Any treatment	.68	1.9	.75	.061	-.014
p-value ($H_0 : \text{Any treatment} = 0$)	.62	.99	.68	.00049	.29
R^2	.0006	.00044	.0001	.0039	.00091
N. of obs.	3,329	3,267	3,140	3,328	3,339

Notes: Standard errors, clustered at the school level, are in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

B Qualitative Insights from Focus Groups

To complement the quantitative analysis, we conducted a series of focus groups to gather qualitative insights on the implementation and effects of the intervention. These discussions aimed to capture the perspectives of school staff on the usefulness of the training and guide, changes in their practices, barriers to implementation, and their views on the challenges faced by students and families in the transition to secondary school.

The qualitative study consisted of 12 focus group discussions conducted over multiple days across different regions, ensuring representation from both urban and rural areas. In total, 97 individuals participated, including 63 teachers and principals from treatment schools and 34 trainers who facilitated the training sessions. The focus groups lasted between 90 to 120 minutes and followed a semi-structured format, allowing participants to openly discuss their experiences while ensuring that key themes were systematically explored. The sessions were conducted in Spanish, recorded, and subsequently transcribed for analysis. Participants also completed short, paper-based questionnaires during these sessions, providing additional individual perspectives. This section is based on the report prepared by José Alejandro Sosa Cortez, who was hired by the World Bank as a consultant to carry out the qualitative analysis.

Below, we summarize key themes from these focus groups. We highlight how the intervention influenced school priorities, how it was perceived as a signal of government commitment and the possible mechanisms behind its short-term success (and why impacts faded). To illustrate these insights, we include selected quotes from participants (in Spanish with inline English translations).

B.1 Dropout Prevention as a School Priority

Teachers reported that the program increased the importance of preventing student dropout within their daily practice. Many teachers noted that they had informally been trying to keep at-risk students in school even before the intervention, but the ENTRE program gave these efforts greater emphasis and structure. Importantly, teachers did not view the time spent on ENTRE activities as an extra burden, but rather as part of their professional duty to their students. As one teacher explained, “Porque realmente todas las estrategias que nos sugieren, pues prácticamente las habíamos hecho desde antes. Ahorita yo creo [que la guía] nos viene a fortalecer, para no dejarlo de hacer, viene a decir que tenemos que seguir luchando...yo creo que esta guía sí nos viene a fortalecer y volver a decirte que sí tenemos que seguir trabajando” (“Because really all the strategies they suggest, we had practically done them before. Now I think [the guide] comes to strengthen us so we do not stop doing it; it comes to tell us that we have to keep fighting...I think this guide really comes to strengthen us and tell us again that indeed we have to keep working”). This sentiment — that the program reinforced and legitimized dropout-prevention as a core responsibility — was common across focus groups.

Principals and teachers described being more attentive to students showing signs of being at risk of dropping out. Several participants admitted that previously they tended to treat all students the same, but the intervention made them more proactive in identifying and following up with those who had problems or were on the verge of leaving. “A veces en el salón de clases uno es parejo con todos...pero ya viendo algún problema

de algún niño, entonces ya uno le da seguimiento...se le acompaña un poquito más, le entiende uno algunos problemas que tiene...entonces creo que ha sido bastante de ayuda para el docente" ("Sometimes in the classroom you treat everyone equally...but upon seeing a problem with a particular child, you start to follow up...you accompany them a bit more, you understand some of the problems they have...so I think it has been quite helpful for us as teachers"). In this way, the program sharpened educators' focus on the needs of individual at-risk students, making dropout prevention a more concrete priority in day-to-day teaching practice.

B.2 The Intervention as a Signal of Government Commitment

"Ahora sí hay un compromiso" ("there is a commitment"): The ENTRE program's rollout was widely viewed as a clear sign that the government was prioritizing dropout. One trainer remarked that school dropout was "realmente...un problema al que el Ministerio no le había entrado nunca" ("really...a problem the Ministry had never tackled"). The new guide and training changed that: "...pero ya con la Guía, pues ya como que hay un compromiso...ya nos está indicando qué es lo que se haga" ("...but now with the Guide, it's as if there is a commitment...it's already indicating to us what must be done"). Thus, the intervention served as a credible signal that reducing dropouts was a national priority. Teachers in the focus groups echoed that having an official manual and workshop made the cause "feel real" — they now had direct instructions and felt the Ministry was watching, which motivated them to act. Several participants noted that before this program, any dropout-prevention efforts were at the discretion of individual schools or teachers ("ahí si quiero lo hago y si no, no" — "if I felt like doing it, I did, and if not, I didn't"), whereas the ENTRE strategy introduced a sense of obligation and formality.

B.3 Mechanisms of Short-Term Success vs. Long-Term Fade

Focus group discussions suggest that the program's short-term success in reducing dropouts stemmed from a combination of heightened awareness and concrete supportive actions by schools. Teachers spoke of using the guide's techniques to motivate students (for example, emphasizing the non-monetary benefits of education, as many students had become cynical about schooling when they saw unemployed graduates). They implemented classroom activities and counseling that reframed continuing education as valuable for personal growth and family well-being, not just for getting a job. Teachers and principals also increased outreach to families — holding meetings with parents of 6th graders, making home visits in some cases, and enlisting community support — all to encourage student transition to 7th grade. These efforts led to early "wins". In one school, a teacher proudly reported that after applying the strategies, "yo tengo el cien por ciento de mis estudiantes que...dicen que sí van a seguir" (English: "I have 100% of my students who...say they will continue [to secondary school]"). In several instances, participants noted that students who had initially planned to drop out were convinced to enroll in básico (lower secondary) thanks to the intervention's activities. Such immediate results underscore the program's effectiveness at generating enthusiasm and commitment in the short run.

At the same time, educators were realistic that these gains might not last without additional support. The teacher who celebrated 100% of her students' intent to continue

was quick to add, “...pero como decíamos, veremos, verdad, porque...en el transcurso del tiempo también pueden pasar varias cosas” (“...but as we were saying, we will see, because...over time many things can happen”). This cautious outlook was common. Participants understood that a student’s stated intention to remain in school could be derailed by future events — for example, financial hardships, family decisions, or other unforeseen barriers in the months and years ahead. In essence, the mechanism of impact was largely motivational and immediate, which helped in the short term but did not resolve deeper issues that might emerge later. Indeed, some teachers admitted that after the initial push, their schools reverted to “business as usual.” Not every school fully institutionalized the guide’s practices; some executed a few activities or talks with students and parents “de manera poco sistemática” (unsystematically), similar to what they had always done, rather than continuing a new structured approach in subsequent years. Furthermore, the program’s design offered limited follow-up: after the one-day training and initial implementation in 2018, there was little in the way of ongoing reinforcement. Trainers noted that monitoring largely consisted of receiving emails or official letters asking schools to complete a survey in the Ministry’s information system, but there was no on-site supervision or continued coaching.

One focus group recommendation was to extend ENTRE activities into the secondary grades — for instance, having a planned dropout-prevention activity for each year of lower secondary — to sustain the positive effects. In summary, the short-term success was driven by increased motivation, targeted attention, and community engagement, whereas the fade-out of impacts can be attributed to the absence of structural change or ongoing support to carry those initial gains forward.

B.4 Perspectives on Program Effectiveness in Practice

Teachers in the focus groups shared a range of practical insights about what worked in implementing ENTRE. Overall, they found the intervention’s toolkit useful and adaptable. “Pues todo ha sido útil, porque todas son herramientas que le ayudan al maestro” (“Well, everything has been useful, because these are all tools that help the teacher”), one teacher affirmed. If a suggested activity did not fit or was not (perceived as) effective, teachers felt empowered to “buscar sus propios métodos...para poder llegar a los alumnos” (“find their own methods...to reach their students”).

Educators stressed that working closely with parents was essential to keep children in school: “Entonces, se trabaja mucho con los padres de familia...Hay muchos papás muy interesados” (“So, we work a lot with parents...There are many parents who are very interested”). Specifically, schools organized parent meetings to discuss the importance of continuing education, sometimes even securing commitments from parents to support their child’s schooling. Teachers described using personal anecdotes and community examples to persuade parents, especially those who were skeptical about the value of further schooling. In the classroom, teachers used the guide’s activities (such as goal-setting exercises and letters about students’ aspirations) to inspire students. They also reported adapting the guide to their local context — for instance, translating abstract concepts into the local language or framing them in terms of familiar community experiences — which made the activities more relatable and effective for students. These

on-the-ground adjustments by teachers and principals were cited as a strength of the program, allowing its core ideas to take hold even in diverse settings.

Focus group sessions with the trainers (facilitators who delivered the workshops and supported schools) reinforced many of these points. Trainers viewed the ENTRE strategy as a useful, systematic resource for schools, and they saw evidence of its positive impact. One trainer recounted how, after the trainings, several principals reached out for help with students who had dropped out: "...me fueron llamando directores para que les acompañara en el proceso...visité varias aldeas y caseríos...y gracias a Dios se logró el objetivo, porque los niños regresaron" ("...principals started calling me to accompany them in the process...I visited several villages and hamlets...and thank God the objective was achieved, because the children returned [to school]").

Trainers did identify some areas for improvement to enhance effectiveness. For instance, they suggested simplifying or refocusing certain content in the guide: some activities were seen as too "scripted" or not sufficiently tailored to local realities. In particular, the trainers felt the guidance around financial support (scholarships and transport subsidies) was problematic (as discussed in the next section on structural challenges), and they recommended encouraging schools to seek local solutions (e.g., engaging NGOs or community funds) rather than relying on ministry-provided scholarships.

Overall, however, the feedback (from both teachers and trainers) was that the program equipped schools with valuable strategies and sprang a renewed commitment to prevent dropout.

B.5 Structural Challenges to Long-Run Sustainability

Despite the generally positive reception and the short-term successes, participants unanimously pointed to structural barriers that limited the program's long-term impact. These challenges often lay outside the immediate control of the school or the scope of the ENTRE intervention, and they help explain why keeping students in school through graduation remains difficult.

Poverty was cited as the most pervasive obstacle. Many families lack the resources to support continued schooling. Several teachers mentioned that the guide encouraged them to help families find scholarships or financial aid for students — but this well-intentioned advice proved frustrating in practice due to the low availability of these funding sources. One trainer bluntly noted, "...detectamos un problema con el tema económico...lo previmos porque el Ministerio de Educación no tiene becas...en los municipios focalizados nadie tiene becas...daba un poco de pena manejarlo, porque lo primero [que nos dijeron] fue: '¿para qué me dice si no hay?'" ("...we identified a problem regarding the economic issue...we anticipated it because the Ministry of Education has no scholarships...in the targeted municipalities nobody has scholarships...it was a bit embarrassing to deal with, because the first thing [parents] said to us was: 'why are you telling me this if there aren't any?'"). This highlights the gap between the program's recommendations and the reality on the ground. In the absence of financial assistance, the economic pressure on families remained. As another participant explained, "para un padre de familia que tiene...7 hijos, para él es mejor que vaya el niño que salió de sexto a trabajar porque le va a ayudar al sostenimiento del hogar" ("for a parent with...7

children, it's better that the one who finished sixth grade goes to work because he will help support the household"). In poor households, the opportunity cost of schooling is high: sending a 12- or 13-year-old to work can significantly contribute to family income or subsistence. These economic realities undercut the long-term effectiveness of the intervention — a motivated student might still drop out in a year or two if their family cannot afford secondary education or needs them to earn money.

Even when families and students are committed to continuing, the supply of accessible schools can be a limiting factor. Teachers from rural areas voiced concern about the lack of nearby básico schools (lower secondary schools) or limited slots in existing schools. In one community, so many students competed to enroll in 7th grade that "al final de cuentas no podían inscribirse...el cupo se cerraba" ("in the end they couldn't register...the slots filled up"). Such capacity constraints mean some graduates of primary school are unable to transition simply because there is nowhere for them to go — a structural issue well beyond the influence of a guidance program. Distance is another barrier: transportation costs and safety concerns become prohibitive if the nearest secondary school is far away, especially for girls. One focus group participant noted that many parents "no tenían el suficiente dinero para trasladar a sus hijos a los centros educativos que no quedaban muy cerca...eso...les generaba pagar pasajes...y era muy riesgoso" ("did not have enough money to transport their children to schools that were not very close by...that meant paying fares...and it was very risky [for the students to travel]").

The focus groups also brought up sociocultural factors that limited the impact of the program. For example, in some rural indigenous communities, traditional gender roles lead families to prioritize boys' education over girls'. Several educators observed that if resources are scarce, parents often choose to send the son to school and not the daughter, believing that a girl's role is at home. Early marriage or teen pregnancy was mentioned as another factor pulling adolescent girls out of school. Additionally, large family size (as noted above) means older siblings are expected to contribute to the household, either through work or by caring for younger siblings, which can cut short their schooling. While the ENTRE intervention tried to change mindsets by emphasizing the importance of education for all children, deeply entrenched attitudes and economic necessity often persisted, limiting how many students ultimately stayed through secondary school.

Finally, there were structural challenges related to the education system and the implementation of the program itself. Coordination across levels was one issue: teachers felt that without the active involvement of school supervisors and district education authorities, their dropout-prevention efforts could be undermined (for instance, if secondary schools did not accommodate incoming at-risk students or if officials did not reinforce the initiative in subsequent years). Focus group participants argued that "tienen que estar incluidos todos, supervisores y...autoridades" ("everyone has to be involved — supervisors and authorities") for the strategy to truly take root.

B.6 More quotes from around specific themes

Below, we add additional quotes around specific themes.

B.6.1 Usefulness of the guide

- “Porque realmente todas las estrategias que nos sugieren, pues prácticamente la habíamos hecho desde antes, y ahorita yo creo nos viene a fortalecer, para no dejarlo de hacer; viene a decir que tenemos que seguir luchando; porque la hemos hecho. Yo pienso que todos los compañeros maestros que estamos aquí, tal como dice el maestro, conoce la realidad social y hace de todo, de todas las profesiones. Empíricamente lo hemos hecho; yo creo que esta guía sí nos viene a fortalecer y volver a decirte que sí tenemos que seguir trabajando”. (English: “Because really all the strategies they suggest, we had practically done them before, and now I think it comes to strengthen us so we don’t stop doing it; it comes to tell us that we have to keep fighting. Because we have done it, I think that all of us teacher colleagues here, just as the teacher says, he knows the social reality and does everything, takes on all professions. Empirically we’ve done it; I think this guide really does strengthen us and reminds us that we have to keep working.”)
- “Será que le prestaran un poquito más de atención a estos niños, porque a veces en el salón de clases, pues, uno es parejo con todos, ¿verdad? Pero ya viendo algún problema de algún niño, entonces ya uno le da seguimiento a este niño, ¿verdad? Así trabajamos de cerca; ya se le acompañan un poquito más; le entiende uno algunos problemas que tiene, ¿verdad? Entonces creo que ha sido bastante de ayuda para el docente”. (English: “It might be that we give a little more attention to those children, because sometimes in the classroom, well, you treat everyone the same, right? But upon seeing some problem with a child, then you follow up with that child, right? That’s how we work closely; you accompany them a little more; you understand some problems they have, right? So, I think it has been quite helpful for the teacher.”)
- “Tal vez yo, desde mi punto de vista, a mí me pareció muy bonita la guía, vea, porque al final, como decían, son puntos que se hablan un poquito más técnicos, y un proceso un poquito más ordenado de los pasos de que, como decíamos anteriormente, que ya lo hemos por mucho tiempo; ahora tal vez faltaría, ¿verdad? ...” (English: “Perhaps, from my point of view, I found the guide very nice, you see, because in the end, as they were saying, these are points that are discussed a little more technically, and a slightly more orderly process of the steps that, as we said before, we have been doing for a long time; now maybe something is missing, right? ...”)
- “Es la integración de todo ...o sea, los incisos que nos hablan de las barreras se integran y se presentan como ya posibles ...una realidad, va, de que tenemos las diferentes barreras, y lo ... Nosotros lo ... Como decía mi compañero, lo hemos vivido, lo sabemos, y aquí ya lo podemos ver. Clasificar las realidades de la comunidad, siento yo, en cinco barreras, porque prácticamente en cinco barreras está la realidad de nuestras comunidades”. (English: “It is the integration of everything ...I mean, the sections that talk about the barriers are integrated and are presented as something possible ... a reality, you know, that we have the different barriers, and it ... We ... like my colleague was saying, we have lived it, we know it, and here we can see it. Classifying the realities of the community, I feel, into five barriers, because practically in five barriers lies the reality of our communities.”)

- "...este, yo pienso de que bueno, en nuestro caso como darnos más opciones a qué hacer, porque teníamos ...sabíamos los problemas, pero tal vez aquí nos dio un poquito más, nos abrió un poquito más a qué poder hacer, y ¿por qué no? ...y a lo mejor implementar algo más que no está ahí". (English: "...well, I think that in our case it gave us more options for what to do, because we had ... we knew the problems, but maybe here it gave us a little more, it opened us up a little more to what we can do, and why not? ... and maybe to implement something else that is not in there.")
- "Se tiene que hacer así, que llegue a todas las escuelas. En nuestra experiencia es importante y nos fortalece." (English: "It has to be done that way, so that it reaches all schools. In our experience it's important, and it strengthens us.")

B.6.2 Integration of guide activities

- "En el caso mío, podría comentar, como le mencionaba, yo lo tomé como parte de mi área, porque en la medida que salía el contenido que me ayudaba a fomentar un poco más esto, pues yo le daba". (English: "In my case, I could say, as I mentioned to you, I took it as part of my subject area, because as content came up that helped me promote this a bit more, well, I would cover it.")
- "Fíjense que, en cuanto a la Guía, yo creo que hoy voy a ser el patito feo, porque yo la verdad no he ... no he hecho ninguna actividad de estas que están en la Guía. ¿Por qué razón? Fíjense que a mí me dejaron con los seis grados, y además soy el director de los dos niveles, primaria y preprimaria, pues a mí no me da chance como para tomar la Guía y decir 'bueno, voy a ver qué hago de la Guía', porque tengo que estar pendiente de la alimentación; tengo que estar solo con primer grado ... que no son muchos, pero solo con primer grado se me va la mañana, yo ya no tengo tiempo para los demás grados. Pues voy haciendo lo que puedo, porque no me queda de otra ...". (English: "You see, regarding the Guide, I think today I'm going to be the odd one out, because honestly I haven't ... I haven't done any of those activities that are in the Guide. Why is that? You see, they left me with all six grades, and I'm also the principal for both levels, primary and pre-primary, so I don't get a chance to pick up the Guide and say 'well, let's see what I can do from the Guide', because I have to be in charge of the food program; I have to be alone with first grade ... there aren't many of them, but just with first grade my whole morning is gone, I no longer have time for the other grades. So I do what I can, because I have no other choice ...")
- "Sí, de igual manera yo estoy trabajando con tres grados, y tengo 54 niños, ¿verdad? O sea, estoy trabajando con cuarto, quinto y sexto; tengo 54 niños, 17 de sexto, ¿verdad? O sea, esa sería una de las limitantes para uno como maestro". (English: "Yes, in the same way I am teaching three grades, and I have 54 children, right? I mean, I'm teaching fourth, fifth and sixth; I have 54 children, 17 of them in sixth, right? I mean, that would be one of the limitations for someone as a teacher.")
- "Ahí está la diferencia: la Guía para nosotros tuvo éxito, porque en nuestro caso tenemos 40 niños; yo tengo dos maestros de sexto grado. Entonces, la Guía se puede aplicar muy bien; aparte de eso, tenemos todas las oportunidades aquí en el sector urbano. A mí casi todos los sectores que llegaron me ofrecieron becas; ahí hice

cuentas, yo dije: ‘a 20 vamos a becar’, ¿verdad?”. (English: “There’s the difference: the Guide was successful for us, because in our case we have 40 children; I have two sixth-grade teachers. So the Guide can be applied very well; besides that, we have all the opportunities here in the urban area. Almost all the organizations that came offered me scholarships; I did the math, and I said: ‘we will give scholarships to 20 of them’, right?”)

B.6.3 Structural barriers not addressed by the guide (social norms, economic barriers, lack of scholarships)

- “En el área rural, se da mucho de que se les da la opción de estudiar a los varones, más no a las niñas, ¿verdad?”. (English: “In rural areas, it often happens that boys are given the option to study, but not girls, right?”)
- “A las hembras no les daba estudio porque él tenía su mentalidad: que la hembra el esposo se la llevaba y él la tenía que mantener. ‘No desquita, no ayudan mucho’, me decía. ‘Es rara la que llega a 15 años’, decía. ‘En cambio los varones’, decía, ‘esos sí sirven’, decía, y los ponía (los pone todavía) a trabajar a la par de él; les saca el jugo, ¿va? ...”. (English: “He wouldn’t allow the girls to get schooling because he had this mindset: that if it’s a girl, her husband would take her away and he would have to support her. ‘They’re not worth it, they don’t help much,’ he used to tell me. ‘It’s rare for one to reach 15 years old,’ he would say. ‘But boys,’ he’d say, ‘those are useful,’ he’d say, and he put them to work next to him (in fact, he still does); he gets as much out of them as he can, you know? ...”)
- “Porque los ponen a trabajar, tanto a los niños como a las niñas. O sea, aunque nosotros hicimos la misma técnica que la seño (que llamamos a los padres de familia para concientizarlos y hacerles ver), ellos son tan cerrados que ellos no ... No ceden, no. Ellos, lo más importante es que el hijo trabaje y que aporte al hogar”. (English: “Because they put them to work, both boys and girls. I mean, even though we used the same technique as the other teacher (we called the parents in to make them aware and help them see), they are so narrow-minded that they just won’t ... They won’t give in, no. For them, the most important thing is that the child works and contributes to the household.”)
- “En la telesecundaria ese es el problema, que no hay una ayuda ... eso lo planteamos en la capacitación: que no hay becas para telesecundaria, ¿vea? No hay lo que es esa ayuda, ¿verdad? y no se podría gestionar ahí”. (English: “In the tele-secondary school that’s the problem, that there is no support ... we brought that up in the training: that there are no scholarships for tele-secondary, you see? There is no such assistance, right? And you couldn’t arrange anything there.”)
- “...hay muchos motivos por qué, ¿verdad?, de que el alumno ya no puede seguir estudiando: por falta de economía, por falta de interés, pero principalmente yo siento que ... principalmente es la falta de economía”. (English: “...there are many reasons why, right?, a student can no longer continue studying: for lack of money, for lack of interest, but mainly I feel that ... mainly it’s the lack of money.”)

- "... y proponerles algunas situaciones, como las becas, que ellos pueden tener acceso para poder estudiar. Pero además de eso, hay cuestiones – incluso ellos no tienen para comer en el día – ¡imagínese cómo este niño va a querer pensar en estudiar, si no tiene ni siquiera para comer!". (English: "... and propose some options, such as scholarships that they can have access to in order to study. But aside from that, there are issues – they don't even have food to eat in the day – imagine how that child is going to even want to think about studying, if he doesn't even have anything to eat!")
- "Donde es corte de café se van todos, desde el más grande hasta el más chiquito, cuando es temporada ... Por ejemplo, los niños de tercero primaria a sexto: nosotros el 2 de enero estamos en la escuela, el 15 comienzan las clases; aquí nosotros sin niños, sin varones. Ajá, con las niñas trabajamos casi que hasta febrero – la primera semana de febrero – y usted allí, como maestro, tiene que adaptarse a eso porque no lo van a cambiar. Si no, le dicen a uno: 'ya no estudie', ¿verdad?". (English: "In places where it's coffee harvesting season, everyone leaves, from the oldest to the youngest, when it's that time ... For example, the boys from third to sixth grade: on January 2nd we are at school, classes start on the 15th; and here we are with no children, no boys. Uh-huh, with the girls we work almost until February – the first week of February – and there you, as a teacher, have to adapt to that because it's not going to change. Otherwise, they tell you, 'don't study anymore,' right?")
- "Y como me decía un señor un día: 'Yo no estudié, pero contar puedo, mirá'. (Sacó un saco de billetes). 'Mirá, ahorita yo no estudié; no puedo leer', me dijo, 'pero mirá el camión que fui a comprar ahorita a la agencia, nuevito'. Y le digo yo: haceme un favor, lléname este cheque. 'No puedo', me dijo. 'Yo no puedo escribir', me dijo, 'ni leer', me dijo, 'pero aquí lo compenso', me dijo, 'mirá'. (Enseñando un fajo de billetes). (English: "And as a man told me one day, 'I didn't study, but I can count, look.' (He pulled out a bag of bills.) 'Look, now I didn't study; I can't read,' he told me, 'but look at the truck I just went and bought at the dealership, brand new.' And I said to him: Do me a favor, fill out this check for me. 'I can't,' he told me. 'I can't write,' he said. 'Or read,' he said. 'But I make up for it with this,' he told me, 'look.' (Showing a wad of bills).")
- "Y ahí fue el primer topón: la mamá tenía la idea de que ella es mujer y no puede seguir estudiando; la otra limitante es el padrastro, porque él no la ve como una hija." (English: "And that's where we hit the first roadblock: the mother thought that because the girl is female, she cannot continue studying; the other obstacle is the stepfather, because he doesn't see her as his daughter.")
- "En mi caso, le ayudé a una niña a conseguir una beca, ¿y qué pasó? La desintegración familiar afectó mucho. La familia se desintegró; entonces la niña ya no quiso seguir estudiando ni con su beca." (English: "In my case, I helped a girl get a scholarship, and what happened? The breakdown of her family had a big impact. The family fell apart, so the girl no longer wanted to continue studying, not even with her scholarship.")

B.6.4 Scholarships

- “El MINEDUC ofrece becas, pero no todos tienen acceso. En años anteriores, los padres de familia interesados hacen todo su trámite y al final no les ha salido la beca. Hay un esfuerzo de los papás, pero en el MINEDUC cuesta; a algunos les sale y a otros no.” (English: “The Ministry of Education offers scholarships, but not everyone has access to them. In past years, parents complete all the paperwork and in the end the scholarship never comes through for them. Parents do make the effort, but with the Ministry of Education it’s difficult — some manage to get it and others don’t.”)
- “Se ofrecieron becas, pero pedían un promedio de 80, y los patojos no llegan a eso. Nos dimos cuenta de que no eran para ese nivel.” (English: “They offered scholarships, but they required an average grade of 80, and the kids don’t reach that. We realized those scholarships were not intended for that level.”)
- “Durante 15 años, nosotros visitamos ONGs que pudieran becar a los niños; sí hubo instituciones que becaron a niñas y niños. Pero hoy en día las instituciones se están alejando; financiamiento ya no hay.” (English: “For 15 years we visited NGOs that could provide scholarships for the children; indeed there were organizations that sponsored some girls and boys. But today those institutions are pulling out — there is no longer any funding.”)

B.6.5 Expanding to other grades

- “Sí es necesario hacerlo. Les dimos fotocopias de la Guía a cada maestro para que la apliquen desde primero hasta sexto grado; así se implementó en la escuela.” (English: “It is necessary to do so. We gave photocopies of the Guide to each teacher so they can apply it from first through sixth grade; that’s how it was implemented in our school.”)
- “Cuando vimos el libro pensamos: ‘¡Ah, esto ya lo hemos hecho!’. Pero eso nos generó otra idea: que no solo haya una transición exitosa de sexto a básico, sino de grado a grado. Hay que tener herramientas puntuales, principalmente de primero a segundo grado, donde hay más repetismo.” (English: “When we saw the book we thought, ‘Oh, we’ve already done this!’ But it gave us another idea: not only to ensure a successful transition from sixth grade to secondary school, but from each grade to the next. We need to have specific tools, especially from first to second grade, where there is a higher rate of grade repetition.”)
- “Yo lo enfocaría no solo a sexto primaria, sino a toda la primaria. Para mí, no solo es para sexto sino también para todos los alumnos.” (English: “I would focus it not only on sixth grade, but on the entire primary level. For me, it’s not just for sixth grade but for all students.”)
- “Desde primero hasta sexto grado, para que tengan el éxito de pasar a básico. Porque si los niños en quinto y sexto ya están desanimados de seguir estudiando, hay que comenzar desde primero primaria.” (English: “From first through sixth grade, the aim is for them to successfully move on to secondary school. If by fifth or sixth grade they are already discouraged from continuing their studies, then we have to start in first grade.”)

- “Hay que irles creando esa idea de transición en cada año, que la escuela sea algo agradable para el niño. Al terminar un ciclo, que vea que lo que sigue es mejor que lo que dejó.” (English: “We have to instill the idea of transition in them at each grade, so that school is something enjoyable for the child. When one cycle is completed, they should see that what comes next is better than what they left behind.”)

B.6.6 Usefulness of the list of students at risk of dropping out

- “Fíjese que nosotros visualizamos: hay algunos que los integramos al listado, y hay otros que aparecen aquí de que sí van a continuar”. (English: “Notice that what we observed is that there are some whom we include on the list, and there are others who show up here that yes, they are going to continue.”)
- “...este todo el grado ...lo único que, de igual manera ...no venía de otra escuela, sino que estaba una niña que se retiró el año anterior; entonces esta niña no apareció en ese listado, pero de igual manera entonces, nosotros tenemos 17 en total ...venían 16 en el listado”. (English: “...in this case the whole class ... the only thing is, likewise ...she didn’t come from another school, rather there was a girl who dropped out the previous year; so that girl did not appear on that list, but anyway, we have 17 in total ...16 came on the list.”)
- “Sí, solo nosotros ... solo nosotros tenemos problema con una niña; en el paquete venía un niño, pero con el niño se logró trabajar ... se le preguntó, se trabajó con la Guía, y él ...lo económico no era el factor, sino que él ya no muy quería. Pero se le logró motivar y sí va a seguir estudiando”. (English: “Yes, only us ...only we have an issue with one girl; in the packet a boy came up, but with that boy we managed to work ... we talked to him, we worked with the Guide, and for him ...the economic issue wasn’t the factor, rather he just didn’t really want to continue. But we managed to motivate him and he is indeed going to keep studying.”)
- “Pues en mi caso son 10 alumnos ...especialmente ahí sí es difícil porque a nadie le gusta seguir. Y entonces convocamos a los padres de familia, a los alumnos ...les dimos unas charlas; pero de igual manera, de los 10 sí se rescataron entre 5 y 6”. (English: “Well, in my case there are 10 students ...it’s particularly difficult because none of them want to continue studying. So we called the parents in, and the students ...we gave them some talks; but in the same way, out of those 10 we managed to salvage about 5 or 6.”)

B.7 Quotes from facilitators

B.7.1 Perceptions of program implementation

- “Considero que el programa se implementó muy bien en general; vimos que la mayoría de maestros aplicaron los talleres tal como se les enseñó.” (English: “I consider that the program was implemented very well overall; we saw that most teachers carried out the workshops exactly as they were taught.”)
- “En mi experiencia, el programa sí se implementó de forma adecuada en las escuelas. Los docentes hicieron un trabajo excelente y los estudiantes mostraron interés.”

(English: "In my experience, the program was indeed implemented properly in the schools. The teachers did an excellent job and the students showed interest.")

- "La verdad, la implementación del programa resultó bastante exitosa en la mayoría de los casos. Hubo mucho compromiso por parte de los capacitadores y de los docentes." (English: "Honestly, the program's implementation turned out to be quite successful in most cases. There was a lot of commitment from both the facilitators and the teachers.")
- "Yo creo que el programa se llevó a cabo bien. Tuvimos algunos contratiempos pequeños al inicio, pero luego todo fluyó según lo planificado." (English: "I think the program was carried out well. We had a few small setbacks at the beginning, but then everything flowed as planned.")
- "En general, la ejecución del programa fue buena; en algunas escuelas costó un poco al principio, pero en otras funcionó sin problemas desde el inicio." (English: "Overall, the program's execution was good; at some schools it was a bit difficult at first, but at others it worked without problems from the start.")

B.7.2 Challenges observed in implementation

- "Uno de los principales desafíos que observamos fue la falta de tiempo en el horario escolar para dedicarle al programa." (English: "One of the main challenges we observed was the lack of time in the school schedule to dedicate to the program.")
- "Vimos que algunos docentes tenían demasiadas responsabilidades, y eso dificultaba que pudieran implementar todas las actividades del programa." (English: "We saw that some teachers had too many responsibilities, which made it difficult for them to implement all of the program's activities.")
- "Un reto grande fue que en ciertas escuelas no contábamos con los materiales suficientes, entonces tocó improvisar con lo que teníamos." (English: "A big challenge was that in certain schools we didn't have enough materials, so we had to improvise with what we had.")
- "Otro desafío que enfrentamos fue la resistencia inicial de algunos directores o padres, que no entendían bien de qué se trataba el programa." (English: "Another challenge we faced was the initial resistance from some principals or parents, who didn't really understand what the program was about.")
- "A veces costaba coordinar con el personal de la escuela para fijar los horarios de las sesiones; esa logística fue un desafío constante." (English: "Sometimes it was hard to coordinate with the school staff to set the session schedules; that scheduling logistics was a constant challenge.")

B.7.3 Recommendations for program improvement

- "Yo recomendaría brindar más acompañamiento y seguimiento a los docentes durante la implementación, para apoyarlos cuando encuentren dificultades." (English: "I would recommend providing more coaching and follow-up to the teachers during the implementation, to support them when they encounter difficulties.")

- “Sería bueno incluir talleres de refuerzo o recapitulación, porque algunos maestros pidieron más tiempo para dominar el contenido.” (English: “It would be good to include refresher or recap workshops, because some teachers asked for more time to master the content.”)
- “Sugiero que se adapten ciertos contenidos del programa a la realidad de cada comunidad; ajustar el material culturalmente ayudaría mucho.” (English: “I suggest adapting certain contents of the program to each community’s reality; adjusting the material culturally would help a lot.”)
- “Creo que para mejorar el programa se podría involucrar más a los padres de familia, para que apoyen lo que los niños aprenden en clase.” (English: “I think to improve the program they could involve parents more, so they support what the children are learning in class.”)
- “Recomendaría proporcionar material didáctico adicional, como folletos o guías visuales, para facilitarle a los maestros la enseñanza del programa.” (English: “I would recommend providing additional teaching materials, like booklets or visual guides, to make it easier for teachers to teach the program.”)

C Guide for teachers and principals on how to prevent dropout (Component 1)

GUÍA PARA PROMOVER LA TRANSICIÓN EXITOSA DE SEXTO PRIMARIA A PRIMER GRADO DEL CICLO BÁSICO

Entre los muchos desafíos que enfrenta nuestro sistema educativo, uno de los más importantes es la deserción escolar. Cada año, cerca de 50 mil estudiantes del Ciclo de Educación Básica abandonan los estudios antes de tiempo para enfrentar las dificultades de la vida sin haber adquirido las habilidades y la madurez necesarias. A pesar de que la Constitución Política de la República de Guatemala (Artículo 74)¹ exige que todos los niños sean educados hasta concluir el Ciclo de Educación Básica, perdemos a un tercio de los estudiantes durante la transición del sexto grado de primaria a primero básico. No podemos seguir aceptando esto como nuestra realidad. Cada niño independientemente de su sexo, etnia o situación económica, debe completar al menos el Ciclo de Educación Básica, como lo requiere la Ley y por nuestros valores como Nación.



Como director y como docente, usted sabe que existen innumerables factores que conducen a la deserción escolar, y que los centros educativos no pueden cambiarlos por sí solos. Sin embargo, hay mucho que podemos hacer como educadores y líderes para mantener a nuestros estudiantes en el centro educativo. Por lo tanto, el Ministerio de Educación ha iniciado una nueva «Estrategia Nacional para la Transición Exitosa -ENTRE-», y **su centro educativo ha sido seleccionado para ser pionero en esta estrategia.**

¹Artículo 74.- Educación obligatoria. Los habitantes tienen el derecho y la obligación de recibir la educación inicial, preprimaria, primaria y básica, dentro de los límites de edad que fije la ley. La educación impartida por el Estado es gratuita. El Estado proveerá y promoverá becas y créditos educativos. La educación científica, la tecnológica y la humanística constituyen objetivos que el Estado deberá orientar y ampliar permanentemente. El Estado promoverá la educación especial, la diversificada y la extra escolar.

Es importante que se una al esfuerzo del Ministerio de Educación para prevenir la deserción escolar, trabajando para aumentar el número de estudiantes que terminan sexto grado y que pasarán al Ciclo de Educación Básica el próximo año. Solicitamos su valioso apoyo y liderazgo para la implementación de acciones que minimicen la deserción escolar en su centro educativo.

Para hacer esto, **el Ministerio de Educación ha elaborado esta guía con acciones específicas y prácticas que usted debe tomar en los próximos meses para ayudar a sus estudiantes de sexto grado a hacer la transición hacia primero básico.**

Estas acciones requieren poco tiempo, pero tienen el impacto de cambiar las vidas de sus estudiantes.

Para ello, le solicitamos que lea la guía, hable con el resto de docentes que imparten sexto grado, el director y supervisor educativo e **implemente por lo menos una de las acciones sugeridas antes de que finalice el ciclo escolar. Si conoce otra acción comente al director.** El Ministerio de Educación seguirá el progreso de su centro educativo y el de otros establecimientos educativos pioneros de la «Estrategia Nacional para la Transición Exitosa -ENTRE-». Agradecemos su compromiso con la meta crítica de reducir el abandono escolar. El futuro de los estudiantes depende de su compromiso con este proceso de prevención.

Costo social de la deserción escolar

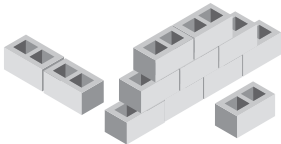
Cometer crímenes	Uso de sustancias nocivas para la salud	Embarazo en niñas y adolescentes	Pobreza	Pocas oportunidades laborales, migración ilegal

Barreras que afectan y algunas intervenciones prácticas para apoyar a los estudiantes

El problema del abandono escolar es muy complejo y tiene causas variadas. Estas forman una barrera para que los estudiantes continúen con sus estudios. Algunas causas como la falta de centros educativos del nivel medio cercanos, requieren de planes a mediano plazo e inversiones considerables, quedando fuera del accionar de los directores y docentes. Sin embargo, existen algunas barreras sobre las cuales ambos pueden tener influencia en el corto plazo.

Principales barreras del abandono escolar:

- Falta de motivación o interés
- Falta de recursos económicos
- Falta de apoyo familiar
- Bajo rendimiento
- Indiferencia



¡MANOS A LA OBRA!

A continuación incluimos algunas intervenciones prácticas que los docentes pueden implementar para derribar las barreras que impiden que sus estudiantes continúen con sus estudios.

Barrera: Falta de motivación o interés

Uno de los principales motivos del abandono escolar en el país es la falta de interés o motivación para estudiar. Dicha falta puede explicarse por distintos factores; algunos de ellos incluyen: baja autoestima, acceso a información limitada sobre el valor real de la educación, entre otros.

Acción de apoyo 1:

Ejercicios cortos y prácticos de autoafirmación

Algunos jóvenes piensan que la educación media no es para ellos, o que no tienen la capacidad para poder enfrentar los retos del nivel medio. Por ello, es importante realizar ejercicios de autoafirmación, o que resalten el impacto significativo que el esfuerzo puede tener en la capacidad de aprender.

Ejercicios sugeridos:

a. Yo estoy orgulloso de mí mismo

Propósito: Propiciar el autoconocimiento y desarrollar expectativas de éxito.

Solicite a los estudiantes que piensen en alguna experiencia que los hizo sentirse orgullosos de sí mismos y exitosos. Formar equipos de trabajo de 4-5 estudiantes. Cada estudiante hablará sobre su experiencia personal por 2-3 minutos al resto del grupo.

Con el grupo de estudiantes, el docente seleccionará una experiencia que ilustre un hábito o un valor que aporte al alcance de una tarea con éxito; a partir de la experiencia seleccionada, el docente motivará a cada estudiante para que exprese cómo este buen hábito o valor contribuye al éxito escolar.

b. Mis valores personales²

Propósito: Propiciar el establecimiento de metas a corto y mediano plazo.

Entregue una hoja a cada estudiante de sexto grado con la información que se detalla a continuación, o bien, escriba en el pizarrón los siguientes valores, actitudes o cualidades:

Lea la siguiente lista de valores, actitudes o cualidades:

Aceptación	Confianza	Felicidad	Lealtad
Amistad	Confianza	Franqueza	Paz / no violencia
Amor	Creatividad	Generosidad	Perseverancia
Armonía	Crecimiento personal	Gratitud	Respeto al ambiente
Autoconfianza	Curiosidad	Honestidad	Responsabilidad
Ayudar a otros	Descubrimiento	Independencia	Sabiduría
Buen humor	Disciplina	Innovación	Servicial
Colaboración	Divertido	Integridad	Sencillez
Compasión	Eficiencia	Inteligencia	Solución de problemas
Complicancia	Entusiasmo	Justicia	Trabajador
Compromiso	Excelencia	Libertad	Esfuerzo
Comunidad	Fe	Liderazgo	

²Este ejercicio fue adaptado del ejercicio de Cook, Purdie-Vaughns, García, & Cohen, 2012.

1. Pregunte: ¿cuáles valores, actitudes o cualidades te motivan más? ¿Con cuáles te identificas más? Elige tres en el orden de preferencia. Si algo viene a la mente que no está en esta lista, anótalo.
2. ¿Qué cualidades aprecias más de ti mismo?
3. Escribe una carta dirigida a ti mismo de tu futuro. Describe tus valores y cualidades. Incluye cómo te gustaría estar el próximo año y en los próximos 5 años. ¿Qué metas te gustaría haber alcanzado en tus estudios? ¿Qué te gustaría hacer al finalizar tus estudios?
4. El docente le pedirá al estudiante que le entregue su carta, la cual volverá a leer dos semanas después y la devolverá al docente. Al mes o dos meses puede volver a leerla para recordar sus metas. Al finalizar el ciclo escolar, el docente se la devolverá al estudiante.

Sugerencia de aplicación:

Planifique este ejercicio de reflexión a partir de recibir la guía y la inducción para la aplicación en el aula.

Otros ejercicios que puede trabajar para propiciar el establecimiento de metas en sus estudiantes son los sugeridos en el Anexo 1 de las páginas 22 y 23 de esta guía.



Acción de apoyo 2:

Ejercicio sugerido:

Proveer información concreta sobre beneficios económicos de la educación media

- Muchas veces las familias y estudiantes de contextos más vulnerables desconocen lo que implica para su futuro el no tener estudios del nivel medio.
- **¿Sabía que Guatemala es uno de los países de Latinoamérica con mayores retornos económicos de la educación media?** Esto significa que finalizar los estudios del nivel medio mejora significativamente la calidad de vida de sus estudiantes, de su familia y de sus futuros hijos.
- Se le recomienda compartir los siguientes **ejemplos para proveer información concreta sobre retornos económicos de la educación media:**

«Atención estudiantes, inscribanse en el nivel medio. No solo es importante para el desarrollo del país sino para su futuro».

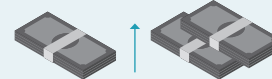
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Estrategia Nacional para la Transición Exitosa -ENTRE-

¿Sabías que los beneficios salariales de la educación media de Guatemala están entre los más altos de Latinoamérica?

Una persona que termina el nivel medio en promedio gana 1,797 quetzales más por mes que una que solo terminó la primaria. Es decir, un 81 % más.

¡Un 81 % MÁS es casi el doble!



Recuerda que vas a trabajar muchos años. A lo largo de tu vida laboral la diferencia acumulada asciende a **más de 800 mil quetzales**. Es decir:

- Mucho más de lo que puede costar **UNA CASA**
- Equivale a **más de 350 televisores de alta resolución**
- Equivale a **11 carros importados**
- Al estudiar podrás optar a trabajos que requieren habilidades más especializadas en matemática, lectura y escritura. **La calidad de los empleos a los que acceden los graduados del nivel medio es superior al de los empleos de quienes solo terminaron la primaria.**
- Piense en ejemplos de empleos disponibles en su comunidad que le sean visibles y cercanos: locales de comercio, bancos, telefonías, etc.
- Por ejemplo, mientras muchos egresados de sexto primaria deben soportar el calor y las inclemencias del clima en sus trabajos, los graduados del nivel medio tienen horarios más cómodos y realizan un menor esfuerzo físico en sus empleos.
- La educación no solo es importante durante la vida laboral, sino también al momento de retirarse.
- En Guatemala:
 - 1 de cada 10 trabajadores que solo terminaron la primaria tienen derecho a jubilación.
 - 5 de cada 10 trabajadores que terminaron el nivel medio tienen una jubilación al retirarse.



¡Tú sí puedes!

¡Finalizar la educación media puede cambiar toda tu vida!

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Estrategia Nacional para la Transición Exitosa -ENTRE-

Consejo práctico: esta información puede compartirse de distintas formas. Algunas ideas:

- Visite el aula de sexto grado para informar que el período de inscripción a primero básico se acerca y use la información de la página anterior para motivarlos a inscribirse.
- Invite a los estudiantes de sexto grado que elaboren un afiche con esta información y lo dibujen (se puede también organizar un concurso para ver quién hace el mejor afiche).
- Aproveche la organización del gobierno escolar para informar y acompañar la transición de los estudiantes de sexto primaria a primero básico.

Sugerencia de implementación: utilice el material de lectura que se encuentra en el Anexo 2 sobre retornos económicos de la educación media (pp. 24 y 25).



Barrera: Falta de recursos económicos

Muchas familias carecen de los recursos económicos necesarios para poder mandar a sus hijos al Ciclo de Educación Básica. Aun cuando la educación sea gratuita, siempre existen costos adicionales (uniforme, transporte, útiles, etc.) o la necesidad por parte de la familia de que el niño trabaje para generar ingresos.



Acción de apoyo:

Apoyar a las familias en el proceso de postulación para ayuda financiera

El Ministerio de Educación dispone en la actualidad de diversos programas de asistencia financiera:

- Bolsa de estudio
- Becas para estudiantes con discapacidad
- Bono de transporte (solo para la Ciudad de Guatemala)

Sin embargo, muchas veces las familias no cuentan con la información necesaria para que sus hijos opten a una beca, dificultades para entender las instrucciones, o porque los trámites les resultan muy engorrosos. Su valiosa orientación para **el proceso de llenado de formularios puede hacer una GRAN diferencia para las familias más vulnerables**. Los pasos para postular a las becas son relativamente prácticos y se detallan en el Anexo 3. Descripción de apoyos escolares.

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Estrategia Nacional para la Transición Exitosa -ENTRE-

El Anexo contiene:

- a. Los requisitos y las instrucciones básicas para optar a distintos programas de asistencia financiera.
- b. Dónde obtener los formularios para obtener a la ayuda económica (se requieren menos de 30 minutos para llenarlos con ayuda de los padres de familia).
- c. En caso de tener alguna duda puede pedir apoyo a su supervisor.

Adicionalmente, podrá averiguar en la municipalidad, con empresas locales o con ciudadanos, si existen otras becas disponibles para estudiantes de su centro educativo.



Barrera: Falta de apoyo familiar

Para los niños de esta edad, el apoyo familiar es FUNDAMENTAL para poder continuar sus estudios, ya que estas decisiones se toman en conjunto con los padres. Por esto se recomienda tener comunicación permanente con las familias para informar sobre el rendimiento académico de sus hijos.



Acción de apoyo:

Motivar a las familias

- Trate de tener una comunicación fluida con las familias de los estudiantes (incluyendo reuniones periódicas y visitas a los hogares), en las cuales se reconozca y valore el rol de la familia para tomar decisiones. Se ha comprobado que los centros educativos en donde se tiene mayor comunicación con los padres, los resultados en las pruebas nacionales son mejores.
- En las reuniones de padres de familia, **evite** mensajes como «su hijo está en riesgo de abandono». Esto puede tener el efecto opuesto al deseado. En su lugar se recomienda utilizar frases como «su hijo tiene capacidad de continuar con sus estudios, si se le apoya lo suficiente en el hogar».
- **Es más efectivo concentrarse en los beneficios de que se inscriban en la educación del nivel medio**, más que sobre los «riesgos» de los estudiantes. Por ejemplo, puede utilizar la información sobre los beneficios personales, profesionales, familiares y económicos de la educación media para motivar no solo a los niños sino también a sus padres.
- Además en las reuniones de padres de familia puede explicarles las opciones y el proceso de inscripción para el Ciclo de Educación Básica.

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PRECAUCIÓN: EVITE ESTIGMATIZAR A LOS ESTUDIANTES

- Los padres a quienes se les dice que su hijo está «en riesgo» pueden interpretar negativamente lo que esto significa, que no vale la pena el esfuerzo de apoyarlos para que permanezcan en el centro educativo.
- Los docentes nunca deben darse por vencidos en apoyar a los estudiantes aún cuando otras personas los consideren «causa perdida».
- Los estudiantes pueden interiorizar mensajes positivos o negativos... la estigmatización puede convertir el riesgo en realidad!

Enfoque su comunicación en los beneficios de permanecer aprendiendo en el centro educativo y sobre el potencial para que cada estudiante logre el éxito escolar.



Barrera: Bajo rendimiento

No todos los estudiantes aprenden al mismo ritmo, identifique los que necesitan mayor atención en su aprendizaje para alcanzar las competencias de cada área y subárea. Cuando note que algún estudiante requiere de ayuda adicional, es necesario encontrar formas para apoyarlo.



Acción de apoyo 1:

Implementar adecuadamente el proceso de mejoramiento de los aprendizajes.

Revise el plan de mejoramiento de los estudiantes en riesgo y determine si este es pertinente para las necesidades del estudiante o si es posible considerar otras actividades de apoyo para el adecuado desarrollo de la competencia.

Considere los siguientes pasos para la revisión del proceso de mejoramiento de los aprendizajes:

1. Identifique el indicador de logro y los contenidos correspondientes a la actividad de evaluación.
2. Determine los aprendizajes en los que el estudiante demuestra dificultad con lo esperado en el indicador de logro y los contenidos.
3. Verifique que el plan del proceso de mejoramiento de los aprendizajes haya logrado que el estudiante reafirmara el aprendizaje esperado.

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4. En caso de que el plan del proceso de mejoramiento de los aprendizajes no haya sido el adecuado, según las necesidades del estudiante y lo establecido en el indicador de logro y los contenidos, replantee la actividad para que el estudiante mejore el aprendizaje.

Recuerde que el proceso de mejoramiento de los aprendizajes busca que los estudiantes desarrollen conocimientos, habilidades y actitudes que le sirvan para la vida.



Acción de apoyo 2:

Implementar tutorías

Para los estudiantes que se quedan atrás en el desarrollo de las competencias, organice un proceso para implementar tutorías y asistencia grupal o personalizada.

Las tutorías pueden ofrecerse a todos los estudiantes, pero se recomienda que sean obligatorias para los de más bajo rendimiento. Estas pueden ofrecerse o realizarse durante el tiempo de recreo, antes o después del horario escolar.

- Se recomienda tener al menos tutorías en las áreas de Comunicación y Lenguaje y Matemática.
- Fomente la solidaridad entre estudiantes de alto rendimiento con los estudiantes de bajo rendimiento.
- Gestione alianzas estratégicas (municipalidades, instituciones privadas, organismos internacionales, entre otros), con el propósito de conseguir un lugar para trabajar con los estudiantes, o para proveer un docente tutor, materiales, entre otros.



Barrera: Indiferencia

Una de las razones que explican el abandono escolar es la indiferencia. La indiferencia es entendida en este contexto como la actitud de desapego que se manifiesta con insensibilidad ante una situación. De la misma forma, en ocasiones tomamos decisiones que cambian la vida. Por ejemplo, el simple hecho de tomar la decisión correcta de inscribirse en el Ciclo de Educación Básica, puede cambiar el futuro de muchos jóvenes.

Para romper la barrera de la indiferencia, se debe sensibilizar a los padres de familia, comprometer a los docentes y motivar a los estudiantes para que terminen la educación media.



Acción de apoyo:

Apoyo en el proceso de inscripción

1. Organice reuniones para explicar a los padres de familia y estudiantes de sexto grado el proceso de inscripción del Ciclo Básico.
 - Las fechas de inscripción:
 - Generalmente el proceso de preinscripción empieza en octubre-noviembre de cada año, debiendo los estudiantes completar su papelería en enero.
 - La documentación requerida para inscribirse es: (i) certificado de primero a sexto grado de primaria (con calificaciones); (ii) diploma de sexto grado; (iii) certificado de nacimiento extendido por Renap y, (iv) fotocopia de DPI del padre, madre o encargado.
 - Investigue en su entorno las opciones de centros educativos cercanos del Ciclo de Educación Básica y preséntelas en las reuniones.
2. Invite a los padres de familia y estudiantes a reuniones de seguimiento para la transición de sexto a primero básico.
3. Invite a los directores de centros educativos del Ciclo de Educación Básica a las sesiones informativas para padres o encargados y estudiantes, de la transición de los estudiantes de sexto grado a primero básico.
4. Organice visitas con los estudiantes para que conozcan los centros educativos del Ciclo de Educación Básica de la localidad, y así motivar la inscripción en el mismo.
5. Invite a estudiantes exitosos del Ciclo Básico de diversas modalidades, para que compartan su experiencia con los de sexto primaria.
6. Si es necesario, considere reunirse con los supervisores y directores de los establecimientos del nivel medio para informarse sobre cómo se lleva a cabo el proceso de inscripción de los estudiantes.

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Estrategia Nacional para la Transición Exitosa -ENTRE-

Vistazo a las recomendaciones

Barrera a la educación	Recomendación de acción
Falta de motivación o interés	Actividades prácticas para propiciar el autoconocimiento y desarrollar expectativas de éxito.
Falta de recursos económicos	Apoyar a las familias en el proceso de postulación para ayuda financiera. Proveer información concreta sobre beneficios económicos de la educación media.
Falta de apoyo familiar	Motivar a las familias para buscar alternativas en beneficio de la superación de sus hijos.
Bajo rendimiento	Asegurar la adecuada implementación del proceso de mejoramiento de los aprendizajes. Implementar tutorías.
Indiferencia	Apoyo en el proceso de inscripción y dotar de información a los estudiantes y padres de familia de los centros del Ciclo de Educación Básica.

Ahora, a alcanzar una meta de permanencia escolar...

El director y los docentes fijarán una meta conjunta de permanencia escolar.

La meta es reducir al mínimo, el número de estudiantes que están en riesgo de abandonar los estudios durante la transición a primer año del Ciclo Básico.

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Estrategia Nacional para la Transición Exitosa -ENTRE-

Para construir un indicador al respecto, se sugiere el siguiente procedimiento:

- a. Identifique el número de estudiantes que egresaron de sexto grado durante el anterior año escolar.
- b. Establezca un aproximado de estudiantes que se inscribieron en el Ciclo de Educación Básica en el presente año.
- c. ¿Qué porcentaje representa los estudiantes no inscritos en el Ciclo de Educación Básica respecto de quienes hablan egresado de sexto grado el año anterior? **Este es el porcentaje que se buscará reducir a la mitad.**

Por ejemplo, asuma que las siguientes cifras son las de su centro educativo:

			Coloque los datos de su establecimiento
Estudiantes egresados de sexto grado	octubre 2017	70 estudiantes	
Estudiantes que se inscribieron en 1.º básico	enero 2018	45 estudiantes	
Número de estudiantes no inscritos en el 2018	octubre 2017 a enero 2018	70 - 45 = 25 estudiantes	
% de deserción (no inscritos en básicos)	enero 2018	25/70 *100 = 36 %	
Meta (mitad de % no inscritos)	enero 2019	36/2= 18 %	



Yo me comprometo a reducir el abandono escolar
a: _____ %

ANEXOS

Anexo 1. Ejercicios para establecer metas a corto y mediano plazo

Mis valores personales

Lea la siguiente lista de valores, actitudes o cualidades:

- Aceptación
- Amistad
- Amor
- Armonía
- Autoconfianza
- Ayudar a otros
- Buen humor
- Colaboración
- Compasión
- Complacencia
- Compromiso
- Confiabilidad
- Confianza
- Creatividad
- Crecimiento personal
- Curiosidad
- Descubrimiento
- Disciplina
- Divertido
- Eficiencia
- Entusiasmo
- Excelencia
- Fe
- Felicidad
- Franqueza
- Generosidad
- Gratitude
- Honestidad
- Independencia
- Innovación
- Integridad
- Inteligencia
- Justicia
- Libertad
- Liderazgo
- Lealtad
- Paz / no violencia
- Perseverancia
- Respeto al ambiente
- Responsabilidad
- Sabiduría
- Servicial
- Sencillez
- Solución de problemas
- Trabajador
- Esfuerzo

Luego de leer el listado, responde:

1. ¿Cuáles valores, actitudes o cualidades te motivan más? Señálos encerrándolos en un círculo.
2. ¿Con cuáles te identificas más? Subráyalos.
3. Elige tres en el orden de preferencia. Si algo te viene a la mente que no está en esta lista, anótalo:

Con paso firme: visualiza tu transición al Ciclo Básico

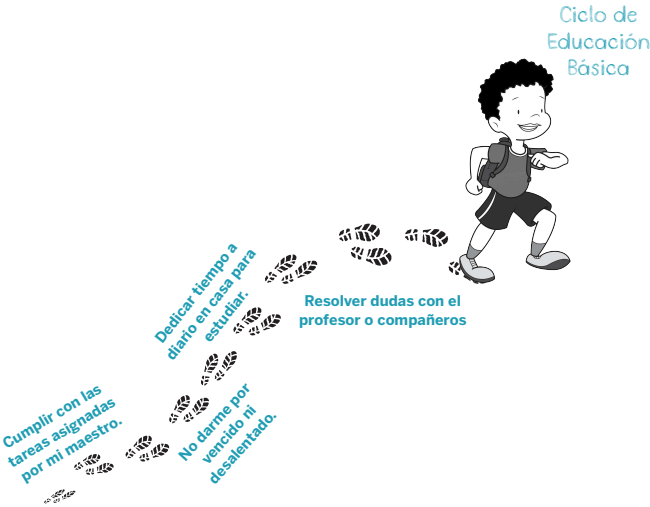
Invite a los estudiantes que tracen varias veces las plantas de sus pies en material de reciclaje y las recorten a manera de tener varias huellas para formar con ellas un recorrido. Motíuelos a que escriban alrededor de las huellas las acciones que deben realizar para lograr la transición exitosa al siguiente nivel educativo.

Sugerencia de implementación:

Utilice el aula como un escenario de aspiraciones de sus estudiantes, en donde se les refuerce que con buenos hábitos, valores y comportamientos lograrán sus metas.

1. ¿Qué cualidades aprecias más de ti mismo?

2. Escribe una carta dirigida a ti mismo de tu futuro. Describe tus valores y cualidades. Incluye cómo te gustaría estar el próximo año y en los próximos 5 años. ¿Qué metas te gustaría haber alcanzado en tus estudios?

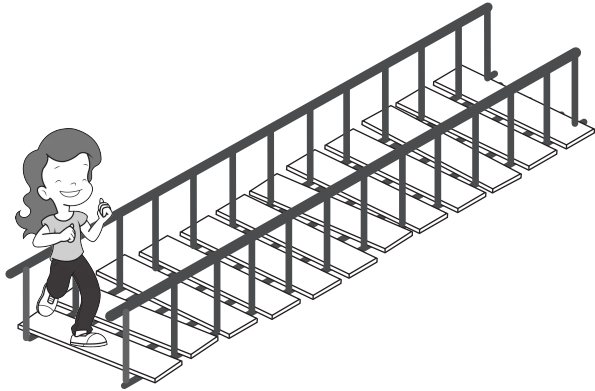


Pasemos al Ciclo de Educación Básica

Los estudiantes de la clase del profe/de la seño _____
_____ irán al Ciclo de
Educación Media.

Construya con material de reciclaje un puente, con el propósito de que los estudiantes puedan ir colocando en cada uno de los tabloncillos, las acciones que les permitirán alcanzar el Ciclo de Educación Media.

Primero
Básico



Sexto
primaria

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Estrategia Nacional para la Transición Exitosa -ENTRE-

Anexo 2. Beneficios económicos de la educación media

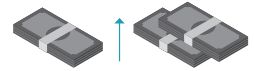
«Atención estudiantes,
inscribanse en el nivel medio.

No solo es importante para el desarrollo del país, sino
para su futuro»

¿Sabías que Guatemala es uno de los países de Latinoamérica con mayores retornos económicos de la educación media? Esto significa que finalizar el nivel medio mejora significativamente tu calidad de vida, la de tu familia, y la de tus futuros hijos.

Una persona que termina el nivel medio en promedio
gana 1,797 quetzales más por mes que una que solo
terminó la primaria. Es decir, un 81 % más.

iUn 81 % MÁS es casi el doble!



Escribe tres ideas sobre cuáles serían los beneficios que lograrías en el futuro al estudiar la media.

Seguramente en los próximos años, iniciarás tu vida laboral. Al acumular años de trabajo, este incremento puede ascender a **más de 800 mil quetzales. Es decir:**

- Mucho más de lo que puede costar **UNA CASA**
- Equivale a **más de 350 televisores de alta resolución**
- Equivale a **11 carros importados**



Al estudiar podrás optar a trabajos que requieren habilidades más especializadas en matemática, lectura y escritura. **La calidad de los empleos a los que acceden los graduados del nivel medio es superior al de los empleos de quienes solo terminaron la primaria.**

Piensa y escribe ejemplos de empleos disponibles en tu comunidad que sean visibles y cercanos. Agrega también qué formación crees que deberían tener las personas para realizarlos.

Por ejemplo, mientras muchos egresados de sexto primaria deben soportar el calor y las inclemencias del clima en sus trabajos, en su mayoría los graduados del nivel medio tienen horarios más cómodos y realizan un menor esfuerzo físico en sus empleos.

Recuerda:

La educación no solo es importante durante la vida laboral, sino también al momento de retirarse.

En Guatemala:

- 1 de cada 10 trabajadores que solo terminaron la primaria tienen derecho a jubilación.
- 5 de cada 10 trabajadores que terminaron el nivel medio tienen una jubilación al retirarse.

Comparte cómo crees que finalizar la educación media puede beneficiarte:

iTú si puedes! iFinalizar la educación media puede
cambiar toda tu vida!

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Estrategia Nacional para la Transición Exitosa -ENTRE-

Anexo 3. Descripción de beneficios escolares

Bolsa de estudio

Requisitos:

- Ser estudiantes en el Nivel de Educación Media en algún centro educativo público (Instituto de Educación Básica por Cooperativa, Telesecundaria, Instituto de Educación Básica INEB).
- Ser guatemalteco y estar entre los 12 y 16 años de edad para iniciar estudios en el Ciclo de Educación Básica.
- Para el primer grado del Ciclo de Educación Básica, haber aprobado sexto grado de educación primaria.
- Haber aprobado todas las asignaturas del último año cursado con un promedio general mínimo de 70 puntos.
- Ser estudiante de escasos recursos económicos, plenamente comprobados.
- Ser soltero/soltera.
- No tener más hermanos que estén gozando del beneficio de la bolsa de estudio.
- No estar gozando de otra beca que brinde el Estado de Guatemala.
- No ser estudiante repitente.

Instrucciones básicas para optar a una bolsa de estudio:

- Llenar la «Solicitud de bolsa de estudio» con la información general y socioeconómica, con declaración jurada del padre, madre o encargado (original), debidamente firmada.
- Carta de compromiso de los padres o encargados (original).
- Obtener el certificado de nacimiento del estudiante emitido por el Registro Nacional de las Personas (Renap) de reciente emisión (original).
- Certificado de calificaciones del último grado aprobado, sexto grado de primaria (fotocopia).
- Constancia de inscripción en el primer grado del Ciclo de Educación Básica del nivel medio.
- Documento Personal de Identificación (DPI) del padre, madre o encargado (fotocopia).
- Constancia de buena conducta extendida por el director del centro educativo público donde estudió sexto primaria.
- Si es huérfano: fotocopia del certificado de defunción de los padres de familia extendido por el Registro Nacional de las Personas (Renap).

Formularios: disponibles en la Dirección Departamental de Educación.

- Teléfono para pedir asistencia en caso de duda: contacte al supervisor.

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Estrategia Nacional para la Transición Exitosa -ENTRE-

Becas para estudiantes con discapacidad

El Programa de Becas para Estudiantes con Discapacidad en centros educativos oficiales es parte del sistema de becas del Ministerio de Educación, como una de las obligaciones del Estado frente al acceso a la educación, contenida en el Artículo 33, numeral 18; asimismo, como uno de los derechos fundamentales de los educandos, de acuerdo al Artículo 39, numeral 8, reafirmado por el Artículo 85, todos de la Ley de Educación Nacional, Decreto Legislativo 12-91. Fue creado en el año 2007 a través del Acuerdo Ministerial N.º 2539-2007 y fortalecido por las reformas contenidas en los Acuerdos Ministeriales N.º 428-2009 y 3276-2011. Para su operatividad se creó su reglamento con el Acuerdo Ministerial N.º 826-2009 y su reforma con el Acuerdo Ministerial N.º 2987-2011.

Tiene como finalidad otorgar ayuda financiera por medio de una beca a estudiantes con discapacidad física, auditiva, visual, intelectual o múltiple, cuya formación se ve limitada por su condición de pobreza y consiste en un desembolso económico anual de actualmente Q 1,000.00.

Requisitos:

- 1. Estar inscrito en el sector oficial.
- 2. Demostrar la necesidad del apoyo económico.
- 3. Tener la condición de discapacidad (física, visual, auditiva, intelectual o múltiple).
- 4. Cumplir con el mínimo de asistencia según el Reglamento de Evaluación de los Aprendizajes 1171-2010.
- 5. Promover el grado con los ajustes necesarios (adecuaciones curriculares).

Instrucciones básicas para optar a la beca: En cada Dirección Departamental de Educación en donde se hace la solicitud de la beca, hay un comité que se encarga de ejecutar la convocatoria, seleccionar a los becarios, renovar, finalizar o suspender la misma.

Formularios: El Ministerio de Educación cuenta con un instructivo y formularios estandarizados que se encuentran publicados en la página oficial del Sistema de Gestión de la Calidad (SGC).

La normativa de becas permite realizar el pago de los mil quetzales en un solo desembolso o en dos pagos de quinientos quetzales, eso lo deciden las unidades financieras de cada Dirección

Departamental de Educación. Por lo que cada programa define en sus propios tiempos el o los pagos y todo el proceso está sujeto a disponibilidad presupuestaria.

Teléfono para pedir asistencia en caso de duda: contacte a su Dirección Departamental de Educación, en la unidad de educación especial.

Bono de transporte para estudiantes de la Ciudad Capital

Requisitos:

- 1. Estudiantes que residen en la ciudad capital.
- 2. Estudiantes inscritos en centro educativo público.
- 3. Estudiantes que utilizan el servicio de transporte público urbano para desplazarse hacia el centro educativo público y el retorno a su domicilio (determinado por la distancia del centro educativo público y la residencia del estudiante).
- 4. Cumplir con una asistencia mensual del 80 % durante cada mes.

Instrucciones básicas para postular: el Director del Establecimiento Educativo deberá solicitar información a la Dirección Departamental de Educación y manifestar su interés para que sus estudiantes, que cumplan con los requisitos, se beneficien de este pago.

Formularios: disponibles en la Dirección Departamental de Educación.

- Teléfono para pedir asistencia en caso de duda: contacte a su Dirección Departamental de Educación.

INFORMACIÓN DE LAS OPCIONES QUE OFRECE EL MINISTERIO DE EDUCACIÓN

Institutos Nacionales de Educación Básica -INEB-

Los Institutos de Educación Básica -INEB- son una opción que está integrada por tres grados y constituye la fase final de la educación obligatoria básica.

Dirigida a la población estudiantil comprendida entre los 13 a 15 años de edad, atendiendo en plan diario, en las diferentes jornadas establecidas a efecto de proporcionar a los estudiantes una educación integral, respondiendo a las demandas sociales y características regionales del país, tanto para en el área urbana como rural.

La atención que se brinda a los estudiantes es por medio de docentes especializados en cada una de las áreas del Currículo Nacional Base.

Los requisitos para inscribirse son:

- 1. certificado de nacimiento extendido por Renap
- 2. certificados de primero a sexto grado de primaria
- 3. Diploma de sexto grado
- 4. Fotocopia de DPI del padre, madre o encargado.

Programa de Extensión y Mejoramiento de la Enseñanza Media (PEMEM)

Los Institutos Nacionales Experimentales de Educación Básica con Orientación Ocupacional, creados bajo el Programa de Extensión y Mejoramiento de la Enseñanza Media (PEMEM).

La población beneficiada es la egresada del Nivel de Educación Primaria, con una oferta educativa caracterizada por orientaciones ocupacionales, entre las que se destacan las siguientes áreas curriculares:

- Productividad y desarrollo
- Agrícola y agropecuario
- Industrial
- Economía doméstica
- Orientación comercial

Ventajas:

Los Institutos Nacionales Experimentales de Educación Básica con Orientación Ocupacional, presentan las siguientes ventajas para el alumnado:

- Subsidio al transporte
- Edificios propios y acondicionados para las áreas prácticas
- Orientación ocupacional:
 - horticultura
 - floricultura
 - cultivos básicos
 - pecuaria
 - avicultura
 - carpintería
 - reparación y mantenimiento de computadoras
 - panadería
 - belleza
 - entre otras

Todos los servicios están cubiertos por el programa de gratuidad.

Telesecundaria

Es una modalidad de entrega educativa del Ciclo de Educación Básica que presta un servicio formal y escolarizado. Su metodología es constructivista, hace énfasis en el aprendizaje significativo, se apoya en programas audiovisuales con los contenidos de las diferentes áreas curriculares y materiales impresos. Se caracteriza porque un solo docente/facilitador es el responsable del proceso de enseñanza, aprendizaje y evaluación de una sección.

Esta modalidad atiende a jóvenes guatemaltecos de comunidades del área rural, entre las edades de 12 a 15 años, y adultos, de acuerdo al rezago educativo que se haya dado en la comunidad.

Beneficios:

- Los estudiantes cuentan con materiales educativos, autoformativos que propician la formación permanente y motivan a la reflexión. Estos materiales son innovadores, totalmente guatemaltecos, alineados al Currículo Nacional Base (CNB), lo cual redundará en los beneficios de la formación integral de los estudiantes; comprenden guías de aprendizaje y planificadores por cada una de las áreas, además cuentan con el CNB para cada grado.
- Cada establecimiento tiene:
 - Proyector
 - Computadora
 - Textos
 - Bibliotecas móviles
 - Disco duro con programas educativos

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Programa Núcleos Familiares Educativos para el Desarrollo, NUFED

Este programa surge como una alternativa para la formación y capacitación dirigida a jóvenes a través de una formación integral, promueve la participación activa de las familias y otras personas de la comunidad con el propósito de facilitar el desarrollo local. La metodología que implementa el programa es la alternancia. Los estudiantes son atendidos por un solo docente que imparte las áreas del Currículo Nacional Base, del Ciclo de Educación Básica, con una duración de tres años.

Los requisitos para inscribirse en el programa NUFED son:

- Presentar certificado de nacimiento, si es menor de edad, o fotocopia confrontada del Documento Personal de Identificación, (DPI), si es mayor de edad.
- Presentar diploma y certificado de sexto grado de educación primaria aprobada, o en caso de haber egresado del programa PEAC, presentar el certificado de la segunda etapa del programa y el diploma correspondiente.
- Firmar documento de compromiso de estudios y buen comportamiento acorde a la normativa legal de convivencia pacífica (firma el padre y el estudiante).
- Disponibilidad de los padres para apoyar las actividades del centro educativo NUFED (en punto de acta).
- No hay límite de edad, siempre y cuando se cumplan con los requisitos anteriormente expuestos.

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OTROS RECURSOS

Buenas prácticas pedagógicas

- Compendio de experiencias exitosas de participantes y egresados del PADEP/D 2013 https://issuu.com/digeduca/docs/buenas_practicas_padep
- Videos: <http://www.mineduc.gob.gt/digeduca/>
- Experiencias positivas innovadoras de lectura para nivel medio:
- http://www.mineduc.gob.gt/digeduca/documents/investigaciones/2016/PNL_Experiencias_positivas_innovadoras.pdf

Educación especial

- Material de apoyo para docentes regulares: <http://www.mineduc.gob.gt/digeesp/documents/Material%20de%20Apoyo.pdf>
- Reglamento del Programa de becas para estudiantes con discapacidad: http://www.mineduc.gob.gt/digeesp/documents/826_2009_Reglamento_del_Programa_de_Becas_para_Estudiantes_con_Discapacidad.pdf

Guía de adecuaciones curriculares

- http://www.mineduc.gob.gt/digeesp/documents/Manual_de_Adecuaciones_Curriculares.pdf
- Material Pedagógico de la Dirección General de Evaluación e Investigación Educativa –Digeduca:-
- Implementación de CNB en el aula: <http://cnbguatemala.org/>
- USO DE CLAVES DE CONTEXTO. Una estrategia para leer comprensivamente (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%201/Lectura/1_lectura_sexto.pdf
- PREDICCIÓN. Una estrategia para mejorar la comprensión lectora (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%203/Lectura/3_sexto_lectura.pdf

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- DEA PRINCIPAL. Para recrearse y asimilar información cuando se lee (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%202/Lectura/2_Lectura_sexto_.pdf
- DIFERENCIAS Y SIMILITUDES. Para leer comprensivamente (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%204/Lectura%20No.%204/4_sexto_lectura.pdf
- IDENTIFICACIÓN DE LA INTENCIÓN DEL AUTOR. Para comprender un texto (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%205/Lectura/5_sexto_lectura.pdf
- RESOLUCIÓN DE PROBLEMAS (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%201/Matematicas/1_mate_sexto.pdf
- RESOLUCIÓN DE PROBLEMAS CON OPERACIONES BÁSICAS. Para solucionar acontecimientos de la vida cotidiana (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%202/Matematicas/2_sexto_mate.pdf
- LECTURA MATEMÁTICA. Destrezas de comprensión lectora aplicadas a la Matemática (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%203/Matematicas/3_sexto_mate.pdf
- FORMAS, PATRONES Y RELACIONES. Aplicación en las actividades cotidianas (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%204/Matematicas%20No.%204/4_sexto_matematica.pdf
- INTERPRETAR TABLAS Y GRÁFICAS ESTADÍSTICAS. Para hacer inferencias en la vida cotidiana (sexto grado)
- http://www.mineduc.gob.gt/digeduca/documents/cuadernillosPedagogicos/No.%205/Matematicas/5_sexto_mate.pdf

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- El taller del escritor. Redacción para docentes de cuarto, quinto y sexto primaria
- http://www.mineduc.gob.gt/digeduca/documents/taller_del_escritor/EI%20taller%20del%20escritor_4,5,6.pdf
- Material de apoyo para desarrollar la lectura
- El tesoro de la lectura. Textos para desarrollar la lectura comprensiva (primaria 4 a 6 grados)
- http://www.mineduc.gob.gt/digeduca/documents/eltesorodelalectura/3_Lectura_comprendiva.pdf
- El tesoro de la lectura. Material de apoyo para desarrollar la lectura comprensiva (primaria 4 a 6 grados)
- http://www.mineduc.gob.gt/digeduca/documents/eltesorodelalectura/3_Segundo_ciclo.pdf